

Stable nullspace adaptive parameter identification of 6 degree-of-freedom plant and actuator models for underactuated vehicles: Theory and experimental evaluation

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Abstract

Model-based approaches to navigation, control, and fault detection that utilize precise nonlinear models of vehicle plant dynamics will enable more accurate control and navigation, assured autonomy, and more complex missions for such vehicles. This paper reports novel theoretical and experimental results addressing the problem of parameter estimation of plant and actuator models for underactuated underwater vehicles operating in 6 degrees-of-freedom (DOF) whose dynamics are modeled by finite-dimensional Newton-Euler equations. This paper reports the first theoretical approach and experimental validation to identify simultaneously plant-model parameters (parameters such as mass, added mass, hydrodynamic drag, and buoyancy) and control-actuator parameters (control-surface models and thruster models) in 6-DOF. Most previously reported studies on parameter identification assume that the control-actuator parameters are known a priori. Moreover, this paper reports the first proof of convergence of the parameter estimates to the true set of parameters for this class of vehicles under a persistence of excitation condition. The reported adaptive identification (AID) algorithm does not require instrumentation of 6-DOF vehicle acceleration, which is required by conventional approaches to parameter estimation such as least squares. Additionally, the reported AID algorithm is applicable under any arbitrary open-loop or closed-loop control law. We report simulation and experimental results for identifying the plant-model and control-actuator parameters for an L3 OceanServer Iver3 autonomous underwater vehicle. We believe this general approach to AID could be extended to apply to other classes of machines and other classes of marine, land, aerial, and space vehicles.

Keywords

Calibration and identification, adaptive systems, parameter identification, learning and adaptive systems, mechanics, design and control, underactuated robots

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1. Introduction

This paper addresses the problem of simultaneously estimating the plant-model and control-actuator parameters for an underactuated underwater vehicle (UV) operating in 6 degrees-of-freedom (DOF). In particular, we report a new adaptive identification (AID) algorithm, including a novel proof of asymptotic parameter convergence, and associated simulation and experimental results.

To the best of our knowledge, this paper reports the first theoretical approach and experimental validation to identify simultaneously UV dynamical plant-model parameters (for critical terms such as mass, added mass, hydrodynamic drag, and buoyancy) and control-actuator parameters (for control-surface models and thruster models) in 6-DOF. Most previously reported studies on parameter identification assume that the control-actuator parameters are known *a priori*. Moreover, this paper reports the first proof of

convergence of the parameter estimates to the true set of parameters for this class of vehicles when a persistence of excitation condition is satisfied. Most previously reported AID approaches for this class of systems only show stability and boundedness of the parameter estimates.

The performance of a dynamic model in predicting a robotic vehicle's state depends primarily on the accuracy of the model structure and parameters. Accurate estimates of

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用于欠驱动车辆的6自由度工厂和执行器模型的稳定零空间自适应参数识别：理论与实验评估

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摘要

基于模型的导航、控制和故障检测方法，利用精确的非线性车辆动力学模型，将能够实现更精确的控制和导航、可靠的自主性，以及这些车辆执行更复杂任务的能力。本文报告了针对水下欠驱动车辆在六自由度（DOF）下其动力学由有限维牛顿-欧拉方程建模的情况下，进行装置与执行器模型参数估计问题的新理论和实验结果。本文提出了首个理论方法和实验验证，以同时识别六自由度中的装置模型参数（如质量、附加质量、水动力阻力和浮力）和控制执行器参数（控制面模型和推进器模型）。此前大多数关于参数识别的研究假设控制执行器参数是已知的。此外，本文还首次报告了收敛性的证明。

关键词

校准与识别、自适应系统、参数识别、学习与自适应系统、力学、设计与控制、欠驱动机器人

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1. 引言

本文讨论了同时估计植物模型和控制执行器参数的问题，针对在六自由度（DOF）下运行的欠驱动水下航行器（UV）。特别地，我们报告了一种新的自适应辨识（AID）算法，包括渐近参数收敛的新证明，以及相关的仿真和实验结果。

据我们所知，本文首次提出了理论方法并进行了实验验证，以同时识别6自由度下的UV动态植物模型参数（关键项如质量、附加质量、水动力阻力和浮力）以及控制执行器参数（用于控制面模型和推进器模型）。大多数先前报道的参数识别研究假设控制执行器参数是已知的。此外，本文首次报道了……的证明

当满足激励持续性条件时，这类车辆的参数估计收敛到真实的参数集。此前报道的大多数针对这类系统的自适应识别（AID）方法仅显示参数估计的稳定性和有界性。

动态模型在预测机器人车辆状态时的性能主要取决于模型结构和参数的准确性。准确的估计

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these parameters are required for model-based navigation, control, fault detection, and simulation. Indeed, a broad class of nonlinear model-based trajectory-tracking controllers for robot arms developed since the 1980s require exact knowledge of the plant's kinematic and dynamic parameters—examples include nonlinear tracking controllers for second-order rotating plants [Koditschek \(1988\)](#) and exactly linearizing model-based trajectory-tracking controllers for open kinematic chains [Freund \(1983\)](#); [Luh et al. \(1980\)](#).

While the general form of dynamical plant models for underwater vehicles has been understood for decades, [Society of Naval Architects and Marine Engineers \(U.S.\). Technical and Research Committee. Hydrodynamics Subcommittee \(1950\)](#), the dynamic-model parameters—that is, parameters for terms including mass, added mass, hydrodynamic drag, buoyancy, and control actuators—are impossible to determine analytically and are often not provided by manufacturers of the robotic vehicles. In contrast to plant kinematic parameters, which are often easily obtained from design documents and generally do not vary with time, plant dynamic parameters generally must be measured empirically and may change over time. Sources of such changes may include but are not limited to: substantial payload changes to the vehicle or robotic arm; physical changes to the vehicle, such as lengthening or shortening a UV to accommodate different payloads; variation in ballasting and trim conditions; variation in water density which affects vehicle buoyancy; and changes to on-board equipment and instrumentation (both external and internal). Model parameters must be re-estimated whenever the robot physical configuration is significantly altered.

Estimating plant-model parameters is generally more difficult for underactuated, torpedo-shaped UVs than for fully actuated UVs with thrusters because: (i) the reduced actuation available on underactuated UVs limits the plant excitation that can be induced from the control inputs, and (ii) torpedo-shaped UVs and many uncrewed aerial vehicles (UAVs) are often actuated with control surfaces (e.g., fins, wings, elevators, and rudders), which are difficult to characterize independently of the plant-model parameters.

For these reasons, we seek an approach to parameter estimation for underactuated UVs in 6-DOF that simultaneously estimates plant and actuator parameters and can be performed routinely in the field with minimal time and effort by the vehicle operator. We believe the general approach to AID reported here is not limited to underwater vehicles and could be extended to apply to other classes of machines and other classes of marine, land, aerial, and space vehicles.

1.1. Motivation for accurate model identification

Model-based approaches to navigation, control, and fault detection that utilize precise nonlinear models of vehicle dynamics will enable more accurate control and navigation, assured autonomy, and more complex missions for robotic vehicles. In particular:

- (1) Accurately identified model parameters can be utilized in forward simulation to predict vehicle performance and tune gains for various controllers (open-loop and closed-loop), navigation systems, and motion-planning algorithms.
- (2) Accurately identified model parameters can be utilized in model-based state-estimation (navigation) systems that utilize the full UV plant model in their solution [Harris and Whitcomb \(2018\)](#).
- (3) Accurately identified model parameters can enable model-based control algorithms that offer improved performance over traditional controllers. For example, previously reported studies have shown that nonlinear adaptive model-based control (AMBC) can outperform proportional derivative control (PDC) in trajectory tracking for fully actuated UVs [McFarland and Whitcomb \(2014\)](#) and fully actuated robot arms [Craig et al. \(1987\)](#); [Slotine and Li \(1987\)](#); [Sadeh and Horowitz \(1990\)](#); [Whitcomb et al. \(1993\)](#).
- (4) Online approaches for identifying plant parameters may lead to new methods for online fault detection and fault-tolerant control of UVs, for example [Mao and Whitcomb \(2021\)](#).

1.2. Common approaches to parameter identification

The three most common approaches to model-parameter estimation for vehicles with dynamics that cannot be fully characterized analytically, such as UVs, are:

- (1) utilizing data obtained in captive-motion experiments (e.g., towing a vehicle in a hydrodynamic test tank or wind tunnel),
- (2) utilizing data from computational fluid dynamics (CFD) simulations, or
- (3) utilizing data obtained in full-scale experimental trials of a robotic vehicle in controlled free motion (e.g., under open-loop or closed-loop control).

Captive-motion experiments are the standard in naval architecture [Van Manen and Van Ossanen \(1988\)](#); however, they are time consuming, expensive, and difficult to perform properly. Further, the results are valid only for the specific vehicle configuration tested, and it is often impractical to repeat the experiments for every possible configuration of the robotic vehicle. Typically, the experimental setup involves either rotating-arm experiments [Van Manen and Van Ossanen \(1988\)](#) or planar motion mechanism (PMM) experiments, such as towing a vehicle on a carriage with a load cell at a fixed velocity in a hydrodynamic test facility, for example, [Presterio \(2001\)](#). Captive-motion experiments can be quite accurate in certain DOF, but these experiments often require decoupling the DOF, which can lead to model inaccuracies. Typically, separate tests are conducted with and without the control surfaces (fins) installed in order to isolate the fin drag from the body drag, and separate test

这些参数对于基于模型的导航、控制、故障检测和仿真是必需的。实际上,自20世纪80年代以来,为机器人手臂开发的一大类非线性基于模型的轨迹跟踪控制器都需要精确了解装置的运动学和动力学参数——例如用于二阶旋转装置的非线性跟踪控制器(Koditschek, 1988)以及用于开放运动链的精确线性化基于模型的轨迹跟踪控制器(Freund, 1983; Luh 等, 1980)。

虽然水下航行器动力学模型的一般形式已经被理解了几十年,但海军建筑师与海洋工程师学会(美国)技术与研究委员会水动力学分委员会(1950)指出,动力学模型参数——即包括质量、附加质量、水动力阻力、浮力和控制执行器的各项参数——无法通过解析方法确定,并且通常机器人航行器的制造商也不会提供。与通常可以从设计文件中轻松获得且基本不随时间变化的运动学参数不同,系统动力学参数通常必须通过实测获得,并可能随时间变化。这类变化的来源可能包括但不限于:车辆或机械臂的有效载荷发生显著变化;车辆的物理变化,例如为了适应不同的有效载荷而延长或缩短水下航行器;压载和配重条件的变化;水密度的变化

估计植物模型参数通常对于欠驱动的鱼雷形无人水下航行器(UV)比对于具有推进器的全驱动UV更困难,因为:(i)欠驱动UV可用的驱动减少限制了可以从控制输入引发的植物激励,(ii)鱼雷形UV和许多无人航空器(UAV)通常通过控制面(例如,鳍、机翼、副翼和舵)驱动,而这些控制面难以独立于植物模型参数进行表征。

基于这些原因,我们寻求一种用于六自由度欠驱动水下航行器的参数估计方法,该方法能够同时估计系统和执行器参数,并且可以由车辆操作者在现场以最少的时间和精力常规执行。我们认为,这里报道的自适应辨识(AID)通用方法并不限于水下航行器,还可以扩展应用于其他类型的机器以及其他类型的海洋、陆地、空中和空间车辆。

1.1. 精确模型识别的动机

基于模型的导航、控制和故障检测方法,利用车辆动力学的精确非线性模型,将实现更精确的控制和导航、可靠的自主性,以及机器人车辆更复杂的任务。尤其是:

(1) 精确识别的模型参数可以用于前向仿真,以预测车辆性能并调节各种控制器(开环和闭环)、导航系统和运动规划算法的增益。(2) 精确识别的模型参数可以用于基于模型的状态估计(导航)系统,这些系统在其解算中利用完整的UV系统模型 Harris 和 Whitcomb (2018)。(3) 精确识别的模型参数可以启用基于模型的控制算法,从而提供优于传统控制器的性能。例如,以前的研究表明,非线性自适应基于模型的控制(AMBC)在全驱动UV的轨迹跟踪中可以优于比例导数控制(PDC) McFarland 和 Whitcomb (2014) 以及全驱动机器人臂 Craig 等 (1987); Slotine 和 Li (1987); Sadegh 和 Horowitz (1990); Whitcomb 等 (1993)。(4) 在线识别系统参数的方法可能导致

1.2. 参数识别的常用方法

对于动力学无法完全通过解析方法描述的车辆(如无人车),三种最常见的模型参数估计方法是:

(1) 利用在受控运动实验中获得的数据(例如,在水动力测试水槽或风洞中拖曳车辆), (2) 利用来自计算流体动力学(CFD)模拟的数据,或 (3) 利用在控制自由运动中对机器人车辆进行全尺度实验试验中获得的数据(例如,在开环或闭环控制下)。

受控运动实验是船舶结构设计中的标准方法(Van Manen 和 Van Ossanen, 1988);然而,这些实验耗时、昂贵,并且难以正确执行。此外,实验结果仅对所测试的特定车辆配置有效,而且通常难以为机器人车辆的每种可能配置重复实验。通常,实验装置包括旋转臂实验(Van Manen 和 Van Ossanen, 1988)或平面运动机构(PMM)实验,例如在水动力测试设施中以固定速度拖曳车辆并在车辆上安装载荷传感器,例如 Prestero (2001)。受控运动实验在某些自由度上可以非常准确,但这些实验通常需要对自由度进行解耦,这可能导致模型不准确。通常,需要分别进行安装和未安装控制面(鱼鳍)的测试,以便将鱼鳍阻力与车体阻力分开,并进行单独测试

facilities are often required to characterize the fin lift and drag as a function of the angle of attack.

Parameter estimation based on computational fluid dynamics uses numerical models of fluid flow around the vehicle to simulate virtual PMM tests. CFD-based approaches to model identification vary widely in accuracy, time, and cost. Additionally, CFD-based approaches require detailed computer-aided design (CAD) models of the UV, which may be difficult or impossible to obtain from commercial off-the-shelf (COTS) UV manufacturers. If a CAD model is created by the end user for a COTS UV, often the vehicle is modeled as a smooth hull—appendages such as antennas, strobe lights, sensors (e.g., CTD sensors and sonar transducers), and acoustic modem transducers are left unmodeled [Liu et al. \(2020\)](#). An advantage of CFD-based parameter estimation, however, is the ability to inform design choices during the preliminary vehicle-design process [Phillips et al. \(2010\)](#).

While these approaches have advantages, both captive-motion experiments and CFD-based approaches are often infeasible for many UV end users for reasons of cost, time, and practicality.

Parameter identification based on data collected in full-scale experimental trials of UVs in controlled free motion has several advantages over captive-motion experiments and CFD. First, the approach is accessible to any end user who can deploy an existing UV. Additionally, though beyond the scope of this paper, some approaches that utilize full-scale experimental data, such as the approach described herein, can be extended to run in real time during vehicle missions. Such algorithms can also enable online model-based fault detection and fault isolation [Mao and Whitcomb \(2021\)](#).

1.3. Parameter identification of underactuated robotic vehicles

Many robots, such as torpedo-shaped autonomous underwater vehicles (AUVs), aerial drones, and aerospace vehicles are not fully actuated. Instead, these robots are *underactuated*. We adopt the following definition for Underactuated Control Differential Equations from [Tedrake \(2023\)](#): A second-order control differential equation described by the equations $\ddot{q} = f(q, \dot{q}, u, t)$ is *fully actuated* in state $x = [q^T, \dot{q}^T]^T$ and time t if the resulting map f is surjective: for every $\ddot{q}$ there exists a control input u such that $\ddot{q} = f(q, \dot{q}, u, t)$ which produces the desired response. Otherwise it is underactuated (at state x at time t).

The robotic vehicle used to validate our theoretical results, an Iver3 (L3Harris OceanServer, Fall River, MA, USA) AUV, is clearly underactuated. The vehicle has only five control inputs (4 commanded fin angles and 1 commanded main propeller angular velocity), fewer than the UV's 6-DOF. Thus, the Iver3 robotic vehicle is

underactuated by definition, regardless of the particular method used to model the actuators. The Iver3 belongs to the common class of torpedo-shaped AUVs. This class of UV differs significantly from fully-thruster-actuated remotely operated vehicles (ROVs), where the thrusters are often capable of exerting arbitrary 6-DOF forces and moments on the vehicle. Torpedo-shaped UVs are controllable only when in forward motion, are generally incapable of hovering, and are physically unable to track fully general 6-DOF reference trajectories.

Parameter estimation of underactuated UVs presents real theoretical and practical challenges. A main concern is whether the UV parameters are observable. In other words: do the control actuators have sufficient control authority to excite the plant so that the plant model parameters are observable? The authors have previously defined necessary and sufficient conditions for mass and rotational-inertia parameters in rigid-body dynamical plants to be uniformly completely observable (UCO) [Paine and Whitcomb \(2021\)](#). Previously reported simulation studies confirmed that underactuated rigid-body plants can meet these conditions. We conjectured that a similar approach could be used to show UCO of other parameters such as drag coefficients and gravity terms. In this paper, via the equivalent concept of persistence of excitation (PE), we define sufficient conditions for convergence of plant and actuator parameter estimates to a true parameter set.

To the best of our knowledge, with the exception of [Harris et al. \(2018\)](#), [Harris \(2019\)](#), [Paine \(2018\)](#), upon which this work is based, there are no previously reported studies for parameter identification of underactuated robotic vehicles that simultaneously identify plant-model parameters and actuation parameters in 6-DOF.

With few exceptions, namely [McFarland and Whitcomb \(2021\)](#), many previously reported AID methods require model-based adaptive tracking controllers and are not applicable when the UV is operating under any control law other than a specific adaptive tracking controller. On commercially available robotic vehicles, including UVs, the user is often limited to using the proprietary control system provided by the manufacturer, and the user cannot easily replace the manufacturer's proprietary controller with an adaptive trajectory-tracking controller (ATTC). The AID approach reported herein works for robotic plants operating under any control inputs (specifically, it not limited to the case of trajectory-tracking control), and is thus applicable in the common situation of a UV operating with a proprietary, manufacturer-provided controller or in open-loop control.

In summary, the advantages of this AID approach include:

- This AID method simultaneously estimates both plant and actuator parameters that are present in a physics-based model of robot dynamics, either fully actuated or underactuated. Most previously reported studies on

通常需要设施来表征鳍片的升力和阻力随攻角的变化。

基于计算流体力学的参数估计使用车辆周围流体流动的数值模型来模拟虚拟的PMM测试。基于CFD的模型识别方法在精度、时间和成本上差异很大。此外，基于CFD的方法需要详细的计算机辅助设计（CAD）模型，而这些模型可能难以或不可能从商业现成（COTS）无人水下航行器（UV）制造商处获得。如果最终用户为COTS UV创建了CAD模型，通常该车辆会被建模为光滑的船体——天线、闪光灯、传感器（例如CTD传感器和声呐换能器）以及声学调制解调器换能器等附件不会被建模（Liu等，2020）。然而，基于CFD的参数估计的一个优势是能够在初步车辆设计过程中为设计选择提供信息（Phillips等，2010）。

虽然这些方法有其优点，但由于成本、时间和可行性方面的原因，许多紫外线终端用户往往无法使用受控运动实验和基于计算流体力学（CFD）的方法。

基于在受控自由运动中对无人水下航行器进行全尺寸实验试验收集的数据进行参数识别，相较于约束运动实验和计算流体力学（CFD）具有若干优势。首先，该方法对于任何能够部署现有无人水下航行器的最终用户都是可行的。此外，尽管超出了本文的讨论范围，一些利用全尺寸实验数据的方法，例如本文所述的方法，可以扩展到在车辆任务期间实时运行。这类算法还可以实现在线的基于模型的故障检测与故障隔离（Mao和Whitcomb，2021）。

1.3. 弱驱动机器人车辆的参数识别

许多机器人，例如鱼雷形自主水下航行器（AUV）、空中无人机和航天飞行器，并非完全可驱动。相反，这些机器人是欠驱动的。我们采用Tedrake（2023）提出的欠驱动控制微分方程的定义：一个由方程 $\dot{x} = f(x, u, t)$ 描述的 n 阶控制微分方程，如果在状态 $x \in \mathbb{R}^n$ 和时间 t 下，其映射 f 是满射的，则称其为完全驱动：对于每个 x ，存在一个控制输入 u ，使得 $\dot{x} = f(x, u, t)$ 从而产生期望的响应。否则，在时间 t 的状态 x 下，该系统为欠驱动。

用于验证我们理论结果的机器人车辆是一款 Iver3（L3Harris OceanServer，美国马萨诸塞州福尔河）自主水下航行器（AUV），显然是欠驱动的。该车辆只有五个控制输入（4个指令方向舵角和1个指令主推进器角速度），少于该水下车辆的6自由度。因此，Iver3 机器人车辆是

根据定义属于欠驱动，无论用于建模执行器的具体方法如何。Iver3 属于常见的鱼雷形自主水下航行器（AUV）类别。这类水下航行器与完全由推进器驱动的远程操作车辆（ROV）有显著不同，后者的推进器通常能够对车辆施加任意的六自由度（6-DOF）力和力矩。鱼雷形水下航行器只有在前进运动时才可控，通常无法悬停，并且在物理上无法跟踪完全通用的六自由度参考轨迹。

欠驱动无人水下航行器（UV）的参数估计在理论和实际操作上都存在真正的挑战。一个主要问题是UV的参数是否可观测。换句话说：控制执行器是否具有足够的控制能力来激励系统，以使系统模型参数可观测？作者之前已经定义了刚体动力系统中质量和转动惯量参数在均匀完全可观测（UCO）方面的必要和充分条件（Paine和Whitcomb，2021）。之前报道的仿真研究证实，欠驱动刚体系统能够满足这些条件。我们推测，类似的方法也可以用于证明其他参数（如阻力系数和重力项）的UCO。在本文中，通过等效的激励持久性（PE）概念，我们定义了系统和执行器参数估计收敛到真实参数集的充分条件。

据我们所知，除了Harris等人（2018）、Harris（2019）、Paine（2018）这些本研究所基于的工作之外，目前尚无已报道的研究同时识别6自由度欠驱动机器人车辆的植物模型参数和驱动参数，用于参数识别。

除少数例外，即McFarland和Whitcomb（2021）之外，许多先前报道的自适应增益辨识（AID）方法需要基于模型的自适应跟踪控制器，并且在水下无人机（UV）在特定自适应跟踪控制器以外的任何控制律下运行时不适用。在包括水下无人机在内的商用机器人车辆上，用户通常只能使用制造商提供的专有控制系统，且用户无法轻易将制造商的专有控制器替换为自适应轨迹跟踪控制器（ATTC）。本文报道的AID方法适用于在任何控制输入下运行的机器人系统（具体来说，它不限于轨迹跟踪控制的情况），因此适用于UV在使用制造商提供的专有控制器或处于开环控制的常见情况。

总之，这种AID方法的优点包括：

- 这种AID方法同时估计存在于基于物理的机器人动力学模型中的植物和执行器参数，无论是全驱动还是欠驱动。大多数先前报道的研究关于

parameter identification assume that the control-actuator parameters are known *a priori*.

- This AID method is the first reported AID approach with a proof of convergence of the parameter estimates to the true set of parameters for this class of vehicles, provided that a PE condition is satisfied. Most previously reported AID approaches for this class of systems only show stability and boundedness of the parameter estimates.
- This AID method, like other adaptive identifiers, does not require body-relative linear and angular acceleration signals, which is beneficial because body-relative linear acceleration terms are difficult to obtain. Although not discussed in detail here, we find this AID method is more robust to sensor noise when compared to the least-squares-based methods described in [Harris et al. \(2018\)](#).
- This AID method ensures stability and asymptotic convergence of the identification plant velocity error and convergence of the parameter estimates, in contrast to many machine learning (ML) approaches which are generally unable to provide such guarantees.
- The simulation and experimental studies reported here show that AID-estimated parameters can converge with a “single training set” containing on the order of 1000 seconds of free-motion AUV dive data, and the performance of the resulting model was verified in both self-validation and cross-validation.
- This AID method may be performed by a user in the field when the plant dynamics change, unlike captive-motion experiments or CFD.
- This AID method works with any open or closed-loop control law, provided that a PE condition is satisfied. Unlike many previously reported AID methods, this method does not require a trajectory-tracking controller, which is often unsuitable for underactuated vehicles without control authority in all DOF.

2. Literature review

Most previously reported model-based parameter identification methods employ one of two general approaches: (i) least-squares linear regression of experimental data or (ii) adaptive model-based trajectory-tracking control and adaptive identification. Khalil and Dombre provide a detailed overview of these approaches [Khalil and Dombre \(2002\)](#). We separately group machine learning (ML) and neural network (NN) techniques as a third type of method for system identification, because these methods do not explicitly use a dynamic model for all or part of the plant. Instead, these methods largely map plant input-to-output behavior. The following section describes recent work in these three areas and compares the different approaches.

2.1. Least squares and Kalman filtering

A variety of previously reported studies have employed least-squares, total least-squares, or extended Kalman filters to identify plant parameters entering linearly into the plant

equations of motion for robot manipulators [Khosla and Kanade \(1985\)](#); [Atkeson et al. \(1986\)](#); [An et al. \(1988\)](#); [Armstrong et al. \(1986\)](#); [Swevers et al. \(2007\)](#), uncrewed underwater vehicles (UUVs) [Caccia et al. \(2000\)](#); [Alessandri et al. \(1998\)](#); [Martin and Whitcomb \(2014\)](#); [Harris et al. \(2018\)](#), and spacecraft [Norman et al. \(2011\)](#); [Keim et al. \(2006\)](#). Most methods require the time derivative of joint angles or time derivative of 6-DOF body-relative linear and angular velocity, which is difficult to measure directly, and often is numerically estimated by differentiating sensor measurements—a process that is sensitive to sensor noise. Some studies ameliorate the problem of sensor noise through use of signal filtering. However, unless performed using an acausal method, filtering introduces time delay. These issues are (sometimes) adequately handled during offline data processing, but pose a significant challenge for online parameter estimation.

2.2. Adaptive trajectory-tracking control

Adaptive model-based tracking controllers have been proposed for both fully actuated [McFarland and Whitcomb \(2014\)](#), [Sahu and Subudhi \(2014\)](#) and underactuated marine vehicles [Jiang \(2002\)](#), [Aguiar and Hespanha \(2007\)](#), spacecraft [Ahmed et al. \(1998\)](#), [Wong et al. \(2001\)](#), and mobile robots [Fukao et al. \(2000\)](#), [Shojaei et al. \(2011\)](#). However, precise definitions for persistence of excitation (PE) to ensure asymptotic convergence of parameter-estimation error are absent from many studies, such as [Shojaei et al. \(2011\)](#), [Sahu and Subudhi \(2014\)](#), [Wong et al. \(2001\)](#). PE-like conditions were reported in [Aguiar and Hespanha \(2007\)](#), with a focus on stability of the trajectory-tracking error and robustness to parametric uncertainty. Necessary and sufficient PE conditions for asymptotic identification were reported in [Ahmed et al. \(1998\)](#), but only for parameters in the inertia matrix for 3-DOF rotational plants.

In [Craig et al. \(1987\)](#) the authors report a stable adaptive trajectory-tracking algorithm for fully actuated robot arms that require instrumentation of joint acceleration. In [Slotine and Li \(1987\)](#); [Sadegh and Horowitz \(1990\)](#); [Whitcomb et al. \(1993\)](#) the authors report three different stable adaptive trajectory-tracking algorithms for fully actuated robot arms that do not require instrumentation of joint acceleration

2.2.1. Relationship of adaptive identification to adaptive trajectory tracking control. The present study addresses AID; it does not address the problem of simultaneously performing adaptive model identification *and* model-based trajectory tracking control, that is, ATTC. There are numerous previously reported results for ATTC of mechanical systems in which the plant state is controlled to follow a smooth trajectory known *a priori*. ATTC, however, will not work when either (i) the plant is underactuated or (ii) the plant is controlled by an existing controller that cannot be modified, as is often the case with commercially available robot systems, or (iii) the plant is under open-loop control,

参数识别假设控制执行器参数是已知的先验条件。

- 这种自适应识别 (AID) 方法是首次报道的 AID 方法, 对于这类车辆的参数估计收敛到真实参数集合提供了证明, 前提是满足持久激励 (PE) 条件。以往报道的大多数针对这类系统的 AID 方法仅展示了参数估计的稳定性和有界性。
- 这种 AID 方法, 像其他自适应辨识器一样, 不需要与机体相关的线性和角加速度信号, 这很有益, 因为与机体相关的线性加速度项很难获取。尽管此处未详细讨论, 但我们发现, 与 Harris 等人 (2018) 中描述的基于最小二乘法的方法相比, 这种 AID 方法对传感器噪声更具鲁棒性。
- 这种 AID 方法确保了识别过程中的植物速度误差的稳定性和渐近收敛, 以及参数估计的收敛性, 相比之下, 许多机器学习 (ML) 方法通常无法提供此类保证。
- 这里报告的仿真和实验研究表明, AID 估计参数可以通过包含约 1000 秒自由运动自主水下航行器下潜数据的“单一训练集”收敛, 并且所得模型的性能在自我验证和交叉验证中得到了验证。
- 当植物动态发生变化时, 与受限运动实验或计算流体力学 (CFD) 不同, 该 AID 方法可以由现场用户执行。
- 这种 AID 方法适用于任何开环或闭环控制律, 前提是满足 PE 条件。与许多先前报道的 AID 方法不同, 该方法不需要轨迹跟踪控制器, 而轨迹跟踪控制器通常不适用于在所有自由度上都没有控制权的欠驱车辆。

2. 文献综述

大多数先前报道的基于模型的参数识别方法采用两种一般方法之一: (i) 实验数据的最小二乘线性回归, 或 (ii) 自适应基于模型的轨迹跟踪控制和自适应识别。Khalil 和 Dombre 提供了这些方法的详细概述 Khalil 和 Dombre (2002)。我们将机器学习 (ML) 和神经网络 (NN) 技术单独归为系统识别的第三类方法, 因为这些方法并不对全部或部分系统明确使用动态模型。相反, 这些方法主要映射系统输入到输出的行为。以下部分描述了这三个领域的最新工作, 并比较了不同的方法。

2.1. 最小二乘法与卡尔曼滤波

各种先前报道的研究已经采用最小二乘法、总最小二乘法或扩展卡尔曼滤波器来识别线性进入系统的植物参数

机器人操作器的运动方程 Khosla 和 Kanade (1985); Atkeson 等 (1986); An 等 (1988); Armstrong 等 (1986); Swevers 等 (2007), 无人水下航行器 (UUVs) Caccia 等 (2000); Alessandri 等 (1998); Martin 和 Whitcomb (2014); Harris 等 (2018), 以及航天器 Norman 等 (2011); Keim 等 (2006)。大多数方法需要关节角的时间导数或 6 自由度 (6-DOF) 体坐标下的相对线速度和角速度的时间导数, 这在测量上十分困难, 通常通过对传感器测量进行数值求导来估计——这一过程对传感器噪声非常敏感。一些研究通过信号滤波来缓解传感器噪声问题。然而, 除非使用非因果方法, 否则滤波会引入时间延迟。这些问题在离线数据处理中 (有时) 可以得到充分解决, 但对在线参数估计构成了重大挑战。

2.2. 自适应轨迹跟踪控制

自适应基于模型的跟踪控制器已被提出用于完全驱动的系统 (McFarland 和 Whitcomb, 2014; Sahu 和 Subudhi, 2014) 以及欠驱动的海洋车辆 (Jiang, 2002; Aguiar 和 Hespanha, 2007)、航天器 (Ahmed 等, 1998; Wong 等, 2001) 和移动机器人 (Fukao 等, 2000; Shojaei 等, 2011)。然而, 许多研究中缺乏用于确保参数估计误差渐近收敛的激励持续性 (PE) 的精确定义, 例如 Shojaei 等 (2011)、Sahu 和 Subudhi (2014)、Wong 等 (2001)。类似 PE 的条件在 Aguiar 和 Hespanha (2007) 中有报道, 其重点是轨迹跟踪误差的稳定性以及对参数不确定性的鲁棒性。Ahmed 等 (1998) 报告了用于渐近辨识的必要和充分 PE 条件, 但仅针对三自由度旋转系统的惯性矩阵中的参数。

在 Craig 等人 (1987) 中, 作者报告了一种用于全驱动机器人手臂的稳定自适应轨迹跟踪算法, 该算法需要对关节加速度进行仪器化。在 Slotine 和 Li (1987); Sadegh 和 Horowitz (1990); Whitcomb 等人 (1993) 中, 作者报告了三种不同的用于全驱动机器人手臂的稳定自适应轨迹跟踪算法, 这些算法不需要对关节加速度进行仪器化

2.2.1. 自适应辨识与自适应轨迹跟踪控制的关系。本研究涉及自适应辨识 (AID); 它并未涉及同时执行自适应模型辨识和基于模型的轨迹跟踪控制 (即 ATTC) 的问题。此前有许多关于机械系统 ATTC 的研究成果, 其中植物状态被控制以跟随事先已知的平滑轨迹。然而, 当出现以下情况时, ATTC 将无法工作: (i) 植物受控不足; (ii) 植物由现有控制器控制且无法修改, 如商业化机器人系统中常见的情况; 或 (iii) 植物处于开环控制状态。

as is often the case for system identification experiments in the early stages of vehicle development.

In general, papers that report novel adaptive model-based tracking controllers prove convergence of the trajectory-tracking error, and prove stability and boundedness (but not convergence) of the parameter-estimation error.

2.3. Adaptive parameter estimation

Adaptive methods for parameter identification of *linear time-invariant plants* are well understood [Narendra and Annaswamy \(1989\)](#), [Sastry and Bodson \(1989\)](#), [Astrom \(1989\)](#), [Tao \(2003\)](#). In [Hsu et al. \(1987\)](#), the authors report an AID algorithm for robot arms that requires instrumentation of joint acceleration, as well as a low-pass filter approach for the parameter update law to avoid this requirement.

Relatively few studies have considered model-parameter identification of underactuated UVs, for example, [Graver et al. \(2003\)](#), or underactuated robot arms, for example, [Ayusawa et al. \(2014\)](#). To the best of our knowledge, only two previous studies, [Paine and Whitcomb \(2018\)](#), [McFarland and Whitcomb \(2021\)](#), have addressed AID of underactuated UVs.

Few UV parameter identification methods which do not require direct instrumentation of acceleration or trajectory-tracking controllers have been reported. The AID algorithm presented here may prove useful in applications with controlled or uncontrolled plants in which reference trajectory tracking is impractical or infeasible.

AIDs for fully actuated multi-DOF UV plant models were first reported by Smallwood and Whitcomb in [Smallwood and Whitcomb \(2003\)](#), but this AID was limited to fully diagonal plant models in which the dynamics of each DOF is fully decoupled and independent from the dynamics of other DOF. McFarland and Whitcomb in [McFarland and Whitcomb \(2013, 2021\)](#) report an AID for fully coupled 6-DOF UV plant models; the AID algorithm applies to fully actuated or underactuated plants, and the experiments reported utilized 6-DOF actuation. That study is the foundation for the AID reported here. In [Paine and Whitcomb \(2018\)](#), Paine and Whitcomb reported an extension of the AID reported in [McFarland and Whitcomb \(2013\)](#) to 3-DOF underactuated UV plant models and includes a simulation study with Gaussian noise.

2.4. Machine learning

In their pioneering paper [Narendra and Parthasarathy \(1990\)](#), Narendra and Parthasarathy developed the theoretical foundations for the application of neural networks (NNs) to perform system identification of nonlinear dynamical plants. More recently, experimental results using ML and NN-based identification methods have been reported for underactuated UVs [Wehbe et al. \(2017\)](#), [Wehbe and Krell \(2017\)](#), [Karras et al. \(2013\)](#). Broad interest in

using these methods for system identification has led to a variety of different approaches.

In general, ML approaches to model identification appear to be divided on the level to which they utilize physics-based models. Some authors caution against adopting techniques and algorithms from domains void of underlying physics (such as natural language processing) to the natural sciences; instead, these authors argue specifically for increased usage of physics-based models in ML [Willcox et al. \(2021\)](#). An interesting hybrid approach is reported in [Woo et al. \(2018\)](#), wherein the authors decompose the dynamics of a uncrewed surface vessel (USV) into linear and nonlinear terms; they estimate the parameters corresponding to the linear terms using the parameter-estimation technique from [Sonnenburg and Woolsey \(2013\)](#) and use ML for the nonlinear model. A similar hybrid approach for UVs was proposed in [Van de Ven et al. \(2007\)](#). In this case, the underlying structure of the nonlinear dynamic model is maintained via the careful design of separate neural networks to estimate each of the mass, damping, and buoyancy terms. However, other authors in ML, for example, [Bagherzadeh \(2018\)](#), argue against the use of a predetermined structure, saying that the use of predetermined model structure reduces the problem of model identification to one of parameter estimation, which may not fully capture the nonlinearities of the dynamics. Clearly, there is some disagreement in this field, and some researchers report satisfactory results obtained via their particular approach to system identification.

2.5. Approach comparison

We can offer some comparison of these approaches. Least-squares estimation is often performed offline with some signal filtering, whereas the adaptive techniques, including the AID reported herein, can be implemented online. Additionally, least-squares estimation typically requires signals of translational and rotational body-frame acceleration, while AIDs generally require only body-frame velocities. Although some argue that ML and NN methods may be able to perform system identification by capturing complex nonlinear dynamics that are difficult to model, these methods are not guaranteed to be stable or, in some cases, to generalize beyond the training data set. Additionally, in most cases, significant computational time and training data are needed to complete estimates using ML and NN methods. On the other hand, the adaptive techniques can be proven to be stable and to provide asymptotic error convergence, as we will show in Section 5.2 and 5.3.

3. Mathematical conventions

For a matrix $A \in \mathbb{R}^{n \times n}$, we denote eigenvalues ordered from minimum to maximum magnitude as $\lambda_{\min}(A) \leq \lambda_{n-1}(A) \dots \leq \lambda_2(A) \leq \lambda_{\max}(A)$. For a vector $x \in \mathbb{R}^n$, we define the following vector norm:

- Euclidean (L^2) norm:

正如在车辆开发早期阶段的系统辨识实验中常见的情况一样。

一般来说, 报道新型自适应模型跟踪控制器的论文会证明轨迹跟踪误差的收敛性, 并证明参数估计误差的稳定性和有界性(但不证明收敛性)。

2.3. 自适应参数估计

线性时不变系统参数辨识的自适应方法已被充分理解(Narendra 和 Annaswamy, 1989; Sastry 和 Bodson, 1989; Astrom, 1989; Tao, 2003)。在 Hsu 等人(1987)中, 作者报告了一种用于机器人机械臂的自适应辨识(AID)算法, 该算法需要对关节加速度进行仪器测量, 并且提出了一种用于参数更新律的低滤波方法以避免该要求。

相对较少的研究考虑了欠驱动水下航行器的模型参数识别, 例如, Graver 等人(2003), 或欠驱动机器人手臂, 例如, Ayusawa 等人(2014)。据我们所知, 只有两项先前的研究, Paine 和 Whitcomb(2018), McFarland 和 Whitcomb(2021), 涉及了欠驱动水下航行器的自适应识别(AID)。

已经报道的几种紫外线参数识别方法不需要直接对加速度或轨迹跟踪控制器进行检测。本文提出的AID算法在受控或不受控的系统中可能会非常有用, 这些系统中参考轨迹跟踪是不切实际或不可行的。

完全驱动的多自由度水下车辆(UV)系统模型的自适应逆动力学(AID)最早由Smallwood和Whitcomb在Smallwood和Whitcomb(2003)中报道, 但这种AID仅限于完全对角的系统模型, 其中每个自由度的动力学完全解耦, 并且独立于其他自由度的动力学。McFarland和Whitcomb在McFarland和Whitcomb(2013, 2021)中报道了一种用于完全耦合的6自由度UV系统模型的AID; 该AID算法适用于完全驱动或欠驱动的系统, 并且所报道的实验使用了6自由度的驱动。这项研究是本文所报道的AID的基础。在Paine和Whitcomb(2018)中, Paine和Whitcomb报道了对McFarland和Whitcomb(2013)中报道的AID的扩展, 适用于3自由度的欠驱动UV系统模型, 并包括了带高斯噪声的仿真研究。

2.4. 机器学习

在他们开创性的论文中, Narendra 和 Parthasarathy(1990)发展了神经网络(NN)在非线性动力系统识别中的应用理论基础。最近, 利用基于机器学习(ML)和神经网络的识别方法对欠驱动无人潜水器(UVs)的实验结果已有报道, 参见 Wehbe 等人(2017)、Wehbe 和 Krell(2017)、Karras 等人(2013)。广泛的兴趣在于

使用这些方法进行系统识别已经导致了各种不同的方法。

一般来说, 机器学习(ML)在模型识别的方法似乎在利用基于物理的模型的程度存在分歧。一些作者警告不要将来自缺乏基础物理的领域(例如自然语言处理)的技术和算法应用于自然科学; 相反, 这些作者特别主张在机器学习中增加基于物理模型的使用(Willcox 等, 2021)。Woo 等(2018)报告了一种有趣的混合方法, 作者将无人水面艇(USV)的动力学分解为线性项和非线性项; 他们使用 Sonnenburg 和 Woolsey(2013)的参数估计技术来估计线性项对应的参数, 并对非线性模型使用机器学习。Van de Ven 等(2007)提出了针对无人水下航行器(UV)的类似混合方法。在这种情况下, 通过精心设计的独立神经网络来估计每个参与非线性动态模型的结构, 从而保持了非线性动态模型的基础结构。

2.5. 方法比较

我们可以对这些方法进行一些比较。最小二乘估计通常在离线情况下进行, 并伴随一定的信号滤波, 而自适应技术, 包括本文报道的AID, 可以实现在线应用。此外, 最小二乘估计通常需要平移和旋转体坐标系加速度的信号, 而AID通常只需要体坐标系速度。尽管有人认为最大似然(ML)和神经网络(NN)方法可能通过捕捉难以建模的复杂非线性动态来进行系统辨识, 但这些方法不能保证稳定性, 或者在某些情况下不能超越训练数据集进行泛化。此外, 在大多数情况下, 使用ML和NN方法进行估计需要大量的计算时间和训练数据。另一方面, 自适应技术可以被证明是稳定的, 并且能够提供渐近误差收敛, 正如我们将在第5.2节和5.3节中展示的那样。

3. 数学约定

对于矩阵 $A \in \mathbb{R}^{n \times n}$, 我们将按从最小到最大幅值排序的特征值记为 $\lambda_{\min}(A) \leq \lambda_{n-1}(A) \leq \lambda_2(A) \leq \lambda_{\max}(A)$ 。对于向量 $x \in \mathbb{R}^n$, 我们定义如下向量范数:

- 欧几里得 (L^2) 范数:

$$\|x\|_2 = (x^T x)^{1/2} \quad (1)$$

We also define the following norms on the matrix A :

- Spectral norm:

$$\|A\|_2 = \max_{\|x\|_2=1} \|Ax\|_2 \quad (2)$$

- Frobenius norm:

$$\|A\|_F = \left(\sum_{i=1}^n \sum_{j=1}^n a_{ij}^2 \right)^{1/2} \quad (3)$$

noting that $\|A\|_2 \leq \|A\|_F$. When not specified, the $\|\cdot\|_2$ norm is implied for both vectors and matrices.

For a vector $x \in \mathbb{R}^n$, we also define the functions:

- The diagonal matrix operator $\text{diag}(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ such that $\forall x \in \mathbb{R}^n$,

$$\text{diag}(x) = \begin{bmatrix} x_1 & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & x_n \end{bmatrix} \quad (4)$$

- The skew-symmetric operator $(\cdot)^\wedge : \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 3}$ such that $\forall x = [x_1 \ x_2 \ x_3]^T \in \mathbb{R}^3$,

$$x^\wedge = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} \quad (5)$$

- The $se(3)$ adjoint operator $ad(\cdot) : \mathbb{R}^6 \rightarrow \mathbb{R}^{6 \times 6}$ such that $\forall v = [v^T \ \omega^T]^T \in \mathbb{R}^6$,

$$ad_{se(3)}(v) = \begin{bmatrix} \omega^\wedge & 0_{3 \times 3} \\ v^\wedge & \omega^\wedge \end{bmatrix} \quad (6)$$

4. Finite-dimensional dynamical plant models for 6-DOF vehicles

We consider the class of plants which can be modeled by the Newton-Euler equations in body-relative coordinates. We assume the inertial terms include the mass of the body used in classical rigid-body dynamics, and also include the additional “added mass” or “hydrodynamic mass” arising from the accelerating/decelerating submerged body’s interaction with surrounding fluid. Furthermore, we consider that such plants can be subject to quadratic drag forces and net gravitational forces and moments.

4.1. State and control input representation

- $\eta(t) = [x(t)^T \ \varphi(t)^T]^T \in \mathbb{R}^6$ is the position and orientation of the body frame with respect to an inertial frame of reference, which contains the position of the body-frame origin $x(t) \in \mathbb{R}^3$ and its orientation in Euler angles

$\varphi(t) \in \mathbb{R}^3$. We assume that roll and pitch are bounded to avoid the kinematic singularities arising in the Euler angle representation.

- $v(t) = [v(t)^T \ \omega(t)^T]^T = [v_1 \ v_2 \ v_3 \ v_4 \ v_5 \ v_6]^T \in \mathbb{R}^6$ is the body-frame velocity, which contains the linear body velocity, $v(t) = [v_1 \ v_2 \ v_3]^T \in \mathbb{R}^3$, and angular body velocities, $\omega(t) = [v_4 \ v_5 \ v_6]^T \in \mathbb{R}^3$.
- $\dot{v}(t) = [\dot{v}_1 \ \dot{v}_2 \ \dot{v}_3 \ \dot{v}_4 \ \dot{v}_5 \ \dot{v}_6]^T \in \mathbb{R}^6$ is the time derivative of body-frame velocity.
- $\xi(t) \in \mathbb{R}^{k+i}$ is the control input vector, with k propellers and i hydrodynamic control surfaces (e.g., fins).

The control inputs $\xi(t)$ for the specific vehicle studied in this paper are defined in Section 4.3 and [Appendix A](#).

4.2. Commonly accepted plant model

This section describes the commonly accepted finite-dimensional nonlinear equations of motion for 6-DOF vehicles subject to quadratic drag and gravitational forces. We will provide a brief history of the development of this model for 6-DOF *underwater* vehicles. However, we argue that the nonlinear dynamics of many 6-DOF plants, including terrestrial, aerospace, and underwater vehicles, can be modeled by these same equations of motion, which are simply the Newton-Euler equations with additional terms to represent drag, buoyancy, and control forces.

4.2.1. History of the development of plant models for underwater vehicles. The dynamics of a rigid-body underwater vehicle includes the finite-dimensional dynamics of the rigid-body vehicle body itself and the infinite-dimensional dynamics of the fluid surrounding the vehicle. While the former can be described by a finite-dimensional ordinary differential equation (ODE), the latter is described by the incompressible Navier–Stokes equation. With the exception of a few special cases, the infinite-dimensional Navier–Stokes equation has no closed-form solution and, for the case of marine vehicles, generally cannot be solved in real time ([Larsson et al. 1998](#)).

The most commonly accepted finite-dimensional models for submarine vehicles trace their lineage to studies beginning in the late 1950s and early 1960s at the U.S. Navy’s David Taylor Model Basin, ([Goodman, 1960](#), [Imlay, 1961](#)), which referred to mass of the system resulting from immersion in a fluid as “added mass” and “derivative mass.” In [Gertler and Hagen \(1967\)](#), [Feldman \(1979\)](#) the authors expanded upon these equations, which have since become widely known as the “Standard equations of motion for submarine simulation” ([Gertler and Hagen, 1967](#)) and the “DTNSDC revised standard submarine equations of motion” ([Feldman, 1979](#)). More recently, Fossen in [Fossen \(1994, 2002\)](#) built upon these earlier studies to express in matrix form the equations of motion of an immersed body subject to non-conservative hydrodynamic forces. The matrix model is now commonly used throughout the literature.

$$\|x\|_2 = (x^T x)^{1/2} \quad (1)$$

我们还在矩阵 A 上定义以下范数：• 谱范数：

$$\|A\|_2 = \max_{\|x\|_2=1} \|Ax\|_2 \quad (2)$$

• 弗罗贝尼乌斯范数：

$$\|A\|_F = \left(\sum_{i=1}^n \sum_{j=1}^n a_{ij}^2 \right)^{1/2} \quad (3)$$

注意 $kAk_2 \leq kAk_F$ 。当未指定时， $k \cdot k_2$ 范数对向量和矩阵都适用。

对于向量 $x \in \mathbb{R}^n$ ，我们还定义以下函数：

- 对角矩阵算子 $\text{diag} \delta \cdot \mathbb{P} : \mathbb{R}^n \rightarrow \rightarrow \mathbb{R}^{n \times n}$ ，使得 $\forall x \in \mathbb{R}^n$ ，

$$\text{diag}(x) = \begin{bmatrix} x_1 & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & x_n \end{bmatrix} \quad (4)$$

- 斜对称算子 $\delta \cdot \mathbb{P}^\wedge : \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 3}$ ，使得 $\forall x \begin{bmatrix} 1/4 & 1/2 x_1 & x_2 & x_3 \end{bmatrix}^T \in \mathbb{R}^3$ ，

$$x^\wedge = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} \quad (5)$$

- $\text{SE}(3)$ 的伴随算子 $\text{ad} \delta \cdot \mathbb{P} : \mathbb{R}^6 \rightarrow \rightarrow \mathbb{R}^{6 \times 6}$ ，使得 $\forall v = \begin{bmatrix} 1/2 v^T & \omega^T \end{bmatrix}^T \in \mathbb{R}^6$ ，

$$\text{ad}_{\text{se}(3)}(v) = \begin{bmatrix} \omega^\wedge & 0_{3 \times 3} \\ v^\wedge & \omega^\wedge \end{bmatrix} \quad (6)$$

4. 用于6自由度车辆的有限维动力学系统模型

我们考虑可以用体坐标下的牛顿-欧拉方程建模的植物类。我们假设惯性项包括经典刚体动力学中使用的物体质量，还包括由于加速/减速的浸入水体的物体与周围流体相互作用而产生的额外“附加质量”或“流体动力质量”。此外，我们认为这些植物可能会受到二次阻力以及净重力和力矩的作用。

4.1. 状态与控制输入表示

- $\eta \delta \mathbb{P} = [x \delta \mathbb{P}^T \varphi \delta \mathbb{P}^T]^T \in \mathbb{R}^6$ 是相对于惯性参考系的机体坐标系的位置和方向，其中包含机体坐标系原点的位置 $x \delta \mathbb{P} \in \mathbb{R}^3$ 及其欧拉角表示的方向

$\varphi \delta \mathbb{P} \in \mathbb{R}^3$ 。我们假设滚转角和俯仰角是有界的，以避免欧拉角表示中出现的运动学奇异性。

- $v \delta \mathbb{P} = \begin{bmatrix} 1/2 v \delta \mathbb{P}^T \omega \delta \mathbb{P}^T \end{bmatrix}^T = \begin{bmatrix} 1/2 v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{bmatrix}^T \in \mathbb{R}^6$ 是机体坐标系下的速度，它包含线性机体速度， $v \delta \mathbb{P} = \begin{bmatrix} 1/2 v_1 & v_2 & v_3 \end{bmatrix}^T \in \mathbb{R}^3$ ，以及角速度， $\omega \delta \mathbb{P} = \begin{bmatrix} 1/2 v_4 & v_5 & v_6 \end{bmatrix}^T \in \mathbb{R}^3$ 。
- $\dot{v} \delta \mathbb{P} \begin{bmatrix} 1/4 & 1/2 & \dots & \dots & \dots & \dots \end{bmatrix}^T \in \mathbb{R}^6$ 是机体坐标系速度的时间导数。
- $\xi \delta \mathbb{P} \in \mathbb{R}^{k+i}$ 是控制输入向量，包含 k 个推进器和 i 个水动力控制面（例如，鳍）。

本文研究的特定车辆的控制输入 $\xi(t)$ 在第4.3节和附录A中定义。

4.2. 常用的植物模型

本节描述了常被接受的有限维非线性运动方程，这些方程适用于受二次阻力和重力作用的六自由度（6-DOF）车辆。我们将简要回顾该模型在六自由度水下车辆发展中的历史。然而，我们认为，包括陆地、航空航天和水下车辆在内的许多六自由度系统的非线性动力学，可以通过这些相同的运动方程来建模，这些方程只是牛顿-欧拉方程，并增加了表示阻力、浮力和控制力的项。

4.2.1. 水下航行器植物模型发展的历史。刚体水下航行器的动力学包括刚体车辆本身的有限维动力学以及环绕车辆的流体的无限维动力学。前者可以用有限维常微分方程（ODE）描述，而后者则用不可压缩的纳维-斯托克斯方程描述。除少数特殊情况外，无限维的纳维-斯托克斯方程没有闭式解，并且在海洋航行器的情况下，通常无法实时求解（Larsson 等, 1998）。

用于潜艇车辆的最常见的有限维模型，其渊源可追溯到20世纪50年代末和60年代初美国海军大卫·泰勒模型水池的研究（Goodman, 1960, Imlay, 1961），该研究将系统因浸入流体而产生的质量称为“附加质量”和“导数质量”。在Gertler和Hagen（1967）、Feldman（1979）中，作者扩展了这些方程，这些方程后来被广泛称为“潜艇仿真的标准运动方程”（Gertler和Hagen, 1967）和“DTNSDC修订标准潜艇运动方程”（Feldman, 1979）。最近，Fossen在Fossen（1994, 2002）中基于这些早期研究，将浸入体在受非保守流体动力作用下的运动方程以矩阵形式表示。该矩阵模型现在已被文献广泛采用。

4.2.2 Plant model. The commonly accepted second-order finite-dimensional dynamic model of this class of dynamical plants, as described in a body-relative frame coincident with the center of mass, is given by

$$\tau(v, \varphi, \xi) = M\dot{v} + C(v)v + D_q(v)v + \mathcal{G}(\varphi) \quad (7)$$

where

- $\tau(v, \varphi, \xi) \in \mathbb{R}^6$ is the vector of control-actuator forces and moments acting on the rigid body,
- $M \in \mathbb{R}^{6 \times 6}$ is the diagonal positive definite symmetric (PDS) inertia matrix,
- $C(v) \in \mathbb{R}^{6 \times 6}$ is the Coriolis matrix,
- $D_q(v) \in \mathbb{R}^{6 \times 6}$ is the quadratic hydrodynamic drag matrix, and
- $\mathcal{G}(\varphi) \in \mathbb{R}^6$ is the vector of buoyancy or gravitational forces and moments.

The body-frame angular velocity vector $v(t) \in \mathbb{R}^6$ and the time derivative of the body-frame position and attitude vector $\dot{\eta}(t) \in \mathbb{R}^6$ are related by the kinematic equation

$$\dot{\eta}(t) = J(\eta(t))v(t) \quad (8)$$

where $J(\eta) \in \mathbb{R}^{6 \times 6}$ is the usual kinematic Jacobian relation between $v(t)$ and $\dot{\eta}(t)$, which remains bounded for bounded pitch, as described in Section 2.2.1 of Fossen (1994).

Rearranging (7) to find the time derivative of the body velocity, $\dot{v}(t)$, yields

$$\dot{v} = M^{-1}(-C(v)v - D_q(v)v - \mathcal{G}(\varphi) + \tau(v, \varphi, \xi)). \quad (9)$$

In the sequel, for convenience we will use the term “plant acceleration” to denote $\dot{v}(t)$, the time derivative of the body velocity defined in (9).

We assume the mass matrix is diagonal and can have different inertia quantities in each translational degree of freedom, that is, each diagonal entry of M can be unique. This is a common way to model hydrodynamic “added mass,” that is

$$M = \text{diag}([m_{11} \ m_{22} \ m_{33} \ m_{44} \ m_{55} \ m_{66}]^T) \quad (10)$$

$$= \text{diag}(\theta_m) \quad (11)$$

where $\theta_m \triangleq [m_{11} \ m_{22} \ m_{33} \ m_{44} \ m_{55} \ m_{66}]^T \in \mathbb{R}^6$ is the mass parameter vector. Because the mass parameters enter linearly into the vector-valued function $M\dot{v} \in \mathbb{R}^6$, we can factor $M\dot{v}$ such that

$$M\dot{v} = W_m(\dot{v})\theta_m, \quad (12)$$

where $W_m(\dot{v}) \in \mathbb{R}^{6 \times 6}$ is the mass regressor matrix-valued function.

We parameterize the Coriolis matrix $C(v)$ from M ,

$$C(v) = \begin{bmatrix} 0 & -(M_{11}v)^\wedge \\ -(M_{11}v)^\wedge & -(M_{22}\omega)^\wedge \end{bmatrix} \quad (13)$$

where $(\cdot)^\wedge$ is the skew-symmetric operator defined in (5), $M_{11} \triangleq \text{diag}([m_{11} \ m_{22} \ m_{33}]^T)$ and $M_{22} \triangleq \text{diag}([m_{44} \ m_{55} \ m_{66}]^T)$. Equivalently, we can write

$$C(v)v = \text{ad}_{\text{se}(3)}(v)Mv, \quad (14)$$

and because the mass parameters enter linearly into (14), we have the factorization

$$C(v)v = \text{ad}_{\text{se}(3)}(v)W_m(v)\theta_m \quad (15)$$

where $W_m(v) \in \mathbb{R}^{6 \times 6}$ is the same mass regressor matrix as in (12), with the velocity v instead of the plant acceleration $\dot{v}$ as its argument, and θ_m is the same mass parameter vector.

We assume the quadratic drag matrix, $D_q(v)$, is diagonal positive semidefinite such that

$$D_q(v) = \text{diag}(|v|) \text{diag}([d_{11} \ d_{22} \ d_{33} \ d_{44} \ d_{55} \ d_{66}]^T) \quad (16)$$

$$= \text{diag}(|v|)\text{diag}(\theta_d) \quad (17)$$

with $\theta_d \triangleq [d_{11} \ d_{22} \ d_{33} \ d_{44} \ d_{55} \ d_{66}]^T \in \mathbb{R}^6$ containing all of the drag parameters. We can again factor the linearly-entering parameters from (17), yielding

$$D_q(v)v = \text{diag}(|v|)\text{diag}(\theta_d)v \quad (18)$$

$$= \text{diag}(|v|)\text{diag}(v)\theta_d \quad (19)$$

$$= W_d(v)\theta_d \quad (20)$$

where $W_d(v) \in \mathbb{R}^{6 \times 6}$ is the drag regressor matrix-valued function.

We define the function $\mathcal{G}(\varphi) \in \mathbb{R}^{6 \times 1}$, which represents the gravitational forces and moments arising from net effective mass and righting moments,

$$\mathcal{G}(\varphi) = \begin{bmatrix} R(\varphi)^T e_3 & 0_{3 \times 3} \\ 0_{3 \times 1} & (R^T(\varphi)e_3)^\wedge \end{bmatrix} \begin{bmatrix} g \\ b \end{bmatrix}, \quad (21)$$

where $e_3 = [0 \ 0 \ 1]^T$, $g = g_c(m - \rho \nabla) \in \mathbb{R}$ is the net effective buoyant force (in Newtons) with gravitational acceleration constant g_c , dry mass m , and displacement volume ∇ , which displaces fluid with density ρ . The vector $b = g_c \bar{b} \rho \nabla \in \mathbb{R}^3$ represents the net righting torque (in Newton-meters) resulting from a center of buoyancy not coincident with the center of mass, where $\bar{b} \in \mathbb{R}^3$ is the vector from the center of mass to the center of buoyancy. We note that this formulation assumes the origin of the body frame is at the center of mass of the UV. Factoring the linearly-entering parameters from (21), we have

$$\mathcal{G}(\varphi) = W_g(\varphi)\theta_g, \quad (22)$$

where $W_g(\varphi) \in \mathbb{R}^{6 \times 4}$ is the gravitational regressor matrix-valued function and $\theta_g = [g \ b^T]^T \in \mathbb{R}^4$ is the gravitational parameter vector, which is composed of both the buoyancy and righting-moment parameters.

4.2.2 植物模型。该类动力植物的常用二阶有限维动态模型，如在与质心重合的物体相对坐标系中所描述，给出如下

$$\tau(\mathbf{v}, \varphi, \xi) = M\dot{\mathbf{v}} + C(\mathbf{v})\mathbf{v} + D_q(\mathbf{v})\mathbf{v} + \mathcal{G}(\varphi) \quad (7)$$

哪里

- $\tau \in \mathbb{R}^6$ 是作用于刚体的控制执行器力和力矩向量,
- $M \in \mathbb{R}^{6 \times 6}$ 是对角正定对称(PDS)惯性矩阵,
- $C(\mathbf{v}) \in \mathbb{R}^{6 \times 6}$ 是科里奥利矩阵,
- $D_q(\mathbf{v}) \in \mathbb{R}^{6 \times 6}$ 是二次水动力阻力矩阵,
- $\mathcal{G}(\varphi) \in \mathbb{R}^6$ 是浮力或重力作用力和力矩向量。

机体坐标系角速度向量 $\dot{\boldsymbol{\theta}} \in \mathbb{R}^3$ 与机体坐标系位置 and 姿态向量的时间导数 $\dot{\boldsymbol{\eta}} \in \mathbb{R}^6$ 之间通过运动学方程相关联

$$\dot{\boldsymbol{\eta}}(t) = J(\boldsymbol{\eta}(t))\mathbf{v}(t) \quad (8)$$

其中 $J \in \mathbb{R}^{6 \times 6}$ 是 $\mathbf{v}(t)$ 和 $\dot{\boldsymbol{\eta}}(t)$ 之间的常用运动学雅可比关系, 对于有界的俯仰角保持有界, 如 Fossen (1994) 的第 2.2.1 节所述。

将公式 (7) 重新排列以求得物体速度 $\mathbf{v}(t)$ 的时间导数, 得到

$$\dot{\mathbf{v}} = M^{-1}(-C(\mathbf{v})\mathbf{v} - D_q(\mathbf{v})\mathbf{v} - \mathcal{G}(\varphi) + \tau(\mathbf{v}, \varphi, \xi)). \quad (9)$$

在后续内容中, 为方便起见, 我们将使用术语“植物加速度”来表示 $\dot{\mathbf{v}}$, 即在公式(9)中定义的身体速度的时间导数。

我们假设质量矩阵是对角的, 并且在每个平移自由度上可以有不同的惯性量, 也就是说, M 的每个对角线项都可以是独特的。这是一种常见的模拟流体动力学“附加质量”的方法, 也就是说

$$M = \text{diag}([m_{11} \ m_{22} \ m_{33} \ m_{44} \ m_{55} \ m_{66}]^T) \quad (10)$$

$$= \text{diag}(\boldsymbol{\theta}_m) \quad (11)$$

其中 $\boldsymbol{\theta}_m \in \mathbb{R}^6$ 是质量参数向量。因为质量参数线性地进入向量值函数 $M \in \mathbb{R}^{6 \times 6}$, 我们可以将 M 因式分解, 使得

$$M\dot{\mathbf{v}} = W_m(\dot{\mathbf{v}})\boldsymbol{\theta}_m, \quad (12)$$

其中 $W_m \in \mathbb{R}^{6 \times 6}$ 是质量回归矩阵值函数。

我们从 M 参数化科里奥利矩阵 $C(\mathbf{v})$

$$C(\mathbf{v}) = \begin{bmatrix} 0 & -(M_{11}\mathbf{v})^\wedge \\ -(M_{11}\mathbf{v})^\wedge & -(M_{22}\boldsymbol{\omega})^\wedge \end{bmatrix} \quad (13)$$

其中 $(\cdot)^\wedge$ 是在 (5) 中定义的斜对称算子, $M_{11} \in \mathbb{R}^{3 \times 3}$ 和 $M_{22} \in \mathbb{R}^{3 \times 3}$ 等价地, 我们可以写成

$$C(\mathbf{v})\mathbf{v} = \text{ad}_{\text{se}(3)}(\mathbf{v})M\mathbf{v}, \quad (14)$$

并且因为质量参数线性地进入式(14), 我们有因式分解

$$C(\mathbf{v})\mathbf{v} = \text{ad}_{\text{se}(3)}(\mathbf{v})W_m(\mathbf{v})\boldsymbol{\theta}_m \quad (15)$$

其中 $W_m \in \mathbb{R}^{6 \times 6}$ 是与 (12) 中相同的质量回归矩阵, 只是将速度 $\mathbf{v}$ 作为其自变量, 而不是系统加速度 $\dot{\mathbf{v}}$, 而 $\boldsymbol{\theta}_m$ 是相同的质量参数向量。

我们假设二次阻力矩阵 $D_q(\mathbf{v})$ 是对角半正定的, 以便

$$D_q(\mathbf{v}) = \text{diag}(|\mathbf{v}|) \text{diag}([d_{11} \ d_{22} \ d_{33} \ d_{44} \ d_{55} \ d_{66}]^T) \quad (16)$$

$$= \text{diag}(|\mathbf{v}|)\text{diag}(\boldsymbol{\theta}_d) \quad (17)$$

包含所有阻力参数的 $\boldsymbol{\theta}_d \in \mathbb{R}^6$ 。我们可以再次将线性进入的参数从 (17) 中提出, 从而得到

$$D_q(\mathbf{v})\mathbf{v} = \text{diag}(|\mathbf{v}|)\text{diag}(\boldsymbol{\theta}_d)\mathbf{v} \quad (18)$$

$$= \text{diag}(|\mathbf{v}|)\text{diag}(\mathbf{v})\boldsymbol{\theta}_d \quad (19)$$

$$= W_d(\mathbf{v})\boldsymbol{\theta}_d \quad (20)$$

其中 $W_d \in \mathbb{R}^{6 \times 6}$ 是阻力回归矩阵值函数。

我们定义函数 $\mathcal{G}(\varphi) \in \mathbb{R}^6$, 它表示由净有效质量和复位力矩产生的重力和力矩,

$$\mathcal{G}(\varphi) = \begin{bmatrix} R(\varphi)^T \mathbf{e}_3 & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 1} & (R^T(\varphi)\mathbf{e}_3)^\wedge \end{bmatrix} \begin{bmatrix} g \\ b \end{bmatrix}, \quad (21)$$

其中 $\mathbf{e}_3 \in \mathbb{R}^3$ 是净有效浮力 (以牛顿为单位), 其中重力加速度常数为 g , 干质量为 m , 排开体积为 ∇ , 排开密度为 ρ 的流体。向量 $\mathbf{b} \in \mathbb{R}^3$ 表示由浮心不与质心重合产生的净复位力矩 (以牛顿·米为单位), 其中 $\mathbf{b} \in \mathbb{R}^3$ 是从质心指向浮心的向量。我们注意到, 这种表述假设机体坐标系的原点位于无人水下潜航器的质心。从 (21) 式中因式分解线性进入的参数, 我们得到

$$\mathcal{G}(\varphi) = W_g(\varphi)\boldsymbol{\theta}_g, \quad (22)$$

其中 $W_g \in \mathbb{R}^{6 \times 4}$ 是重力回归矩阵值函数, 且 $\boldsymbol{\theta}_g \in \mathbb{R}^4$ 是重力参数向量, 由浮力参数和恢复力矩参数组成。

4.2.3 Actuator model. The vector $\tau(v, \phi, \xi) \in \mathbb{R}^6$ represents the model of control-actuator forces and moments that are a result of body velocity v , attitude of the vehicle ϕ , and the control inputs $\xi \in \mathbb{R}^{k+i}$ to the control actuators, such as fin angle and propeller angular velocity, with parameters $\theta_a \in \mathbb{R}^a$. If the actuation parameters θ_a enter linearly into $\tau(v, \phi, \xi)$, as is true for the UV actuator model described explicitly in [Appendix A](#), then we may factor the control vector such that

$$\tau(v, \phi, \xi) = W_a(v, \phi, \xi)\theta_a, \quad (23)$$

where $W_a(v, \phi, \xi) \in \mathbb{R}^{6 \times a}$ is the (usually) nonlinear actuator regressor matrix and $\theta_a \in \mathbb{R}^a$ is the parameter vector that contains the minimum number of parameters required to fully characterize the control actuators.

The AID in this study is applicable to actuator models of the general form (23), with linearly-entering parameters and the additional requirement that $W_a(v, \phi, \xi)$ be uniformly continuous in v, ϕ, ξ and bounded for bounded v, ϕ, ξ .

4.3. Combined parameter vector

4.3.1. Plant parameters. In summary, our goal is to identify the following plant parameters:

- Mass parameters:

$$\theta_m = [m_{11} \ m_{22} \ m_{33} \ m_{44} \ m_{55} \ m_{66}]^T \in \mathbb{R}^6 \quad (24)$$

- Drag parameters:

$$\theta_d = [d_{11} \ d_{22} \ d_{33} \ d_{44} \ d_{55} \ d_{66}]^T \in \mathbb{R}^6 \quad (25)$$

- Buoyant-force and righting moment parameters:

$$\theta_g = [g \ b_1 \ b_2 \ b_3]^T \in \mathbb{R}^4 \quad (26)$$

For convenience, we define the combined plant parameter vector

$$\theta_p \triangleq [\theta_m^T \ \theta_d^T \ \theta_g^T]^T \in \mathbb{R}^p. \quad (27)$$

4.3.2. Control-actuator parameters. Our goal is to identify the following fin and thruster actuator parameters, described in greater detail in [Appendix A](#). For convenience, we define the control-actuator parameter vector

$$\theta_a \triangleq [\theta_{a_1} \ \dots \ \theta_{a_a}]^T \in \mathbb{R}^a. \quad (28)$$

4.3.3. Plant and actuator parameter vector. We define the full plant and actuator parameter vector as

$$\theta \triangleq [\theta_p^T \ \theta_a^T]^T \in \mathbb{R}^{(p+a)}. \quad (29)$$

For our specific parameterization of the Iver3 AUV used for experimental validation, the dimensionality of θ is

$\mathbb{R}^{(p+a)} = \mathbb{R}^{(16+6)} = \mathbb{R}^{22}$. The details regarding the $a = 6$ actuator parameters for the Iver3 AUV are described in [Appendix A](#). However, in the sequel we continue to use $\theta \in \mathbb{R}^{(p+a)}$ where possible for generality.

4.4. Regressor formulation of plant and actuator dynamics

The equations of motion (7) can be written in regressor form by utilizing (12), (15), (20), (22), and (23), which yields

$$W_a(v, \phi, \xi)\theta_a = [W_m(\dot{v}) + ad_{se(3)}(v)W_m(v)]\theta_m + W_d(v)\theta_d + W_g(\phi)\theta_g \quad (30)$$

$$W_a(v, \phi, \xi)\theta_a = W_p(\dot{v}, v, \phi)\theta_p \quad (31)$$

where the plant regressor $W_p(\dot{v}, v, \phi) \in \mathbb{R}^{6 \times p}$ is defined as

$$W_p(\dot{v}, v, \phi) \triangleq \begin{bmatrix} (W_m(\dot{v}) + ad_{se(3)}(v)W_m(v))^T \\ W_d(v)^T \\ W_g(\phi)^T \end{bmatrix}^T. \quad (32)$$

Rearranging (31) to combine the plant and control-actuator dynamics yields

$$0 = [W_p(\dot{v}, v, \phi), -W_a(v, \phi, \xi)] \begin{bmatrix} \theta_p \\ \theta_a \end{bmatrix} \quad (33)$$

$$= \mathbb{W}(\dot{v}, v, \phi, \xi)\theta \quad (34)$$

where $\mathbb{W}(\dot{v}, v, \phi, \xi) \in \mathbb{R}^{6 \times (p+a)}$ is the (known) time-varying combined plant-and-actuator regressor matrix-valued function, defined as

$$\mathbb{W}(\dot{v}, v, \phi, \xi) \triangleq [W_p(\dot{v}, v, \phi), -W_a(v, \phi, \xi)] \quad (35)$$

$$= \begin{bmatrix} (W_m(\dot{v}) + ad_{se(3)}(v)W_m(v))^T \\ W_d(v)^T \\ W_g(\phi)^T \\ -W_a(v, \phi, \xi)^T \end{bmatrix}^T. \quad (36)$$

4.5. Regressor nullspace properties

4.5.1. Non-uniqueness of the parameter vector. To the best of our knowledge, previous studies of UV model identification, with the exception of [Harris et al. \(2018\)](#) and [Paine \(2018\)](#), assume that the actuator parameters are known. When the actuator parameters are known and the plant model is minimally parameterized, the true plant parameters represent a single, unique point in the linear vector space of possible plant parameters.

In contrast, the AID addressed herein identifies plant and actuator parameters simultaneously. From (34), we note that $\theta \in \mathbb{R}^{(p+a)}$ is in the nullspace of the combined regressor $\mathbb{W}(\dot{v}, v, \phi, \xi)$ for an evolution of $\dot{v}(t), v(t), \phi(t), \xi(t)$ that satisfies the plant equation of motion (7).

4.2.3 执行器模型。向量 $\tau \delta v, \varphi, \xi \in \mathbb{R}^6$ 表示由机体速度 v 、车辆姿态 φ 以及控制输入 $\xi \in \mathbb{R}^{k+i}$ 对控制执行器（如鳍角和螺旋桨角速度）产生的控制执行力和力矩的模型，其中包含参数 $\theta_a \in \mathbb{R}^a$ 。如果执行参数 θ_a 在 $\tau(v, \varphi, \xi)$ 中呈线性关系，如附录 A 中明确描述的水下车辆执行器模型所示，则我们可以将控制向量因式分解为

$$\tau(v, \varphi, \xi) = W_a(v, \varphi, \xi)\theta_a, \quad (23)$$

其中 $W_a \delta v, \varphi, \xi \in \mathbb{R}^{6 \times a}$ 是（通常）非线性执行器回归矩阵， $\theta_a \in \mathbb{R}^a$ 是包含完全描述控制执行器所需的最少参数的参数向量。

本研究中的AID适用于一般形式(23)的执行器模型，其参数线性进入，并额外要求 $W_a(v, \varphi, \xi)$ 在 v, φ, ξ 中是统一连续的，并且对于有界的 v, φ, ξ 是有界的。

4.3. 组合参数向量

4.3.1. 系统参数。总之，我们的目标是识别以下系统参数：

- 质量参数：

$$\theta_m = [m \quad \frac{1}{2}m_{11} \quad m_{22} \quad m_{33} \quad m_{44} \quad m_{55} \quad m_{66}]^T \in \mathbb{R}^6 \quad (24)$$

- 拖动参数：

$$\theta_d = [d_{11} \quad d_{22} \quad d_{33} \quad d_{44} \quad d_{55} \quad d_{66}]^T \in \mathbb{R}^6 \quad (25)$$

- 浮力和矫正力矩参数：

$$\theta_g = [g \quad b_1 \quad b_2 \quad b_3]^T \in \mathbb{R}^4 \quad (26)$$

为了方便，我们定义了组合植物参数向量

$$\theta_p \triangleq [\theta_m^T \quad \theta_d^T \quad \theta_g^T]^T \in \mathbb{R}^p. \quad (27)$$

4.3.2. 控制执行器参数。我们的目标是识别以下鳍和推进器执行器参数，这些参数在附录A中有更详细的描述。为了方便起见，我们定义控制执行器参数向量

$$\theta_a \triangleq [\theta_{a_1} \quad \dots \quad \theta_{a_a}]^T \in \mathbb{R}^a. \quad (28)$$

4.3.3. 植物和执行器参数向量。我们将完整的植物和执行器参数向量定义为

$$\theta \triangleq [\theta_p^T \quad \theta_a^T]^T \in \mathbb{R}^{(p+a)}. \quad (29)$$

对于我们用于实验验证的 Iver3 AUV 的特定参数化， θ 的维度是

$\mathbb{R}^{(16+6)} = \mathbb{R}^{22}$ 。关于 Iver3 AUV 的 $a=6$ 执行器参数的详细信息在附录 A 中描述。然而，在后续内容中，我们尽可能继续使用 $\theta \in \mathbb{R}^{(p+a)}$ 以保持普遍性。

4.4. 植物及回归器的公式化执行器动力学

运动方程 (7) 可以通过利用 (12)、(15)、(20)、(22) 和 (23) 写成回归形式，从而得到

$$W_a(v, \varphi, \xi)\theta_a = [W_m(\dot{v}) + ad_{se(3)}(v)W_m(v)]\theta_m + W_d(v)\theta_d + W_g(\varphi)\theta_g \quad (30)$$

$$W_a(v, \varphi, \xi)\theta_a = W_p(\dot{v}, v, \varphi)\theta_p \quad (31)$$

其中植物回归器 $W_p \delta v, v, \varphi \in \mathbb{R}^{6 \times p}$ 定义为

$$W_p(\dot{v}, v, \varphi) \triangleq \begin{bmatrix} (W_m(\dot{v}) + ad_{se(3)}(v)W_m(v))^T \\ W_d(v)^T \\ W_g(\varphi)^T \end{bmatrix}^T. \quad (32)$$

将 (31) 重新排列以结合植物和控制器-执行器动态得到

$$0 = [W_p(\dot{v}, v, \varphi), -W_a(v, \varphi, \xi)] \begin{bmatrix} \theta_p \\ \theta_a \end{bmatrix} \quad (33)$$

$$= \mathbb{W}(\dot{v}, v, \varphi, \xi)\theta \quad (34)$$

其中 $W \delta v, v, \varphi, \xi \in \mathbb{R}^{6 \times (p+a)}$ 是（已知的）时变组合植物与执行器回归矩阵值函数，定义为

$$\mathbb{W}(\dot{v}, v, \varphi, \xi) \triangleq [W_p(\dot{v}, v, \varphi), -W_a(v, \varphi, \xi)] \quad (35)$$

$$= \begin{bmatrix} (W_m(\dot{v}) + ad_{se(3)}(v)W_m(v))^T \\ W_d(v)^T \\ W_g(\varphi)^T \\ -W_a(v, \varphi, \xi)^T \end{bmatrix}^T. \quad (36)$$

4.5. 回归器零空间属性

4.5.1. 参数向量的非唯一性。据我们所知，先前对UV模型识别的研究，除了Harris等（2018）和Paine（2018）之外，都假设执行器参数是已知的。当执行器参数已知且被控对象模型是最小参数化时，真实的被控对象参数在可能的被控对象参数的线性向量空间中代表一个唯一的点。

相比之下，此处讨论的自适应辨识（AID）同时识别植物和执行器参数。从 (34) 式中，我们注意到 $\theta \in \mathbb{R}^{(p+a)}$ 位于组合回归矩阵 $W(\dot{v}, v, \varphi, \xi)$ 的零空间中，对于一个满足植物运动方程 (7) 的 $\dot{v}(t), v(t), \varphi(t), \xi(t)$ 的演化。

A consequence of this nullspace structure is that many different parameter vectors satisfy the same equations of motion, even when the model is minimally parameterized. As an example, any nonzero scalar multiple of θ equivalently satisfies (34) and thus also satisfies (7).

Given a “true” parameter vector $\theta \in \mathbb{R}^{p+a}$ and all possible bounded evolutions of $\dot{v}, v, \varphi, \xi$ that satisfy the plant equation of motion (7), and thus also satisfy the nullspace relationship (34) for this θ , there is then also a set $P(\theta) \subset \mathbb{R}^{(p+a)}$ of all nonzero parameter vectors θ^* that equivalently satisfy (7) for the same evolutions of $\dot{v}, v, \varphi, \xi$ and (34) for the regressor matrix-valued function $\mathbb{W}(\dot{v}, v, \varphi, \xi)$ of these arguments. This set $P(\theta)$ is the “persistent nullspace” of the regressor, defined as

$$P(\theta) \triangleq \{\theta^* : \theta^* \neq 0 \text{ and } \forall t, \dot{v}, v, \varphi, \xi \text{ s.t.} \\ \mathbb{W}(\dot{v}, v, \varphi, \xi)\theta = 0, \mathbb{W}(\dot{v}, v, \varphi, \xi)\theta^* = 0\} \quad (37)$$

Thus, the “true” parameter vector is not only a unique single vector in the parameter space $\mathbb{R}^{p+a}$, but any element of the set $P(\theta)$. Note that the set $P(\theta)$ does not contain the origin, and that the set $P(\theta) \cup 0$ is a linear vector subspace of $\mathbb{R}^{(p+a)}$.

In Section 4.5.2, we explore the identical input-output plant dynamics that arise from different parameter vectors in $P(\theta)$. These properties will be utilized in Section 5, where we report a novel adaptive identifier for θ and show that under a persistence of excitation (PE) condition, the estimate of θ converges to the true parameter set $P(\theta)$.

4.5.2. Acceleration input-output dynamics. We now prove our earlier claim that the input-output dynamics of (7), namely the resulting plant acceleration given control inputs and parameters, are the same for any parameter vector belonging to the true parameter set $P(\theta)$.

From the plant acceleration (9), after making the appropriate regressor substitutions (15), (20), and (22), as well as writing the true mass M as a function of the parameter vector $\theta \in P(\theta)$ from which it arises, we obtain

$$\dot{v} = M(\theta)^{-1}(-ad_{se(3)}(v)W_m(v)\theta_m - W_d(v)\theta_d \\ - W_g(\varphi)\theta_g + W_a(v, \varphi, \xi)\theta_a). \quad (38)$$

Observing the structure of (38), we define $W_{\dot{v}}(v, \varphi, \xi) \in \mathbb{R}^{6 \times (p+a)}$ as

$$W_{\dot{v}}(v, \varphi, \xi) \triangleq \begin{bmatrix} -ad_{se(3)}(v)W_m(v)^T \\ -W_d(v)^T \\ -W_g(\varphi)^T \\ W_a(v, \varphi, \xi)^T \end{bmatrix} \in \mathbb{R}^{6 \times (p+a)} \quad (39)$$

$$= [W_m(\dot{v}) 0_{6 \times (p+a-6)}] - \mathbb{W}(\dot{v}, v, \varphi, \xi), \quad (40)$$

which allows us to write the plant acceleration $\dot{v}$ (38) as a function of the underlying parameters θ :

$$\dot{v} = \dot{v}(\theta) \triangleq M(\theta)^{-1}W_{\dot{v}}(v, \varphi, \xi)\theta. \quad (41)$$

Now we choose another vector $\theta^* \in P(\theta)$. By definition of $P(\theta)$ (37), we have the nullspace equation

$$\mathbb{W}(\dot{v}, v, \varphi, \xi)\theta^* = 0, \quad (42)$$

which, substituting (40), results in

$$([W_m(\dot{v}), 0_{n \times (p+a-6)}] - W_{\dot{v}}(v, \varphi, \xi))\theta^* = 0 \quad (43)$$

$$M(\theta^*)\dot{v} - W_{\dot{v}}(v, \varphi, \xi)\theta^* = 0 \quad (44)$$

With further rearranging, we have

$$M(\theta^*)\dot{v} = W_{\dot{v}}(v, \varphi, \xi)\theta^* \quad (45)$$

$$\dot{v} = M(\theta^*)^{-1}W_{\dot{v}}(v, \varphi, \xi)\theta^* \quad (46)$$

and thus obtain an equivalent expression to (9) for the plant acceleration $\dot{v}$, where (46) can also be written as a function of its parameter vector

$$\dot{v}(\theta^*) \triangleq M(\theta^*)^{-1}W_{\dot{v}}(v, \varphi, \xi)\theta^* \quad (47)$$

$$= \dot{v} \quad (48)$$

$$= \dot{v}(\theta) \quad (49)$$

which verifies our claim in Section 4.5.1 that any parameter vector $\theta \in P(\theta)$ yields the same input–output behavior.

It is important to note that a nonzero parameter vector $\tilde{\theta} \notin P(\theta)$ by definition does not satisfy $\mathbb{W}(\dot{v}(\theta), v, \varphi, \xi)\tilde{\theta} = 0 \forall t$ and over all $\dot{v}(\theta), v, \varphi, \xi$ that satisfy (7, 34) for the underlying true parameter vector θ . However, given $\tilde{\theta} \notin P(\theta)$ we can define a corresponding acceleration signal arising from these parameters

$$\dot{v}(\tilde{\theta}) \triangleq M(\tilde{\theta})^{-1}W_{\dot{v}}(v, \varphi, \xi)\tilde{\theta}. \quad (50)$$

The term $\dot{v}(\tilde{\theta})$, (50), represents the time derivative of the body velocity given the same velocity v , attitude φ , and control inputs ξ as before, but for the different set of parameters $\tilde{\theta}$.

We can obtain an analogous nullspace relationship to (34) for $\dot{v}(\tilde{\theta})$ by rearranging (50) and substituting the regressor form of the inertia term defined in (12), which yields

$$M(\tilde{\theta})\dot{v}(\tilde{\theta}) = W_{\dot{v}}(v, \varphi, \xi)\tilde{\theta} \quad (51)$$

$$W_m(\dot{v}(\tilde{\theta}))\tilde{\theta}_m = W_{\dot{v}}(v, \varphi, \xi)\tilde{\theta}, \quad (52)$$

and applying (40), we obtain

$$W_m(\dot{v}(\tilde{\theta}))\tilde{\theta}_m - W_{\dot{v}}(v, \varphi, \xi)\tilde{\theta} = 0 \quad (53)$$

$$([W_m(\dot{v}(\tilde{\theta})) 0_{6 \times (p+a-6)}] - W_{\dot{v}}(v, \varphi, \xi))\tilde{\theta} = 0 \quad (54)$$

$$\mathbb{W}(\dot{v}(\tilde{\theta}), v, \varphi, \xi)\tilde{\theta} = 0. \quad (55)$$

这种零空间结构的一个结果是，即使模型使用最少参数化，也有许多不同的参数向量满足相同的运动方程。例如， θ 的任何非零标量倍数同样满足 (34)，因此也满足 (7)。

给定一个“真实”的参数向量 $\theta \in \mathbb{R}^{p+a}$ 以及满足植物运动方程 (7) 的所有可能有界演化的 $\dot{v}$ 、 v 、 φ 、 ξ ，因此也满足对于此 θ 的零空间关系 (34)，那么也存在一个集合 $P(\theta) \subseteq \mathbb{R}^{(p+a)}$ ，包含所有非零参数向量 θ^* ，它们在相同的 $\dot{v}$ 、 v 、 φ 、 ξ 演化下也等效地满足 (7) 以及对于这些参数的回归矩阵值函数 $W(\dot{v}, v, \varphi, \xi)$ 的 (34)。该集合 $P(\theta)$ 被定义为回归器的“持久零空间”。

$$P(\theta) \triangleq \{\theta^* : \theta^* \neq 0 \text{ and } \forall t, \dot{v}, v, \varphi, \xi \text{ s.t.} \\ \mathbb{W}(\dot{v}, v, \varphi, \xi)\theta = 0, \mathbb{W}(\dot{v}, v, \varphi, \xi)\theta^* = 0\} \quad (37)$$

因此，“真实”的参数向量不仅是在参数空间 $\mathbb{R}^{p+a}$ 中的唯一单个向量，而且是集合 $P(\theta)$ 的任意元素。请注意，集合 $P(\theta)$ 不包含原点，并且集合 $P(\theta)$ 是 $\mathbb{R}^{(p+a)}$ 的线性向量子空间。

在第4.5.2节中，我们探讨了在 $P(\theta)$ 中由不同参数向量产生的相同输入-输出系统动态。这些特性将在第5节中使用，我们将在该节中报告一个针对 θ 的新型自适应识别器，并展示在持久激励 (PE) 条件下， θ 的估计会收敛到真实参数集 $P(\theta)$ 。

4.5.2. 加速度输入-输出动力学。我们现在证明之前的论断，即 (7) 的输入-输出动力学，即给定控制输入和参数时产生的系统加速度，对于属于真实参数集 $P(\theta)$ 的任何参数向量都是相同的。

从植物加速度 (9) 出发，在进行适当的回归量替换 (15)、(20) 和 (22) 之后，并将真实质量 M 写成由其产生的参数向量 $\theta \in P(\theta)$ 的函数，我们得到

$$\dot{v} = M(\theta)^{-1}(-ad_{se(3)}(v)W_m(v)\theta_m - W_d(v)\theta_d \\ - W_g(\varphi)\theta_g + W_a(v, \varphi, \xi)\theta_a). \quad (38)$$

观察 (38) 的结构，我们定义 $W_{\dot{v}}(\dot{v}, v, \varphi, \xi) \in \mathbb{R}^{6 \times (p+a)}$ 为

$$W_{\dot{v}}(v, \varphi, \xi) \triangleq \begin{bmatrix} -ad_{se(3)}(v)W_m(v)^T \\ -W_d(v)^T \\ -W_g(\varphi)^T \\ W_a(v, \varphi, \xi)^T \end{bmatrix} \in \mathbb{R}^{6 \times (p+a)} \quad (39)$$

$$= [W_m(\dot{v}) \ 0_{6 \times (p+a-6)}] - \mathbb{W}(\dot{v}, v, \varphi, \xi), \quad (40)$$

这使我们可以将植物加速度 $\dot{v}$ (38) 表示为底层参数 θ 的函数：

$$\dot{v} = \dot{v}(\theta) \triangleq M(\theta)^{-1}W_{\dot{v}}(v, \varphi, \xi)\theta. \quad (41)$$

现在我们选择另一个向量 $\theta^* \in P(\theta)$ 。根据 $P(\theta)$ (37) 的定义，我们得到零空间方程

$$\mathbb{W}(\dot{v}, v, \varphi, \xi)\theta^* = 0, \quad (42)$$

将 (40) 代入，得到

$$([W_m(\dot{v}), 0_{n \times (p+a-6)}] - W_{\dot{v}}(v, \varphi, \xi))\theta^* = 0 \quad (43)$$

$$M(\theta^*)\dot{v} - W_{\dot{v}}(v, \varphi, \xi)\theta^* = 0 \quad (44)$$

经过进一步重排，我们有

$$M(\theta^*)\dot{v} = W_{\dot{v}}(v, \varphi, \xi)\theta^* \quad (45)$$

$$\dot{v} = M(\theta^*)^{-1}W_{\dot{v}}(v, \varphi, \xi)\theta^* \quad (46)$$

从而得到与植物加速度 $\dot{v}$ (9) 等效的表达式，其中 (46) 也可以写成其参数向量的函数

$$\dot{v}(\theta^*) \triangleq M(\theta^*)^{-1}W_{\dot{v}}(v, \varphi, \xi)\theta^* \quad (47)$$

$$= \dot{v} \quad (48)$$

$$= \dot{v}(\theta) \quad (49)$$

这验证了我们在第4.5.1节中的主张，即任何参数向量 $\theta \in P(\theta)$ 都会产生相同的输入-输出行为。

需要注意的是，根据定义，一个非零参数向量 $\tilde{\theta} \in P(\theta)$ 不满足 $W_{\dot{v}}(\dot{v}, v, \varphi, \xi)\tilde{\theta} = 0 \forall t$ ，并且对满足 (7, 34) 的底层真实参数向量 θ 的所有 $\dot{v} \in P(\theta)$ ， $\tilde{\theta}$ 都不成立。然而，给定 $\tilde{\theta} \in P(\theta)$ ，我们可以定义由这些参数产生的相应加速度信号

$$\dot{v}(\tilde{\theta}) \triangleq M(\tilde{\theta})^{-1}W_{\dot{v}}(v, \varphi, \xi)\tilde{\theta}. \quad (50)$$

术语 $\dot{v}(\tilde{\theta})$ (50)，表示在相同速度 v 、姿态 φ 和控制输入 ξ 的情况下，物体速度的时间导数，但对于不同的参数集合 $\tilde{\theta}$ 。

通过重新排列 (50) 并代入在 (12) 中定义的惯性项的回归形式，我们可以获得与 (34) 对 $\dot{v} \in P(\theta)$ 的类似零空间关系，这会产生

$$M(\tilde{\theta})\dot{v}(\tilde{\theta}) = W_{\dot{v}}(v, \varphi, \xi)\tilde{\theta} \quad (51)$$

$$W_m(\dot{v}(\tilde{\theta}))\tilde{\theta}_m = W_{\dot{v}}(v, \varphi, \xi)\tilde{\theta}, \quad (52)$$

并应用 (40)，我们得到

$$W_m(\dot{v}(\tilde{\theta}))\tilde{\theta}_m - W_{\dot{v}}(v, \varphi, \xi)\tilde{\theta} = 0 \quad (53)$$

$$[[W_m(\dot{v}(\tilde{\theta})) \ 0_{6 \times (p+a-6)}] - W_{\dot{v}}(v, \varphi, \xi)]\tilde{\theta} = 0 \quad (54)$$

$$\mathbb{W}(\dot{v}(\tilde{\theta}), v, \varphi, \xi)\tilde{\theta} = 0. \quad (55)$$

In other words, when the plant acceleration in the regressor matrix (34) is replaced by an acceleration resulting from a different parameter class, $\dot{v}(\tilde{\theta})$, we have an analogous nullspace relationship $\tilde{\theta} \in \text{null}(\mathbb{W}(\dot{v}(\tilde{\theta}), v, \varphi, \xi))$, which holds $\forall t$ and over all $\dot{v}(\tilde{\theta}), v, \varphi, \xi$ that satisfy (7,34) given a different parameter class $P(\tilde{\theta})$. We utilize this fact in the design of the adaptive identifier reported in Section 5.

5. Adaptive identifier

This section reports the theoretical derivation and stability analysis of a new adaptive identification (AID) algorithm for plants of the form (7). The AID reported herein significantly expands upon and extends a previous approach for adaptive parameter estimation of UV plant models by McFarland and Whitcomb (McFarland and Whitcomb, 2013, 2021). The specific differences are:

- (1) The AID reported herein performs simultaneous identification of plant *and* actuator model parameters. In contrast, the AID reported in McFarland and Whitcomb (McFarland and Whitcomb, 2013, 2021) identifies only plant parameters.
- (2) This study provides a mathematical proof of stability *and* of parameter-estimate convergence. In contrast, McFarland and Whitcomb (McFarland and Whitcomb, 2013, 2021) provide only a stability proof.

Problem Statement: The goal of this adaptive identification algorithm is to estimate the parameters θ (29) for plants and actuators of the form (7) or, equivalently, of the form (34), using only the available signals of vehicle attitude $\varphi(t)$, linear and angular body velocity $v(t)$, and the control input $\xi(t)$.

Error Coordinates: We define the velocity error $\Delta v(t) \in \mathbb{R}^6$ as the difference between the velocity $\hat{v}(t) \in \mathbb{R}^6$ of an identification plant, whose dynamics are defined in (63), and the actual vehicle velocity, $v(t) \in \mathbb{R}^6$, whose dynamics are defined in (9) (and equivalently in (41)), as

$$\Delta v(t) \triangleq \hat{v}(t) - v(t). \quad (56)$$

We define the vector of parameter estimates as $\hat{\theta}(t) \in \mathbb{R}^{p+a}$ and choose a constant parameter vector $\theta^* \in \mathbb{R}^{p+a}$, $\theta^* \in P(\theta)$, from the set $P(\theta)$ of “true” parameters defined in (37), which results in the error coordinates

$$\Delta\theta(t) \triangleq \hat{\theta}(t) - \theta^*. \quad (57)$$

Task: Our task is to design update laws $\dot{\hat{v}}, \dot{\hat{\theta}}$ for the identification plant and estimated parameters, respectively, such that all signals remain bounded and

$$\lim_{t \rightarrow \infty} \Delta v(t) = 0, \quad (58)$$

$$\lim_{t \rightarrow \infty} \hat{\theta} \in P(\theta). \quad (59)$$

Assumptions: We make the following assumptions.

- $v(t), \eta(t), \xi(t)$ are bounded.
- $\mathbb{W}(\dot{v}(\theta^*), v, \varphi, \xi)$ (35) is locally uniformly continuous in $\dot{v}(\theta^*), v, \varphi, \xi$ and a bounded function of these variables.
- As a result, $W_{\dot{v}}(v, \varphi, \xi)$ (39) is locally uniformly continuous in v, φ, ξ and a bounded function of these variables.
- The control inputs ξ are uniformly continuous in time and bounded.
- The true parameters θ^* are bounded and constant.
- $\hat{v}(t_0) = v(t_0)$, that is $\Delta v(t_0) = 0$
- $\exists \epsilon \in \mathbb{R}^+$ such that $\mathcal{T}(t_0)^{1/2} + \epsilon \leq \lambda_{\min}(M(\theta^*))$, where

$$\mathcal{T}(t_0) \triangleq \lambda_{\max}(\Gamma) \Delta\theta(t_0)^T \Gamma^{-1} \Delta\theta(t_0) \quad (60)$$

and $\Gamma \in \mathbb{R}^{(p+a) \times (p+a)}$ is a parameter adaptation gain matrix which will be defined below.

In the remainder of this section, we present a novel AID algorithm and show that it achieves (58, 59) with all signals bounded. Section 5.1 presents the update laws. Section 5.2 reports a Lyapunov analysis to prove uniform stability with respect to the origin of the error system, which we denote as

$$S^* = \{\Delta v = 0, \Delta\theta = 0\}, \quad (61)$$

convergence of Δv , and boundedness of all signals. Note that $\Delta\theta = 0 \Rightarrow \hat{\theta} = \theta^* \in P(\theta)$, which means S^* is a subset of

$$S = \{\Delta v = 0, \hat{\theta} \in P(\theta)\}. \quad (62)$$

In Section 5.3, we present a novel proof of convergence to the larger set S under a persistence of excitation (PE) condition.

5.1. Update laws and error dynamics

We choose the update laws

$$\dot{\hat{v}} = -\mathcal{A}\Delta v + \dot{v}(\hat{\theta}) \quad (63)$$

$$\dot{\hat{\theta}} = \Gamma \mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)^T \Delta v \quad (64)$$

where (63) defines the adaptive identification plant dynamics, (64) defines the adaptive parameter update law,

$$\mathcal{A} \in \mathbb{R}^{6 \times 6} \quad (65)$$

is a diagonal positive definite symmetric (PDS) gain matrix for the identification plant,

$$\Gamma \in \mathbb{R}^{(p+a) \times (p+a)} \quad (66)$$

is a diagonal PDS gain matrix for the parameter estimates, and $\dot{v}(\hat{\theta})$ is the time derivative of the body velocity arising from the parameter estimate $\hat{\theta}$ instead of the true parameters θ^* , with properties as described in Section 4.5, defined as

换句话说，当回归矩阵 (34) 中的植物加速度被由不同参数类别产生的加速度 $\mathbf{v}\delta\tilde{\theta}\mathbf{P}$ 替代时，我们有一个类似的零空间关系 $\tilde{\theta} \in \text{null}(\mathbf{W}\delta\tilde{\theta}\mathbf{P}, \mathbf{v}, \varphi, \xi\mathbf{P})$ ，它对所有满足 (7,34) 的 $\mathbf{v}\delta\tilde{\theta}\mathbf{P}, \mathbf{v}, \varphi, \xi$ 以及给定不同参数类别 $\mathbf{P}\delta\tilde{\theta}\mathbf{P}$ 的情况成立。我们在第 5 节报告的自适应识别器设计中利用了这个事实。

5. 自适应标识符

本节报告了一种适用于形式为 (7) 的系统的新型自适应辨识 (AID) 算法的理论推导和稳定性分析。本文报道的 AID 在 McFarland 和 Whitcomb (2013, 2021) 提出的 UV 系统模型自适应参数估计的先前方法基础上进行了显著扩展和延伸。具体区别如下：

(1) 本文报道的自适应辨识 (AID) 能够同时识别植物和执行器的模型参数。相比之下，McFarland 和 Whitcomb (McFarland and Whitcomb, 2013, 2021) 报道的自适应辨识仅识别植物参数。(2) 本研究提供了稳定性和参数估计收敛性的数学证明。相比之下，McFarland 和 Whitcomb (McFarland and Whitcomb, 2013, 2021) 仅提供了稳定性证明。

问题陈述：该自适应识别算法的目标是估计形式为 (7) 的植物和执行器的参数 θ (29)，或者等价地，形式为 (34) 的参数，所使用的仅是车辆姿态信号 $\varphi(t)$ 、线性和角速度 $\mathbf{v}(t)$ 以及控制输入 $\xi(t)$ 。

误差坐标：我们将速度误差 $\Delta\mathbf{v}\delta\mathbf{t}\mathbf{P} \in \mathbb{R}^6$ 定义为识别模型的速度 $\mathbf{b}\mathbf{v}\delta\mathbf{t}\mathbf{P} \in \mathbb{R}^6$ 与实际车辆速度 $\mathbf{v}\delta\mathbf{t}\mathbf{P} \in \mathbb{R}^6$ 之间的差，其中识别模型的动力学在 (63) 中定义，实际车辆速度的动力学在 (9) (以及等效的 (41)) 中定义，如下所示

$$\Delta\mathbf{v}(t) \triangleq \hat{\mathbf{v}}(t) - \mathbf{v}(t). \quad (56)$$

我们将参数估计向量定义为 $\mathbf{b}\theta\delta\mathbf{t}\mathbf{P} \in \mathbb{R}^{p+a}$ ，并从 (37) 中定义的“真实”参数集合 $\mathbf{P}(\theta)$ 中选择一个常数参数向量 $\theta^* \in \mathbb{R}^{p+a}$ ， $\theta^* \in \mathbf{P}(\theta)$ ，从而得到误差坐标

$$\Delta\theta(t) \triangleq \hat{\theta}(t) - \theta^*. \quad (57)$$

任务：我们的任务是为识别系统和估计参数分别设计更新律 $\dot{\mathbf{b}}\mathbf{v}$ 、 $\dot{\mathbf{b}}\theta$ ，以确保所有信号保持有界，并

$$\lim_{t \rightarrow \infty} \Delta\mathbf{v}(t) = 0, \quad (58)$$

$$\lim_{t \rightarrow \infty} \hat{\theta} \in \mathbf{P}(\theta). \quad (59)$$

假设：我们做出以下假设。

- $\mathbf{v}(t)$ 、 $\eta(t)$ 、 $\xi(t)$ 是有界的。
- $\mathbf{W}\delta\tilde{\theta}\mathbf{P}$ 、 $\mathbf{v}$ 、 φ 、 $\xi\mathbf{P}$ (35) 在 $\mathbf{v}\delta\tilde{\theta}\mathbf{P}$ 、 $\mathbf{v}$ 、 φ 、 ξ 和这些变量的有界函数中局部一致连续。
- 因此， $\mathbf{W}_v\delta\mathbf{v}$ 、 φ 、 $\xi\mathbf{P}$ (39) 在 $\mathbf{v}$ 、 φ 、 ξ 和这些变量的有界函数中局部一致连续。
- 控制输入 ξ 在时间上均匀连续且有界。
- 真实参数 θ^* 是有界且常数的。
- $\mathbf{b}\mathbf{v}\delta\mathbf{t}\mathbf{P} \frac{1}{4} \mathbf{v}\delta\mathbf{t}\mathbf{P}$ ，即 $\Delta\mathbf{v}(t_0) = 0$ ， $\exists \epsilon \in \mathbb{R}^+$ ，使得 $\|\mathbf{v}\delta\mathbf{t}\mathbf{P}\|^{1/2} \|\mathbf{b}\mathbf{v}\delta\mathbf{t}\mathbf{P}\| \leq \lambda_{\min}(\mathbf{M}\delta\theta^*\mathbf{P}\mathbf{P})$ ，其中

$$\mathcal{T}(t_0) \triangleq \lambda_{\max}(\Gamma)\Delta\theta(t_0)^T\Gamma^{-1}\Delta\theta(t_0) \quad (60)$$

并且 $\Gamma \in \mathbb{R}^{(p+a) \times (p+a)}$ 是一个参数自适应增益矩阵，将在下文定义。

在本节的其余部分，我们提出了一种新型 AID 算法，并展示它在所有信号有界的情况下实现了 (58, 59)。第 5.1 节介绍了更新规律。第 5.2 节报告了李雅普诺夫分析，以证明误差系统关于原点的均匀稳定性，我们将其表示为

$$S^* = \{\Delta\mathbf{v} = 0, \Delta\theta = 0\}, \quad (61)$$

$\Delta\mathbf{v}$ 的收敛性，以及所有信号的有界性。注意 $\Delta\theta \in \mathbf{P}(\theta)$ ，这意味着 S^* 是的一个子集

$$S = \{\Delta\mathbf{v} = 0, \hat{\theta} \in \mathbf{P}(\theta)\}. \quad (62)$$

在第 5.3 节中，我们在持续激励 (PE) 条件下提出了一个收敛到较大集合 S 的新证明。

5.1. 更新法律和误差动态

我们选择更新规律

$$\dot{\hat{\mathbf{v}}} = -\mathcal{A}\Delta\mathbf{v} + \dot{\mathbf{v}}(\hat{\theta}) \quad (63)$$

$$\dot{\hat{\theta}} = \Gamma \mathbb{W}(\dot{\mathbf{v}}(\hat{\theta}), \mathbf{v}, \varphi, \xi)^T \Delta\mathbf{v} \quad (64)$$

其中 (63) 定义了自适应鉴别植物的动力学，(64) 定义了自适应参数更新律，

$$\mathcal{A} \in \mathbb{R}^{6 \times 6} \quad (65)$$

是用于识别植物的对角正定对称 (PDS) 增益矩阵，

$$\Gamma \in \mathbb{R}^{(p+a) \times (p+a)} \quad (66)$$

是参数估计的对角 PDS 增益矩阵，而 $\mathbf{v}\delta\mathbf{b}\theta\mathbf{P}$ 是由参数估计 $\mathbf{b}\theta$ 而非真实参数 θ^* 引起的体速度的时间导数，其性质如第 4.5 节所述，定义为

$$\dot{v}(\hat{\theta}) \triangleq M(\hat{\theta})W_{\dot{v}}(v, \varphi, \xi)\hat{\theta}. \quad (67)$$

Similarly to our usage of the term “plant acceleration” to denote $\dot{v}(\theta^*)$ defined in (9), as noted in Section 4.2.2, in the sequel, for convenience we will use the term “identification plant acceleration” to denote $\dot{v}(\hat{\theta})$ defined in (67).

The rate of change of the velocity error is

$$\Delta\dot{v} = \dot{\hat{v}} - \dot{v} \quad (68)$$

$$\Delta\dot{v} = -\mathcal{A}\Delta v + \dot{v}(\hat{\theta}) - \dot{v}(\theta^*) \quad (69)$$

$$\Delta\dot{v} = -\mathcal{A}\Delta v + M(\theta^*)^{-1}M(\theta^*)\dot{v}(\hat{\theta}) - \dot{v}(\theta^*) \quad (70)$$

Because M is parameterized by θ^* , we can rearrange and use the definitions (12, 35, 47) to find

$$\begin{aligned} \Delta\dot{v} = & -\mathcal{A}\Delta v \\ & + M(\theta^*)^{-1} [W_m(\hat{v}(\hat{\theta})) \quad 0_{6 \times (p+a-6)}] \theta^* \end{aligned} \quad (71)$$

$$-M(\theta^*)^{-1}W_{\dot{v}}(v, \varphi, \xi)\theta^* \quad (72)$$

$$\Delta\dot{v} = -\mathcal{A}\Delta v + M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)\theta^*. \quad (73)$$

Using the fact that $\hat{\theta} \in \text{null}(\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi))$ from Section 4.5.2 and the error coordinate definition (57), we can rearrange the second term in (73) to find

$$M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)\theta^* \quad (74)$$

$$= M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)(\theta^* - \hat{\theta}) \quad (75)$$

$$= -M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)\Delta\theta. \quad (76)$$

Substituting (76) into (73) yields

$$\Delta\dot{v} = -\mathcal{A}\Delta v - M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)\Delta\theta. \quad (77)$$

The parameter error dynamics follow directly from the error coordinates (57), the parameter update law (64), and the assumption that the true parameters are constant, that is, $\dot{\theta}^* = 0$, thus the parameter error dynamics can be written

$$\Delta\dot{\theta} = \Gamma \mathbb{W}^T(\dot{v}(\hat{\theta}), v, \varphi, \xi)\Delta v. \quad (78)$$

By defining the full error state vector for the AID as

$$z(t) = \begin{bmatrix} \Delta v(t) \\ \Delta\theta(t) \end{bmatrix}, \quad (79)$$

we can construct a representation of this error system using the velocity error dynamics (77) and parameter error dynamics (78):

$$\dot{z}(t) = \begin{bmatrix} M^{-1} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & \Gamma \end{bmatrix} \begin{bmatrix} -M\mathcal{A} & -\mathbb{W}(t, \hat{\theta}) \\ \mathbb{W}(t, \hat{\theta})^T & 0_{(p+a) \times (p+a)} \end{bmatrix} z(t) \quad (80)$$

$$= \left(\underbrace{\begin{bmatrix} M^{-1} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & \Gamma \end{bmatrix}}_E \underbrace{\begin{bmatrix} 0_{6 \times 6} & -\mathbb{W}(t, \hat{\theta}) \\ \mathbb{W}(t, \hat{\theta})^T & 0_{(p+a) \times (p+a)} \end{bmatrix}}_{F(t)} + \underbrace{\begin{bmatrix} -\mathcal{A} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & 0_{(p+a) \times (p+a)} \end{bmatrix}}_G \right) z(t) \quad (81)$$

where we adopt the notation

$$\mathbb{W}(t, \hat{\theta}) = \mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi) \quad (82)$$

$$M = M(\theta^*) \quad (83)$$

for readability. Note that the matrix

$$E = E^T = \begin{bmatrix} M^{-1} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & \Gamma \end{bmatrix} \quad (84)$$

is block diagonal PDS,

$$F(t) = -F(t)^T = \begin{bmatrix} 0_{6 \times 6} & -\mathbb{W}(t, \hat{\theta}) \\ \mathbb{W}(t, \hat{\theta})^T & 0_{(p+a) \times (p+a)} \end{bmatrix} \quad (85)$$

is skew-symmetric, and

$$G = G^T = \begin{bmatrix} -\mathcal{A} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & 0_{(p+a) \times (p+a)} \end{bmatrix} \quad (86)$$

is block diagonal and negative semidefinite.

5.2 Convergence of the velocity error

Theorem 1. *The system (80)*

$$\dot{z}(t) = \begin{bmatrix} M^{-1} & 0 \\ 0 & \Gamma \end{bmatrix} \begin{bmatrix} -M\mathcal{A} & -\mathbb{W}(t, \hat{\theta}) \\ \mathbb{W}(t, \hat{\theta})^T & 0 \end{bmatrix} z(t)$$

where $z(t)$, M , Γ , $\mathcal{A}$, $\mathbb{W}(t, \hat{\theta})$ are defined in (79, 11, 66, 65, 82) is locally uniformly stable about the set S^* (61), all signals remain bounded, and $\lim_{t \rightarrow \infty} \Delta v = 0$.

Proof. Consider the following Lyapunov function candidate:

$$V(t) = \frac{1}{2} z(t)^T E^{-1} z(t). \quad (87)$$

Note that $V(t) \in \mathcal{C}^1$ and is radially unbounded in $z(t)$, positive definite since E^{-1} is PDS, and equal to zero if and only if $z(t) = 0$. The time derivative of (87) is

$$\dot{v}(\hat{\theta}) \triangleq M(\hat{\theta})W_{\dot{v}}(v, \varphi, \xi)\hat{\theta}. \quad (67)$$

类似于我们使用术语“植物加速度”来表示在(9)中定义的 $_v\delta\theta^*\mathbf{p}$ ，如第4.2.2节所述，接下来，为了方便起见，我们将使用术语“辨识植物加速度”来表示在(67)中定义的 $_v\delta b\theta\mathbf{p}$ 。

速度误差的变化率是

$$\Delta\dot{v} = \dot{\hat{v}} - \dot{v} \quad (68)$$

$$\Delta\dot{v} = -\mathcal{A}\Delta v + \dot{v}(\hat{\theta}) - \dot{v}(\theta^*) \quad (69)$$

$$\Delta\dot{v} = -\mathcal{A}\Delta v + M(\theta^*)^{-1}M(\theta^*)\dot{v}(\hat{\theta}) - \dot{v}(\theta^*) \quad (70)$$

因为 M 是由 θ^* 参数化的，我们可以重新排列并使用定义 (12, 35, 47) 来找到

$$\Delta\dot{v} = -\mathcal{A}\Delta v + M(\theta^*)^{-1}[W_m(\hat{v}(\hat{\theta})) \quad 0_{6 \times (p+a-6)}]\theta^* \quad (71)$$

$$-M(\theta^*)^{-1}W_{\dot{v}}(v, \varphi, \xi)\theta^* \quad (72)$$

$$\Delta\dot{v} = -\mathcal{A}\Delta v + M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)\theta^*. \quad (73)$$

利用第4.5.2节中的事实 $b\theta 2 \text{ null}\delta W\delta _v\delta b\theta\mathbf{p}, v, \varphi, \xi$ 和误差坐标定义 (57)，我们可以重新排列 (73) 中的第二项来发现

$$M(\theta^*)^{-1} \mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)\theta^* \quad (74)$$

$$= M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)(\theta^* - \hat{\theta}) \quad (75)$$

$$= -M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)\Delta\theta. \quad (76)$$

将 (76) 代入 (73) 得到

$$\Delta\dot{v} = -\mathcal{A}\Delta v - M(\theta^*)^{-1}\mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi)\Delta\theta. \quad (77)$$

参数误差动力学直接来自误差坐标 (57)、参数更新律 (64) 以及真实参数为常数的假设，即 $\dot{\theta}^* = 0$ ，因此参数误差动力学可以写成

$$\Delta\dot{\theta} = \Gamma \mathbb{W}^T(\dot{v}(\hat{\theta}), v, \varphi, \xi)\Delta v. \quad (78)$$

通过定义 AID 的完整误差状态向量为

$$z(t) = \begin{bmatrix} \Delta v(t) \\ \Delta\theta(t) \end{bmatrix}, \quad (79)$$

我们可以使用速度误差动力学(77)和参数误差动力学(78)来构建该误差系统的表示：

$$\dot{z}(t) = \begin{bmatrix} M^{-1} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & \Gamma \end{bmatrix} \begin{bmatrix} -M\mathcal{A} & -\mathbb{W}(t, \hat{\theta}) \\ \mathbb{W}(t, \hat{\theta})^T & 0_{(p+a) \times (p+a)} \end{bmatrix} z(t) \quad (80)$$

$$= \left(\underbrace{\begin{bmatrix} M^{-1} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & \Gamma \end{bmatrix}}_E \underbrace{\begin{bmatrix} 0_{6 \times 6} & -\mathbb{W}(t, \hat{\theta}) \\ \mathbb{W}(t, \hat{\theta})^T & 0_{(p+a) \times (p+a)} \end{bmatrix}}_{F(t)} + \underbrace{\begin{bmatrix} -\mathcal{A} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & 0_{(p+a) \times (p+a)} \end{bmatrix}}_G \right) z(t) \quad (81)$$

我们采用这一符号的地方

$$\mathbb{W}(t, \hat{\theta}) = \mathbb{W}(\dot{v}(\hat{\theta}), v, \varphi, \xi) \quad (82)$$

$$M = M(\theta^*) \quad (83)$$

为了可读性。请注意矩阵

$$E = E^T = \begin{bmatrix} M^{-1} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & \Gamma \end{bmatrix} \quad (84)$$

是块对角PDS，

$$F(t) = -F(t)^T = \begin{bmatrix} 0_{6 \times 6} & -\mathbb{W}(t, \hat{\theta}) \\ \mathbb{W}(t, \hat{\theta})^T & 0_{(p+a) \times (p+a)} \end{bmatrix} \quad (85)$$

是斜对称的，并且

$$G = G^T = \begin{bmatrix} -\mathcal{A} & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & 0_{(p+a) \times (p+a)} \end{bmatrix} \quad (86)$$

是块对角的且半负定。

5.2 Convergence of the velocity error

定理 1. 系统 (80)

$$\dot{z}(t) = \begin{bmatrix} M^{-1} & 0 \\ 0 & \Gamma \end{bmatrix} \begin{bmatrix} -M\mathcal{A} & -\mathbb{W}(t, \hat{\theta}) \\ \mathbb{W}(t, \hat{\theta})^T & 0 \end{bmatrix} z(t)$$

其中 $z(t)$ 、 M 、 Γ 、 $\mathcal{A}$ 、 $\mathbb{W}(t, \hat{\theta})$ 、 $b\theta\mathbf{p}$ 在 (79, 11, 66, 65, 82) 中定义，在集合 S^* (61) 附近是局部一致稳定的，所有信号保持有界，并且 $\lim_{t \rightarrow \infty} \Delta v = 0$ 。

证明。考虑以下李雅普诺夫函数候选：

$$V(t) = \frac{1}{2} z(t)^T E^{-1} z(t). \quad (87)$$

注意， $V \in C^1$ 并且在 $z(t)$ 中径向无界，由于 E^{-1} 是正定对称(PDS)，因此为正定，并且当且仅当 $z(t) \rightarrow 0$ 时等于零。(87) 的时间导数为

$$\begin{aligned}\dot{V}(t) &= \frac{1}{2}z(t)^T(EF(t) + G)^TE^{-1}z(t) \\ &\quad + \frac{1}{2}z(t)^TE^{-1}(EF(t) + G)z(t).\end{aligned}\quad (88)$$

$E = E^T$ and $G = G^T$, so

$$\dot{V}(t) = \frac{1}{2}z^T(F(t)^T + GE^{-1})z + \frac{1}{2}z^T(F(t) + E^{-1}G)z, \quad (89)$$

and since $F(t) = -F(t)^T$ and E^{-1} and G commute, we have

$$\dot{V}(t) = z^TE^{-1}Gz \quad (90)$$

$$\dot{V}(t) = z^T \begin{bmatrix} -MA & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & 0_{(p+a) \times (p+a)} \end{bmatrix} z \leq 0. \quad (91)$$

This expression is negative definite in Δv and negative semidefinite in $\Delta\theta$, satisfying the requirements on a Lyapunov function to show that the system (81) is uniformly stable about the set S^* and that Δv and $\Delta\theta$ are bounded. We note that with $\Delta\theta$ bounded and θ^* constant, it is implied that $\hat{\theta}$ is bounded.

To show that $\Delta\dot{v}$ (77) is bounded, it remains to be shown that $M(\hat{\theta})^{-1}$ is bounded. We adapt the proof in McFarland and Whitcomb (2013). Consider that $\forall t > t_0$, the following hold:

- $\Delta\theta(t)^T\Gamma^{-1}\theta(t) \geq 0$ from PDS of Γ
- $0 \leq V(t) \leq V(t_0)$ from (91)
- $\Delta v(t_0) = 0$ by assumption

These facts can be used to show

$$\frac{1}{\lambda_{\max}(\Gamma)}\|\Delta\theta_m(t)\|_2^2 \leq \Delta\theta(t)^T\Gamma^{-1}\Delta\theta(t) \quad (92)$$

$$\leq 2V(t) \quad (93)$$

$$\leq 2V(t_0) \quad (94)$$

$$\leq \Delta\theta(t_0)^T\Gamma^{-1}\Delta\theta(t_0). \quad (95)$$

This implies

$$\|\Delta\theta_m(t)\|_2^2 \leq \lambda_{\max}(\Gamma)\Delta\theta(t_0)^T\Gamma^{-1}\Delta\theta(t_0) \quad (96)$$

and since, by assumption (60),

$$[\lambda_{\max}(\Gamma)\Delta\theta(t_0)^T\Gamma^{-1}\Delta\theta(t_0)]^{\frac{1}{2}} \leq \lambda_{\min}(M(\theta^*)) - \epsilon, \quad (97)$$

we can state

$$\|\Delta\theta_m(t)\|_2^2 \leq \lambda_{\min}(M(\theta^*)) - \epsilon. \quad (98)$$

Using the Rayleigh-Ritz theorem, we will show that the smallest eigenvalue of $M(\hat{\theta})$ is bounded away from zero as follows:

$$\lambda_{\min}(M(\hat{\theta})) = \min_{\|x\|=1} (x^TM(\hat{\theta})x) \quad (99)$$

$$= \min_{\|x\|=1} (x^TM(\theta^*)x + x^TM(\Delta\theta)x) \quad (100)$$

$$\geq \min_{\|x\|=1} (x^TM(\theta)x) - \max_{\|x\|=1} |x^TM(\Delta\theta)x|. \quad (101)$$

The second term in (101) is equal to the maximum magnitude of the eigenvalues of $M(\Delta\theta)$. Since $M(\Delta\theta)$ is symmetric, this is equivalent to its maximum singular value or $\|M(\Delta\theta)\|_2$. Furthermore, $\|M(\Delta\theta)\|_2 \leq \|M(\Delta\theta)\|_F$, and for diagonal mass matrices, $\|M(\Delta\theta)\|_F = \|\Delta\theta_m\|_2$ by definition. Thus, we have

$$\lambda_{\min}(M(\hat{\theta})) \geq \lambda_{\min}(M(\theta^*)) - \|M(\Delta\theta)\|_2 \quad (102)$$

$$\geq \lambda_{\min}(M(\theta^*)) - \|M(\Delta\theta)\|_F \quad (103)$$

$$\geq \lambda_{\min}(M(\theta^*)) - \|\Delta\theta_m\|_2. \quad (104)$$

Substituting (98), we have

$$\lambda_{\min}(M(\hat{\theta})) \geq \epsilon, \quad (105)$$

thus $M(\hat{\theta})^{-1}$ exists and is bounded.

All signals in the right-hand side of (77) are thus shown to be bounded. This implies the time-derivative of the velocity error, $\Delta\dot{v}$, is bounded. We note that the plant acceleration $\dot{v}$ is also bounded, since all of its components are bounded (41). Since both $\dot{v}$ and $\Delta\dot{v}$ are bounded, the velocity update law $\hat{v}$ is also bounded (68), and from bounded $\hat{v}$, Δv we have bounded $\dot{v}(\hat{\theta})$ (63).

From (79), (87), and (91) it can easily be shown that $\Delta v \in \mathcal{L}_2 \cap \mathcal{L}_\infty$, which together with bounded $\Delta\dot{v}$ implies that $\lim_{t \rightarrow \infty} \Delta v = 0$. Thus we have shown that this AID achieves (58). Regarding the parameter estimates, however, we can only conclude that $\Delta\theta$ is bounded, that the update law $\hat{\theta}$ (64) is bounded, and that $\lim_{t \rightarrow \infty} \Delta\dot{\theta} = \lim_{t \rightarrow \infty} \dot{\hat{\theta}} = 0$ (78). Additional arguments and conditions are needed to show mathematically the convergence of $\hat{\theta}$ to the set $P(\theta)$.

5.3. Convergence of the parameter estimate

Section 5.2 showed that $\lim_{t \rightarrow \infty} \Delta v(t) = 0$ and $\lim_{t \rightarrow \infty} \hat{\theta}(t) = 0$. This argument by itself does not show that the parameter estimate converges to the true parameter set $P(\theta)$, defined in (37). Moreover, because the set of true parameters is not defined by a single point in parameter space but rather by the set $P(\theta)$, the goal is not to show convergence of $\hat{\theta}(t)$ to a single point (i.e., the goal is *not* to show that $\lim_{t \rightarrow \infty} \Delta\theta(t) = 0$). This Section will state conditions for and show the convergence of $\hat{\theta}(t)$ to $P(\theta)$ (59).

$$\begin{aligned}\dot{V}(t) &= \frac{1}{2}z(t)^T(EF(t) + G)^TE^{-1}z(t) \\ &\quad + \frac{1}{2}z(t)^TE^{-1}(EF(t) + G)z(t).\end{aligned}\quad (88)$$

$E = E^T$ 和 $G = G^T$, 所以

$$\dot{V}(t) = \frac{1}{2}z^T(F(t)^T + GE^{-1})z + \frac{1}{2}z^T(F(t) + E^{-1}G)z, \quad (89)$$

并且由于 $F(t) = -F(t)^T$ 与 E^{-1} 以及 G 可交换, 我们有

$$\dot{V}(t) = z^TE^{-1}Gz \quad (90)$$

$$\dot{V}(t) = z^T \begin{bmatrix} -MA & 0_{6 \times (p+a)} \\ 0_{(p+a) \times 6} & 0_{(p+a) \times (p+a)} \end{bmatrix} z \leq 0. \quad (91)$$

这个表达式在 Δv 中是负定的, 在 $\Delta \theta$ 中是负半定的, 满足关于 Lyapunov 函数的要求, 从而表明系统 (81) 关于集合 S^* 是一致稳定的, 并且 Δv 和 $\Delta \theta$ 是有界的。我们注意到, 当 $\Delta \theta$ 有界且 θ^* 为常数时, 这意味着 $b\theta$ 也是有界的。

为了证明 Δv (77) 是有界的, 还需要证明 $M \delta b \theta P^{-1}$ 是有界的。我们采用 McFarland 和 Whitcomb (2013) 中的证明。考虑 $\forall t > t_0$, 以下成立:

- $\Delta \theta(t) \Gamma \Gamma^{-1} \theta(t) \geq 0$ 来自 Γ 的 PDS
- $0 \leq V(t) \leq V(t_0)$ 来自 (91)
- $\Delta v(t_0) = 0$ 根据假设

这些事实可以用来表明

$$\frac{1}{\lambda_{\max}(\Gamma)} \|\Delta \theta_m(t)\|_2^2 \leq \Delta \theta(t)^T \Gamma^{-1} \Delta \theta(t) \quad (92)$$

$$\leq 2V(t) \quad (93)$$

$$\leq 2V(t_0) \quad (94)$$

$$\leq \Delta \theta(t_0)^T \Gamma^{-1} \Delta \theta(t_0). \quad (95)$$

这意味着

$$\|\Delta \theta_m(t)\|_2^2 \leq \lambda_{\max}(\Gamma) \Delta \theta(t_0)^T \Gamma^{-1} \Delta \theta(t_0) \quad (96)$$

并且根据假设 (60),

$$[\lambda_{\max}(\Gamma) \Delta \theta(t_0)^T \Gamma^{-1} \Delta \theta(t_0)]^{\frac{1}{2}} \leq \lambda_{\min}(M(\theta^*)) - \epsilon, \quad (97)$$

我们可以声明

$$\|\Delta \theta_m(t)\|_2^2 \leq \lambda_{\min}(M(\theta^*)) - \epsilon. \quad (98)$$

使用 Rayleigh-Ritz 定理, 我们将如下证明 $M \delta b \theta P$ 的最小特征值与零有界:

$$\lambda_{\min}(M(\hat{\theta})) = \min_{\|x\|=1} (x^T M(\hat{\theta}) x) \quad (99)$$

$$= \min_{\|x\|=1} (x^T M(\theta^*) x + x^T M(\Delta \theta) x) \quad (100)$$

$$\geq \min_{\|x\|=1} (x^T M(\theta) x) - \max_{\|x\|=1} |x^T M(\Delta \theta) x|. \quad (101)$$

(101) 中的第二项等于 $M(\Delta \theta)$ 特征值的最大幅值。由于 $M(\Delta \theta)$ 是对称的, 这等同于其最大奇异值或 $kM(\Delta \theta)k_2$ 。此外, $kM(\Delta \theta)k_2 \leq kM(\Delta \theta)k_F$, 对于对角质矩阵, $kM(\Delta \theta)k_F = k\Delta \theta_m k_2$ 根据定义成立。因此, 我们有

$$\lambda_{\min}(M(\hat{\theta})) \geq \lambda_{\min}(M(\theta^*)) - \|M(\Delta \theta)\|_2 \quad (102)$$

$$\geq \lambda_{\min}(M(\theta^*)) - \|M(\Delta \theta)\|_F \quad (103)$$

$$\geq \lambda_{\min}(M(\theta^*)) - \|\Delta \theta_m\|_2. \quad (104)$$

代入(98), 我们得到

$$\lambda_{\min}(M(\hat{\theta})) \geq \epsilon, \quad (105)$$

因此 $M \delta b \theta P^{-1}$ 存在且有界。

因此, 式 (77) 右侧的所有信号都被证明是有界的。这意味着速度误差的时间导数 Δv 是有界的。我们注意到, 系统加速度 $\ddot{v}$ 也是有界的, 因为它的所有分量都是有界的 (41)。由于 $\ddot{v}$ 和 Δv 都是有界的, 速度更新律 $\dot{b}v$ 也是有界的 (68), 并且由有界的 $\dot{b}v$, Δv 我们得到有界的 $\dot{v} \delta b \theta P$ (63)。

从(79)、(87)和(91)可以很容易地证明 $\Delta v \in L_2 \cap L_\infty$, 这与有界的 Δv 一起意味着 $\lim_{t \rightarrow \infty} \Delta v = 0$ 。因此, 我们已经证明该自适应控制策略实现了(58)。然而, 关于参数估计, 我们只能得出 $\Delta \theta$ 是有界的, 更新律 $\dot{b} \theta$ (64) 是有界的, 并且 $\lim_{t \rightarrow \infty} \Delta \theta = \lim_{t \rightarrow \infty} \dot{b} \theta = 0$ (78)。要在数学上证明 $b \theta$ 收敛到集合 $P(\theta)$, 还需要额外的论证和条件。

5.3. 参数估计的收敛性

第5.2节显示了 $\lim_{t \rightarrow \infty} \Delta v(t) = 0$ 和 $\lim_{t \rightarrow \infty} \dot{b} \theta \delta t P = 0$ 。单凭这个论证并不能说明参数估计收敛到在(37)中定义的真实参数集合 $P(\theta)$ 。此外, 因为真实参数集合不是由参数空间中的单个点定义, 而是由集合 $P(\theta)$ 定义, 所以目标不是显示 $b \theta \delta t P$ 收敛到单个点 (即目标不是显示 $\lim_{t \rightarrow \infty} \Delta \theta(t) = 0$)。本节将陈述条件并展示 $b \theta \delta t P$ 收敛到 $P(\theta)$ (59) 的情况。

The remainder of this Section is organized as follows:

- Section 5.3.1 introduces a change of coordinates to characterize $\hat{\theta}(t)$ as the sum of two components residing in orthogonal subspaces related to $P(\theta)$.
- Section 5.3.2, then defines an equivalent error system using these coordinates.
- Finally, Section 5.3.3 shows a sufficient condition for $\hat{\theta}(t)$ to converge to the true parameter space $P(\theta)$, that is, $\lim_{t \rightarrow \infty} \hat{\theta}(t) \in P(\theta)$.

We note that the problem of parameter convergence with respect to a unique set of true parameters is well understood for a class of linear time-varying systems, as originally reported in [Morgan and Narendra \(1977\)](#). To the best of our knowledge, however, we are unaware of any previously reported convergence proof applicable to the nonlinear system (80) with nullspace parameter structure.

5.3.1. Coordinate transformation. Given the set $P(\theta)$ (37), we can define the linear vector subspace $P_s = P(\theta) \cup 0$, with $\dim(P_s) = r$, and we can define an orthonormal basis $\{p_1, \dots, p_r\}$ for P_s , as well as the matrix

$$\bar{P} \triangleq [p_1, \dots, p_r] \in \mathbb{R}^{(p+a) \times r} \quad (106)$$

whose rank is r , with properties

$$\bar{P}^T \bar{P} = I_{r \times r} \quad (107)$$

$$\mathbb{W}(\dot{\theta}^*, v, \varphi, \xi) \bar{P} = 0_{6 \times r}. \quad (108)$$

Similarly, we can define the linear vector subspace $P_\perp$ as the orthogonal complement of P_s , with orthonormal basis $\{q_1, \dots, q_{(p+a)-r}\}$, as well as the matrix

$$\bar{P}_\perp \triangleq [q_1, \dots, q_{(p+a)-r}] \in \mathbb{R}^{(p+a) \times [(p+a)-r]} \quad (109)$$

whose rank is $(p+a) - r$, with properties

$$\bar{P}_\perp^T \bar{P}_\perp = I_{[(p+a)-r] \times [(p+a)-r]} \quad (110)$$

$$\bar{P}_\perp^T \bar{P} = 0_{[(p+a)-r] \times r}. \quad (111)$$

We note that $\mathbb{R}^{(p+a)}$ is the direct sum of the orthogonal subspaces $P_\perp$ and P_s , that is, $\mathbb{R}^{(p+a)} = P_\perp \oplus P_s$.

We combine $\bar{P}$ and $\bar{P}_\perp$ into the orthogonal matrix

$$Q \triangleq [\bar{P}_\perp \quad \bar{P}] \in \mathbb{R}^{(p+a) \times (p+a)} \quad (112)$$

Using Q , we can perform a change of coordinates on the parameter estimate $\hat{\theta}$ so that

$$\hat{\theta} = Q \bar{\theta} \quad (113)$$

$$\dot{\hat{\theta}} = Q \dot{\bar{\theta}}, \quad (114)$$

where $\bar{\theta} \in \mathbb{R}^{(p+a)}$ can be decomposed into $\bar{\theta}_{P_\perp} \in \mathbb{R}^{(p+a)-r}$ and $\bar{\theta}_P \in \mathbb{R}^r$, the coordinates of $\hat{\theta}$ in $P_\perp$ and P , respectively:

$$\bar{\theta} = \begin{bmatrix} \bar{\theta}_{P_\perp} \\ \bar{\theta}_P \end{bmatrix}. \quad (115)$$

5.3.2. Equivalent error dynamics. We can obtain an expression equivalent to (73) for $\Delta \dot{v}$ by factoring out $M(\hat{\theta})$ instead of $M(\theta^*)$ and substituting (67) for $\dot{v}(\hat{\theta})$ in the derivation (68–73):

$$\Delta \dot{v} = \dot{\hat{v}} - \dot{v} \quad (116)$$

$$= -\mathcal{A} \Delta v + \dot{v}(\hat{\theta}) - \dot{v}(\theta^*) \quad (117)$$

$$= -\mathcal{A} \Delta v + \dot{v}(\hat{\theta}) - M(\hat{\theta})^{-1} M(\hat{\theta}) \dot{v}(\theta^*) \quad (118)$$

$$= -\mathcal{A} \Delta v + M(\hat{\theta})^{-1} W_{\dot{v}}(v, \varphi, \xi) \hat{\theta} - M^{-1}(\hat{\theta}) [W_m(\dot{v}(\theta^*)) \quad 0_{6 \times ((p+a)-6)}] \hat{\theta} \quad (119)$$

$$= -\mathcal{A} \Delta v - M(\hat{\theta})^{-1} ([W_m(\dot{v}(\theta^*)) \quad 0_{6 \times ((p+a)-6)}] - W_{\dot{v}}(v, \varphi, \xi)) \hat{\theta} \quad (120)$$

$$= -\mathcal{A} \Delta v - M(\hat{\theta})^{-1} \mathbb{W}(\dot{v}(\theta^*), v, \varphi, \xi) \hat{\theta}. \quad (121)$$

Using the coordinates $\bar{\theta}_{P_\perp}, \bar{\theta}_P$ (113,115) for $\hat{\theta}$ and the property (108), we have

$$\Delta \dot{v} = -\mathcal{A} \Delta v - M(\hat{\theta})^{-1} \mathbb{W}(\dot{v}(\theta^*), v, \varphi, \xi) Q \bar{\theta} \quad (122)$$

$$= -\mathcal{A} \Delta v - M(\hat{\theta})^{-1} \mathbb{W}(\dot{v}(\theta^*), v, \varphi, \xi) (\bar{P}_\perp \bar{\theta}_{P_\perp} + \bar{P} \bar{\theta}_P) \quad (123)$$

$$= -\mathcal{A} \Delta v - M(\hat{\theta})^{-1} \mathbb{W}(\dot{v}(\theta^*), v, \varphi, \xi) \bar{P}_\perp \bar{\theta}_{P_\perp}, \quad (124)$$

and defining the term

$$W_\perp(t) \triangleq M^{-1}(\hat{\theta}(t)) \mathbb{W}(\dot{v}(\theta^*), v(t), \varphi(t), \xi(t)) \bar{P}_\perp, \quad (125)$$

we can rewrite (124) as

$$\Delta \dot{v} = -\mathcal{A} \Delta v - W_\perp \bar{\theta}_{P_\perp}. \quad (126)$$

We also have the derivative of $\bar{\theta}$, given by (114)

$$\dot{\bar{\theta}} = Q^T \dot{\hat{\theta}} \quad (127)$$

$$\begin{bmatrix} \dot{\bar{\theta}}_{P_\perp} \\ \dot{\bar{\theta}}_P \end{bmatrix} = \begin{bmatrix} \bar{P}_\perp^T \dot{\hat{\theta}} \\ \bar{P}^T \dot{\hat{\theta}} \end{bmatrix} \quad (128)$$

From (64, 126), this results in the full transformed system

$$\begin{bmatrix} \Delta \dot{v} \\ \dot{\bar{\theta}}_{P_\perp} \\ \dot{\bar{\theta}}_P \end{bmatrix} = \begin{bmatrix} -\mathcal{A} & -W_\perp & 0 \\ \bar{P}_\perp^T \Gamma \mathbb{W}(t, \hat{\theta})^T & 0 & 0 \\ \bar{P}^T \Gamma \mathbb{W}(t, \hat{\theta})^T & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta v \\ \bar{\theta}_{P_\perp} \\ \bar{\theta}_P \end{bmatrix} \quad (129)$$

本节的其余部分组织如下：

- 第5.3.1节介绍了一种坐标变换，用以将 $\mathbf{b}\theta$ 表示为驻留在与 $\mathbf{P}(\theta)$ 相关的正交子空间中的两个分量的和。
- 第5.3.2节随后使用这些坐标定义了一个等效的误差系统。
- 最后，第5.3.3节显示了 $\mathbf{b}\theta$ 收敛到真实参数空间 $\mathbf{P}(\theta)$ 的充分条件，即 $\lim_{t \rightarrow \infty} \mathbf{b}\theta = \mathbf{P}\bar{\theta}$ 。

我们注意到，对于一类线性时变系统，关于唯一真实参数集合的参数收敛问题已经得到了很好的理解，这最早在 Morgan 和 Narendra (1977) 中报道。然而，据我们所知，我们未发现任何先前发布的适用于具有零空间参数结构的非线性系统 (80) 的收敛性证明。

5.3.1. 坐标变换。给定集合 $\mathbf{P}(\theta)$ (37)，我们可以定义线性向量子空间 $\mathbf{P}_s = \mathbf{P}(\theta) \cap \mathbf{0}$ ，其维数为 $\dim(\mathbf{P}_s) = r$ ，我们还可以为 $\mathbf{P}_s$ 定义一个正交标准基 $\{\mathbf{p}_1, \dots, \mathbf{p}_r\}$ ，以及矩阵

$$\bar{\mathbf{P}} \triangleq [\mathbf{p}_1, \dots, \mathbf{p}_r] \in \mathbb{R}^{(p+a) \times r} \quad (106)$$

其秩为 r ，具有以下性质

$$\bar{\mathbf{P}}^T \bar{\mathbf{P}} = \mathbf{I}_{r \times r} \quad (107)$$

$$\mathbb{W}(\dot{\theta}^*, v, \varphi, \xi) \bar{\mathbf{P}} = \mathbf{0}_{6 \times r}. \quad (108)$$

同样地，我们可以将线性向量子空间 $\mathbf{P}_\perp$ 定义为 $\mathbf{P}_s$ 的正交补，其标准正交基为 $\{\mathbf{q}_1, \dots, \mathbf{q}_{(p+a)-r}\}$ ，以及矩阵

$$\bar{\mathbf{P}}_\perp \triangleq [\mathbf{q}_1, \dots, \mathbf{q}_{(p+a)-r}] \in \mathbb{R}^{(p+a) \times [(p+a)-r]} \quad (109)$$

其等级为 $(p+a) - r$ ，具有以下属性

$$\bar{\mathbf{P}}_\perp^T \bar{\mathbf{P}}_\perp = \mathbf{I}_{[(p+a)-r] \times [(p+a)-r]} \quad (110)$$

$$\bar{\mathbf{P}}_\perp^T \bar{\mathbf{P}} = \mathbf{0}_{[(p+a)-r] \times r}. \quad (111)$$

我们注意到 $\mathbf{R}^{(p+a)}$ 是正交子空间 $\mathbf{P}_\perp$ 和 $\mathbf{P}_s$ 的直和，即 $\mathbf{R}^{(p+a)} = \mathbf{P}_\perp \bar{\mathbf{A}} \mathbf{P}_s$ 。

我们将 $\mathbf{P}$ 和 $\bar{\mathbf{P}}_\perp$ 组合成正交矩阵

$$\mathbf{Q} \triangleq [\bar{\mathbf{P}}_\perp \quad \bar{\mathbf{P}}] \in \mathbb{R}^{(p+a) \times (p+a)} \quad (112)$$

使用 $\mathbf{Q}$ ，我们可以对参数估计 $\mathbf{b}\theta$ 进行坐标变换，以便

$$\hat{\theta} = \mathbf{Q} \bar{\theta} \quad (113)$$

$$\dot{\hat{\theta}} = \mathbf{Q} \dot{\bar{\theta}}, \quad (114)$$

其中 $\bar{\theta}^T \mathbf{R}^{(p+a)}$ 可以分解为 $\theta_{P_\perp}^T \mathbf{R}^{(p+a)-r}$ 和 $\bar{\theta}_{P^2}^T \mathbf{R}^r$ ， $\mathbf{b}\theta$ 在 $\mathbf{P}_\perp$ 和 $\mathbf{P}$ 中的坐标分别为：

$$\bar{\theta} = \begin{bmatrix} \bar{\theta}_{P_\perp} \\ \bar{\theta}_P \end{bmatrix}. \quad (115)$$

5.3.2. 等效误差动力学。我们可以通过提出 $\mathbf{M} \delta \mathbf{b}\theta$ 而不是 $\mathbf{M}(\theta^*)$ ，并在推导 (68-73) 中将 (67) 代入 $-\mathbf{v} \delta \mathbf{b}\theta$ ，从而得到与 (73) 对 $\Delta \mathbf{v}$ 等效的表达式：

$$\Delta \dot{\mathbf{v}} = \dot{\hat{\mathbf{v}}} - \dot{\mathbf{v}} \quad (116)$$

$$= -\mathcal{A} \Delta \mathbf{v} + \dot{\mathbf{v}}(\hat{\theta}) - \dot{\mathbf{v}}(\theta^*) \quad (117)$$

$$= -\mathcal{A} \Delta \mathbf{v} + \dot{\mathbf{v}}(\hat{\theta}) - \mathbf{M}(\hat{\theta})^{-1} \mathbf{M}(\hat{\theta}) \dot{\mathbf{v}}(\theta^*) \quad (118)$$

$$= -\mathcal{A} \Delta \mathbf{v} + \mathbf{M}(\hat{\theta})^{-1} \mathbf{W}_{\dot{\mathbf{v}}}(\mathbf{v}, \varphi, \xi) \hat{\theta} - \mathbf{M}^{-1}(\hat{\theta}) [\mathbf{W}_m(\dot{\mathbf{v}}(\theta^*)) \quad \mathbf{0}_{6 \times ((p+a)-6)}] \hat{\theta} \quad (119)$$

$$= -\mathcal{A} \Delta \mathbf{v} - \mathbf{M}(\hat{\theta})^{-1} ([\mathbf{W}_m(\dot{\mathbf{v}}(\theta^*)) \quad \mathbf{0}_{6 \times ((p+a)-6)}] - \mathbf{W}_{\dot{\mathbf{v}}}(\mathbf{v}, \varphi, \xi)) \hat{\theta} \quad (120)$$

$$= -\mathcal{A} \Delta \mathbf{v} - \mathbf{M}(\hat{\theta})^{-1} \mathbb{W}(\dot{\mathbf{v}}(\theta^*), \mathbf{v}, \varphi, \xi) \hat{\theta}. \quad (121)$$

使用坐标 $\theta_{P_\perp}, \theta_P$ (113, 115) 对 $\mathbf{b}\theta$ 和属性 (108)，我们有

$$\Delta \dot{\mathbf{v}} = -\mathcal{A} \Delta \mathbf{v} - \mathbf{M}(\hat{\theta})^{-1} \mathbb{W}(\dot{\mathbf{v}}(\theta^*), \mathbf{v}, \varphi, \xi) \mathbf{Q} \bar{\theta} \quad (122)$$

$$= -\mathcal{A} \Delta \mathbf{v} - \mathbf{M}(\hat{\theta})^{-1} \mathbb{W}(\dot{\mathbf{v}}(\theta^*), \mathbf{v}, \varphi, \xi) \begin{pmatrix} \bar{\mathbf{P}}_\perp \bar{\theta}_{P_\perp} + \bar{\mathbf{P}} \bar{\theta}_P \end{pmatrix} \quad (123)$$

$$= -\mathcal{A} \Delta \mathbf{v} - \mathbf{M}(\hat{\theta})^{-1} \mathbb{W}(\dot{\mathbf{v}}(\theta^*), \mathbf{v}, \varphi, \xi) \bar{\mathbf{P}}_\perp \bar{\theta}_{P_\perp}, \quad (124)$$

并定义该术语

$$\mathbf{W}_\perp(t) \triangleq \mathbf{M}^{-1}(\hat{\theta}(t)) \mathbb{W}(\dot{\mathbf{v}}(\theta^*, t), \mathbf{v}(t), \varphi(t), \xi(t)) \bar{\mathbf{P}}_\perp, \quad (125)$$

我们可以将 (124) 重写为

$$\Delta \dot{\mathbf{v}} = -\mathcal{A} \Delta \mathbf{v} - \mathbf{W}_\perp \bar{\theta}_{P_\perp}. \quad (126)$$

我们也有 θ 的导数，由 (114) 给出

$$\dot{\bar{\theta}} = \mathbf{Q}^T \dot{\hat{\theta}} \quad (127)$$

$$\begin{bmatrix} \dot{\bar{\theta}}_{P_\perp} \\ \dot{\bar{\theta}}_P \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{P}}_\perp^T \dot{\hat{\theta}} \\ \bar{\mathbf{P}}^T \dot{\hat{\theta}} \end{bmatrix} \quad (128)$$

从 (64, 126)，这导致了完整的变换系统

$$\begin{bmatrix} \Delta \dot{\mathbf{v}} \\ \dot{\bar{\theta}}_{P_\perp} \\ \dot{\bar{\theta}}_P \end{bmatrix} = \begin{bmatrix} -\mathcal{A} & -\mathbf{W}_\perp & \mathbf{0} \\ \bar{\mathbf{P}}_\perp^T \Gamma \mathbb{W}(t, \hat{\theta})^T & \mathbf{0} & \mathbf{0} \\ \bar{\mathbf{P}}^T \Gamma \mathbb{W}(t, \hat{\theta})^T & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{v} \\ \bar{\theta}_{P_\perp} \\ \bar{\theta}_P \end{bmatrix} \quad (129)$$

in which we have partially isolated the velocity error dynamics $\Delta\dot{v}$ from the coordinates $\bar{\theta}_P$ of $\hat{\theta}$ that already belong to the true parameter set $P(\theta)$, although a nonlinear state dependence remains in $W_\perp$ through $\hat{\theta}$ (125).

5.3.3. Convergence of $\Delta\dot{v}(t)$. Before showing a sufficient condition for $\lim_{t \rightarrow \infty} \hat{\theta} \in P(\theta)$, we state and prove the following useful lemma.

Lemma 1. *Given the system described in Theorem 1 and the transformed system (129), then*

$$\lim_{t \rightarrow \infty} W_\perp(t) \bar{\theta}_{P_\perp}(t) = 0. \quad (130)$$

Proof. One conclusion of Theorem 1 is that $\lim_{t \rightarrow \infty} \Delta v(t) = 0$. We apply this result to the transformed system (129) to find

$$\lim_{t \rightarrow \infty} \Delta\dot{v}(t) = \lim_{t \rightarrow \infty} (-\mathcal{A}\Delta v(t) - W_\perp(t) \bar{\theta}_{P_\perp}(t)) \quad (131)$$

$$= \lim_{t \rightarrow \infty} -W_\perp(t) \bar{\theta}_{P_\perp}(t). \quad (132)$$

By Barbalat's Lemma (Appendix B.1), if $\lim_{t \rightarrow \infty} \Delta v(t)$ is finite and $\Delta\dot{v}(t)$ is uniformly continuous (UC) with respect to time, then $\lim_{t \rightarrow \infty} \Delta\dot{v}(t) = 0$. We show the details regarding UC of $\Delta\dot{v}$ in Appendix B.2. With the conditions of Barbalat's Lemma satisfied for each element in the vector $\Delta v(t)$, we conclude that

$$\lim_{t \rightarrow \infty} \Delta\dot{v}(t) = \lim_{t \rightarrow \infty} -W_\perp(t) \bar{\theta}_{P_\perp}(t) = 0, \quad (133)$$

and this completes the proof of Lemma 1.

Examining the structure of $W_\perp$ (125), we observe that the nullspace of $M(\hat{\theta})^{-1} \mathbb{W}(\dot{\hat{\theta}}^*, v, \varphi, \xi)$ at any time t always contains $P(\theta)$ (37), but that $\bar{P}_\perp$ is by definition a matrix of basis vectors orthogonal to $P(\theta)$; thus $P(\theta)$ is removed from null($W_\perp(t)$). Intuitively, if $W_\perp(t)$ has no persistent non-trivial nullspace, the only way to satisfy (130) is for $\bar{\theta}_{P_\perp}(t)$ to converge to zero. We will use this argument in the following theorem, which presents a sufficient condition for convergence of $\hat{\theta}$ to $P(\theta)$.

5.3.4. Convergence of $\hat{\theta}(t)$ to the true parameter set $P(\theta)$.

Theorem 2. *For the system described by Theorem 1 and $W_\perp(t)$ defined in (125), if there exist $\epsilon_0, \delta_0 > 0$ such that for any $t \geq t_0$ and any unit vector $w \in \mathbb{R}^{(p+a)-r}$,*

$$\left\| \int_t^{t+\delta_0} W_\perp(\sigma) w d\sigma \right\| \geq \epsilon_0, \quad (134)$$

then $\lim_{t \rightarrow \infty} \hat{\theta}(t) \in P(\theta)$.

In other words, if $W_\perp(t)$ is persistently exciting (PE) as defined in (134), then $\hat{\theta}(t)$ converges to the true parameter set $P(\theta)$ (37), and the full state $z(t)$ converges to the set S (62).

Proof. It suffices to show that $\lim_{t \rightarrow \infty} \bar{\theta}_{P_\perp}(t) = 0$, which implies $\lim_{t \rightarrow \infty} \hat{\theta}(t) \in P(\theta)$. We prove this theorem by contradiction. First, we state assumptions and conditions of the theorem in Part 1. In Part 2, we suppose that $\lim_{t \rightarrow \infty} \bar{\theta}_{P_\perp}(t) \neq 0$, which we will later show to be a contradiction. In Part 3, we show that “when Δv is small, $\bar{\theta}_{P_\perp}$ is flat,” using the approach described in Morgan and Narendra (1977). Finally, in Part 4 we use the flatness property and the PE condition (134) find a result that contradicts Lemma 1.

- Let all of the assumptions in Section 5 be satisfied such that Theorem 1 is true, including that $\lim_{t \rightarrow \infty} \Delta v(t) = 0$.
- Let the PE condition in (134) be satisfied.
- Lemma 1 showed that $\lim_{t \rightarrow \infty} W_\perp(t) \bar{\theta}_{P_\perp}(t) = 0$.
- We also define the bounds $b, b_\perp > 0$ such that

$$\forall t \geq t_0, b_\perp \geq \|W_\perp(t)\| \quad (135)$$

$$\forall t \geq t_0, b \geq \|\mathbb{W}(t, \hat{\theta})\|. \quad (136)$$

These bounds exist since $\mathbb{W}(\dot{\hat{\theta}}(t), v, \varphi, \xi)$ is a bounded function of $\dot{\hat{\theta}}(t), v, \varphi, \xi$, whose boundedness has been discussed in Section 5.2, and

$$\|W_\perp(t)\| \leq \|M(\hat{\theta})^{-1}\| \|\mathbb{W}(t, \theta^*)\| \|\bar{P}_\perp\| \quad (137)$$

where $M(\hat{\theta})^{-1}$ and $\dot{v}(\theta^*)$ were shown to exist and be bounded in Section 5.2, $\mathbb{W}(t, \theta^*) = \mathbb{W}(\dot{v}(\theta^*), v, \varphi, \xi)$ is thus a bounded function of bounded functions, and $\|\bar{P}_\perp\| = 1$.

- Let us suppose that $\lim_{t \rightarrow \infty} \bar{\theta}_{P_\perp}(t) \neq 0$. Then there exists an $\epsilon > 0$ and an unbounded sequence of times $\{t_i\}_{i=1, \dots, \infty}$ such that $\|\bar{\theta}_{P_\perp}(t_i)\| \geq \epsilon$.
- We will show that “when Δv is small, $\bar{\theta}_{P_\perp}$ is flat” using an approach included in the proof of Sublemma 2 in Morgan and Narendra (1977). Given $T', \phi > 0, \exists \epsilon' > 0$ such that if $\|\Delta v(t)\| \leq \epsilon' \forall t \in [t_i, t_i + T']$, then

$$\|\bar{\theta}_{P_\perp}(t) - \bar{\theta}_{P_\perp}(t_i)\| \leq \phi \forall t \in [t_i, t_i + T']. \quad (138)$$

To verify this, given a T' and ϕ , we set

$$\epsilon' = \frac{\phi}{T' \lambda_{\max}(\Gamma) b} \quad (139)$$

and observe that, $\forall t \in [t_i, t_i + T']$, using the expression for $\dot{\bar{\theta}}_{P_\perp}$ in (129), the fact that $\|\bar{P}_\perp\| = 1$, and the bound b (136),

$$\|\bar{\theta}_{P_\perp}(t) - \bar{\theta}_{P_\perp}(t_i)\| = \left\| \int_{t_i}^t \dot{\bar{\theta}}_{P_\perp}(\sigma) d\sigma \right\| \quad (140)$$

$$\leq \left\| \int_{t_i}^t \bar{P}_\perp \Gamma \mathbb{W}(\sigma, \hat{\theta}) \Delta v(\sigma) d\sigma \right\| \quad (141)$$

在其中，我们已经部分地将速度误差动力学 Δ_v 从坐标 θ_p 中隔离出来，这些坐标 θ 已经属于真实参数集 $P(\theta)$ ，尽管非线性状态依赖仍然通过 $b\theta$ (125) 存在于 $W_\perp$ 中。

5.3.3. $\Delta_v \delta t$ 的收敛性。在给出 $\lim_{t \rightarrow \infty} b\theta \in P(\theta)$ 的充分条件之前，我们陈述并证明以下有用的引理。

引理 1. 给定定理 1 中描述的系统以及变换后的系统 (129)，则

$$\lim_{t \rightarrow \infty} W_\perp(t) \bar{\theta}_{P_\perp}(t) = 0. \quad (130)$$

证明。定理 1 的一个结论是 $\lim_{t \rightarrow \infty} \Delta v(t) = 0$ 。我们将这一结果应用于变换后的系统 (129) 来发现

$$\lim_{t \rightarrow \infty} \Delta \dot{v}(t) = \lim_{t \rightarrow \infty} (-\mathcal{A} \Delta v(t) - W_\perp(t) \bar{\theta}_{P_\perp}(t)) \quad (131)$$

$$= \lim_{t \rightarrow \infty} -W_\perp(t) \bar{\theta}_{P_\perp}(t). \quad (132)$$

根据巴尔巴拉引理 (附录 B.1)，如果 $\lim_{t \rightarrow \infty} \Delta v(t)$ 有限且 $\Delta_v(t)$ 随时间是均匀连续的 (UC)，那么 $\lim_{t \rightarrow \infty} \Delta_v(t) = 0$ 。我们将在附录 B.2 中展示 Δ_v 的均匀连续性细节。在向量 $\Delta v(t)$ 的每个元素满足巴尔巴拉引理条件的情况下，我们可以得出结论

$$\lim_{t \rightarrow \infty} \Delta \dot{v}(t) = \lim_{t \rightarrow \infty} -W_\perp(t) \bar{\theta}_{P_\perp}(t) = 0, \quad (133)$$

到此为止，完美地证明了引理 1。

通过研究 $W_\perp$ (125) 的结构，我们观察到 $M \delta b \theta^{-1} W \delta _v \delta \theta^* \bar{p}$, $v, \varphi, \zeta \bar{p}$ 在任意时间 t 的零空间总是包含 $P(\theta)$ (37)，但按定义 $P_\perp$ 是与 $P(\theta)$ 正交的基向量矩阵；因此 $P(\theta)$ 从 $\text{null}(W_\perp(t))$ 中被移除。直观地说，如果 $W_\perp(t)$ 没有持久的非平凡零空间，满足 (130) 的唯一方法是 $\theta_{P_\perp} \delta t$ 收敛到零。我们将在下面的定理中使用这个论点，该定理给出了 $b\theta$ 收敛到 $P(\theta)$ 的充分条件。

5.3.4. $b\theta \delta t$ 收敛到真实参数集 $P(\theta)$ 。定理 2. 对于定理 1 描述的系统以及在 (125) 中定义的 $W_\perp(t)$ ，如果存在 $\epsilon_0, \delta_0 > 0$ ，使得对于任意 $t \geq t_0$ 和任意单位向量 $w \in \mathbb{R}^{(p+a)-r}$ ，

$$\left\| \int_t^{t+\delta_0} W_\perp(\sigma) w d\sigma \right\| \geq \epsilon_0, \quad (134)$$

然后 $\lim_{t \rightarrow \infty} b\theta \delta t \in P(\theta)$ 。

换句话说，如果 $W_\perp(t)$ 是如 (134) 中定义的持久激励 (PE)，那么 $b\theta \delta t$ 会收敛到真实参数集 $P(\theta)$ (37)，并且整体状态 $z(t)$ 会收敛到集合 S (62)。

证明。只需证明 $\lim_{t \rightarrow \infty} \theta_{P_\perp} \delta t = 0$ ，这意味着 $\lim_{t \rightarrow \infty} b\theta \delta t \in P(\theta)$ 。我们通过反证法来证明这个定理。首先，在第 1 部分中我们陈述定理的假设和条件。在第 2 部分中，我们假设 $\lim_{t \rightarrow \infty} \theta_{P_\perp} \delta t \neq 0$ ，而我们将稍后证明这是一个矛盾。在第 3 部分中，我们展示“当 Δv 很小时， $\theta_{P_\perp}$ 是平坦的”，使用 Morgan 和 Narendra (1977) 中描述的方法。最后，在第 4 部分中，我们利用平坦性质和 PE 条件 (134) 找到一个与引理 1 矛盾的结果。

一. 让第 5 节中的所有假设都得到满足，以使定理 1 成立，包括 $\lim_{t \rightarrow \infty} \Delta v(t) = 0$ 。假设 (134) 中的 PE 条件得到满足。引理 1 表明 $\lim_{t \rightarrow \infty} W_\perp \delta(t) \theta_{P_\perp}(t) = 0$ 。我们还定义界 $b, b_\perp > 0$ ，使得

$$\forall t \geq t_0, b_\perp \geq \|W_\perp(t)\| \quad (135)$$

$$\forall t \geq t_0, b \geq \|\mathbb{W}(t, \hat{\theta})\|. \quad (136)$$

这些界存在，因为 $W \delta _v \delta \theta^* \bar{p}$, $v, \varphi, \zeta \bar{p}$ 是 $_v \delta \theta^* \bar{p}$, v, φ, ζ 的有界函数，其有界性已在第 5.2 节讨论过，并且

$$\|W_\perp(t)\| \leq \|M(\hat{\theta})^{-1}\| \|\mathbb{W}(t, \hat{\theta}^*)\| \|\bar{P}_\perp\| \quad (137)$$

其中 $M \delta b \theta^{-1}$ 和 $_v \delta \theta^* \bar{p}$ 已在第 5.2 节中显示存在且有界，因此 $W \delta t, \theta^* \bar{p} \propto W \delta _v \delta \theta^* \bar{p}$, $v, \varphi, \zeta \bar{p}$ 是有界函数的有界函数，并且

$$\|\bar{P}_\perp\| = 1.$$

二. 假设 $\lim_{t \rightarrow \infty} \theta_{P_\perp} \delta t \neq 0$ 。那么存在一个 $\epsilon > 0$ 和一个无限的时间序列 $t_i, i=1, \dots, \infty$ 使得 $k \theta_{P_\perp} \delta t_i \geq \epsilon$ 。

3. 我们将使用 Morgan 和 Narendra (1977) 的子引理 2 的证明中包含的一种方法来证明“当 Δv 很小时， $\theta_{P_\perp}$ 是平的”。给定 $T', f > 0, \exists \epsilon' > 0$ ，使得如果 $k \Delta v(t) \leq \epsilon' \forall t \in [t_i, t_i + T']$ ，则

$$\|\bar{\theta}_{P_\perp}(t) - \bar{\theta}_{P_\perp}(t_i)\| \leq \phi \forall t \in [t_i, t_i + T']. \quad (138)$$

为了验证这一点，给定一个 T' 和 f ，我们设置

$$\epsilon' = \frac{\phi}{T' \lambda_{\max}(\Gamma) b} \quad (139)$$

并注意， $\forall t \in [t_i, t_i + T']$ ，使用 (129) 中 $_ \theta_{P_\perp}$ 的表达式， $\|\bar{P}_\perp\| \propto 1$ 的事实，以及界 b (136)，

$$\|\bar{\theta}_{P_\perp}(t) - \bar{\theta}_{P_\perp}(t_i)\| = \left\| \int_{t_i}^t \dot{\bar{\theta}}_{P_\perp}(\sigma) d\sigma \right\| \quad (140)$$

$$\leq \left\| \int_{t_i}^t \bar{P}_\perp \Gamma \mathbb{W}(\sigma, \hat{\theta}) \Delta v(\sigma) d\sigma \right\| \quad (141)$$

$$\leq \int_{t_i}^t \|\bar{P}_\perp\| \|\Gamma\| \|\mathbb{W}(\sigma, \hat{\theta})\| \|\Delta v(\sigma)\| d\sigma \quad (142)$$

$$\leq \int_{t_i}^t \lambda_{\max}(\Gamma) b \epsilon' d\sigma \quad (143)$$

$$\leq (t - t_i) \lambda_{\max}(\Gamma) b \epsilon' \quad (144)$$

$$\leq T' \lambda_{\max}(\Gamma) b \epsilon' \quad (145)$$

$$\leq \phi. \quad (146)$$

4. Since $\lim_{t \rightarrow \infty} \Delta v(t) = 0$, for any $\epsilon' > 0$ we can always find a $t' \geq t_0$ such that $\|\Delta v(t)\| \leq \epsilon' \forall t \geq t'$. Thus using ϵ_0 from the PE condition (134), ϵ from the assumption made in Part 2, and the bound $b_\perp$ (135), we can choose

$$\phi = \frac{\epsilon_0 \epsilon}{2 \delta_0 b_\perp} \quad (147)$$

$$T' = \delta_0 \quad (148)$$

and find a $t' \geq t_0$ such that the flatness property (138) holds for each interval $[t_i, t_i + \delta_0]$ where $t_i \geq t'$.

We will now show that for each $t_i \geq t'$, there is a $t_j \in [t_i, t_i + \delta_0]$ where $\|W_\perp(t_j) \bar{\theta}_{P_\perp}(t_j)\| > 0$. For each $t_i \geq t'$, we use the reverse triangle inequality and the flatness property (138) to show:

$$\left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(t_i) d\sigma \right\| - \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma) d\sigma \right\| \quad (149)$$

$$\leq \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) [\bar{\theta}_{P_\perp}(t_i) - \bar{\theta}_{P_\perp}(\sigma)] d\sigma \right\| \leq \int_{t_i}^{t_i + \delta_0} \|W_\perp(\sigma)\| \|\bar{\theta}_{P_\perp}(t_i) - \bar{\theta}_{P_\perp}(\sigma)\| d\sigma \quad (150)$$

$$\leq \int_{t_i}^{t_i + \delta_0} b_\perp \phi d\sigma \quad (151)$$

$$\leq \delta_0 b_\perp \phi, \quad (152)$$

We rearrange (152) and choose the unit vector in the PE condition (134) to be

$$w \triangleq \frac{\bar{\theta}_{P_\perp}(t_i)}{\|\bar{\theta}_{P_\perp}(t_i)\|}, \quad (153)$$

so that

$$\left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma) d\sigma \right\| \geq \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(t_i) d\sigma \right\| - \delta_0 b_\perp \phi \quad (154)$$

$$\geq \|\bar{\theta}_{P_\perp}(t_i)\| \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) w d\sigma \right\| - \delta_0 b_\perp \phi. \quad (155)$$

We now invoke the lower bound in the PE condition (134), substitute our choice of ϕ (147), and recall the assumption made in Part 2 that $\|\bar{\theta}_{P_\perp}(t_i)\| \geq \epsilon$. This results in:

$$\left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma) d\sigma \right\| \geq \|\bar{\theta}_{P_\perp}(t_i)\| \epsilon_0 - \delta_0 b_\perp \phi \quad (156)$$

$$\geq \|\bar{\theta}_{P_\perp}(t_i)\| \epsilon_0 - \frac{\epsilon_0 \epsilon}{2} \quad (157)$$

$$\geq \frac{\epsilon_0 \epsilon}{2} \quad (158)$$

$$> 0. \quad (159)$$

Furthermore, we have

$$\int_{t_i}^{t_i + \delta_0} \|W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma)\| d\sigma \geq \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma) d\sigma \right\| \quad (160)$$

$$> 0. \quad (161)$$

By the mean value theorem for integrals, there must be a $t_j \in [t_i, t_i + \delta_0]$ where

$$\int_{t_i}^{t_i + \delta_0} \|W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma)\| d\sigma = \|W_\perp(t_j) \bar{\theta}_{P_\perp}(t_j)\| \delta_0, \quad (162)$$

and therefore from (161, 162)

$$\|W_\perp(t_j) \bar{\theta}_{P_\perp}(t_j)\| \delta_0 > 0 \quad (163)$$

$$\|W_\perp(t_j) \bar{\theta}_{P_\perp}(t_j)\| > 0. \quad (164)$$

Thus, we have found an unbounded sequence of times $\{t_j\}_{j=1, \dots, \infty}$ where $\|W_\perp(t_j) \bar{\theta}_{P_\perp}(t_j)\| > 0$, which contradicts Lemma 1 and implies that the assumption $\lim_{t \rightarrow \infty} \bar{\theta}_{P_\perp}(t) \neq 0$ must be false. Therefore $\lim_{t \rightarrow \infty} \bar{\theta}_{P_\perp}(t) = 0$, which is equivalent to $\lim_{t \rightarrow \infty} \hat{\theta}(t) \in P(\theta)$.

Together with the conclusion in the proof of Theorem one that $\lim_{t \rightarrow \infty} \Delta v(t) = 0$, we have shown a sufficient condition for convergence of the full state $z(t)$ to S , thereby achieving the goals (58,59) of this AID.

6. Simulation results

We report simulation results in which the “true” parameters of the simulated plant were chosen to match approximately those of the JHU Iver3 AUV. We include this simulation study to demonstrate that the AID achieves parameter estimate convergence to the true parameter set. Since the true parameter set of the Iver3 is unknown, we are unable to

$$\leq \int_{t_i}^t \|\bar{P}_\perp\| \|\Gamma\| \|\mathbb{W}(\sigma, \hat{\theta})\| \|\Delta v(\sigma)\| d\sigma \quad (142)$$

$$\leq \int_{t_i}^t \lambda_{\max}(\Gamma) b \epsilon' d\sigma \quad (143)$$

$$\leq (t - t_i) \lambda_{\max}(\Gamma) b \epsilon' \quad (144)$$

$$\leq T' \lambda_{\max}(\Gamma) b \epsilon' \quad (145)$$

$$\leq \phi. \quad (146)$$

4. 由于 $\lim_{t \rightarrow \infty} \Delta v(t) = 0$, 对于任意 $\epsilon' > 0$, 我们总能找到一个 $t' \geq t_0$ 使得 $k \Delta v(t) k \leq \epsilon' \forall t \geq t'$. 因此, 使用 PE 条件 (134) 中的 ϵ_0 、第二部分假设中的 ϵ 以及界 $b_\perp$ (135), 我们可以选择

$$\phi = \frac{\epsilon_0 \epsilon}{2 \delta_0 b_\perp} \quad (147)$$

$$T' = \delta_0 \quad (148)$$

并找到一个 $t' \geq t_0$, 使得平坦性性质 (138) 对每个区间 $[t_i, t_i + \delta_0]$ 都成立, 其中 $t_i \geq t'$.

我们现在将证明, 对于每个 $t_i \geq t'$, 存在一个 $t_j \in [t_i, t_i + \delta_0]$, 使得 $\|W_\perp \delta t_j \bar{P} \theta_{P_\perp} \delta t_j \bar{P}\| > 0$. 对于每个 $t_i \geq t'$, 我们使用逆三角不等式和平坦性性质 (138) 来证明:

$$\left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(t_i) d\sigma \right\| - \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma) d\sigma \right\| \quad (149)$$

$$\leq \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) [\bar{\theta}_{P_\perp}(t_i) - \bar{\theta}_{P_\perp}(\sigma)] d\sigma \right\| \leq \int_{t_i}^{t_i + \delta_0} \|W_\perp(\sigma)\| \|\bar{\theta}_{P_\perp}(t_i) - \bar{\theta}_{P_\perp}(\sigma)\| d\sigma \quad (150)$$

$$\leq \int_{t_i}^{t_i + \delta_0} b_\perp \phi d\sigma \quad (151)$$

$$\leq \delta_0 b_\perp \phi, \quad (152)$$

我们重新排列 (152) 并选择 PE 条件 (134) 中的单位向量为

$$w \triangleq \frac{\bar{\theta}_{P_\perp}(t_i)}{\|\bar{\theta}_{P_\perp}(t_i)\|}, \quad (153)$$

以便

$$\left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma) d\sigma \right\| \geq \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(t_i) d\sigma \right\| - \delta_0 b_\perp \phi \quad (154)$$

$$\geq \|\bar{\theta}_{P_\perp}(t_i)\| \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) w d\sigma \right\| - \delta_0 b_\perp \phi. \quad (155)$$

我们现在在 PE 条件 (134) 中调用下界, 代入我们选择的 f (147), 并回顾第 2 部分中做出的假设 $k \theta_{P_\perp} \delta t_j \bar{P} k \geq \epsilon_0$. 这导致了:

$$\left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma) d\sigma \right\| \geq \|\bar{\theta}_{P_\perp}(t_i)\| \epsilon_0 - \delta_0 b_\perp \phi \quad (156)$$

$$\geq \|\bar{\theta}_{P_\perp}(t_i)\| \epsilon_0 - \frac{\epsilon_0 \epsilon}{2} \quad (157)$$

$$\geq \frac{\epsilon_0 \epsilon}{2} \quad (158)$$

$$> 0. \quad (159)$$

此外, 我们有

$$\int_{t_i}^{t_i + \delta_0} \|W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma)\| d\sigma \geq \left\| \int_{t_i}^{t_i + \delta_0} W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma) d\sigma \right\| \quad (160)$$

$$> 0. \quad (161)$$

根据积分中值定理, 必定存在一个 t_j 属于 $[t_i, t_i + \delta_0]$, 使得

$$\int_{t_i}^{t_i + \delta_0} \|W_\perp(\sigma) \bar{\theta}_{P_\perp}(\sigma)\| d\sigma = \|W_\perp(t_j) \bar{\theta}_{P_\perp}(t_j)\| \delta_0, \quad (162)$$

因此从 (161, 162)

$$\|W_\perp(t_j) \bar{\theta}_{P_\perp}(t_j)\| \delta_0 > 0 \quad (163)$$

$$\|W_\perp(t_j) \bar{\theta}_{P_\perp}(t_j)\| > 0. \quad (164)$$

因此, 我们找到了一个无界的时间序列 $\{t_j\}_{j=1, \dots, \infty}$, 其中 $\|W_\perp \delta t_j \bar{P} \theta_{P_\perp} \delta t_j \bar{P}\| > 0$, 这与引理 1 矛盾, 并且表明假设 $\lim_{t \rightarrow \infty} \theta_{P_\perp} \delta t \bar{P} \neq 0$ 必定是错误的。因此 $\lim_{t \rightarrow \infty} \theta_{P_\perp} \delta t \bar{P} = 0$, 这相当于 $\lim_{t \rightarrow \infty} b \theta \delta t \bar{P} = 2 P(\theta)$.

结合定理一证明中的结论, 即 $\lim_{t \rightarrow \infty} \Delta v(t) = 0$, 我们已经证明了使全状态 $z(t)$ 收敛到 S 的充分条件, 从而实现了该 AID 的目标 (58,59)。

6. 仿真结果

我们报告了模拟结果, 其中模拟系统的“真实”参数被选取为大致匹配 JHU Iver3 AUV 的参数。我们包括这项模拟研究是为了证明 AID 能够实现参数估计收敛到真实参数集。由于 Iver3 的真实参数集未知, 我们无法

verify convergence of the parameter estimates to the “true” parameter set in the experimental results.

For this study, we simulated 5000 seconds of plant motion for the AID process. All available control inputs were excited in open loop to achieve the PE condition (134). A summary of the input signals is provided in Table 1. The set of adaptation gains used in the simulation study are listed in Table 2. We note that with different gain values, it may be possible to achieve more rapid parameter convergence, and that the selection of adaptation gains is an open problem in adaptive systems theory.

Gaussian noise was added to the attitude and velocity signals to mimic the expected measurement noise of the JHU Iver3 AUV sensor suite. All sensors were assumed to be operating at the same frequency, 10 Hz, which is the sampling frequency of the JHU Iver3’s actual MicroStrain 3DM-GX5-25 attitude and heading reference sensor (AHRS). The specific noise characteristics used in this simulation study are provided in Table 3. These values were determined from manufacturer-provided data sheets and bench-top testing.

Figure 1, shows that the estimated parameters converge to “true” parameters in simulation, provided that the PE condition (134) defined in Section 5.3 is satisfied. We note that in order to compute a parameter error between the estimated parameters and the non-unique set of “true” parameters, we employed a scaling factor with respect to a single known parameter.

Table 1. Simulation Study Control Inputs.

Actuator	Training data
Propeller	$3.5 + 4 * \sin(0.7 t) N$
R Fin	$5^\circ + 10^\circ * \sin(0.4 t)$
T Fin	$10^\circ * \sin(0.2 t)$
L Fin	$5^\circ + 10^\circ * \sin(0.4 t)$
B Fin	$10^\circ * \sin(0.2 t)$

Table 2. Simulation Study AID Adaptation Gains.

$\mathcal{A}$	$= \mathbb{I}$
Γ_m	$= 1000 \mathbb{I}$
Γ_d	$= 8000 \mathbb{I}$
$\Gamma_{\mathcal{G}}$	$= \text{diag}([1, 20, 20, 20]^T)$
Γ_a	$= 500 \mathbb{I}$

Table 3. Measurement Sources, Accuracies, and Usage in the Reported Results.

State	Source	Measurement	Simulation	Experimental	Experimental
		Standard Deviation	Results	Results (training)	Results (validation)
Attitude	MicroStrain MSF	0.25°	X	X	
Angular velocity	MicroStrain MST	$0.1^\circ/\text{s}$		X	
Angular velocity	KVH 1775 FOG	$0.01^\circ/\text{s}$	X		X
Trans velocity	RDI 600 kHz DVL	1.3 cm/s	X	X	X

7. Experimental results

This section reports the results of three at-sea field trials to evaluate experimentally the adaptive identification (AID) parameter estimator reported above. We utilized experimental data obtained with JHU’s Iver3 AUV (L3 Ocean-Server, Fall River, MA, USA), shown in Figure 2. We deployed the JHU Iver3 AUV in the Chesapeake Bay, MD, on November 11, 2019, and we report data from three dives (Dive 86, 87, and 88), which are described in Section 7.1.

For these experiments, the Iver3 AUV was operating under “waypoint following” control, wherein the user entered a set of pre-programmed waypoints and the Iver3 proprietary controller determined the specific control inputs required to follow the waypoints.

To estimate the parameters for the Iver3 AUV using the AID, we used signals from a MicroStrain 3DM-GX5-25 (Parker LORD MicroStrain, Willston, VT, USA) which is a microelectromechanical systems (MEMS) AHRS, for the attitude and angular rate, and a Teledyne RDI Explorer (Teledyne RD Instruments, Poway, CA, USA) 600 kHz phased-array Doppler velocity log (DVL) for the translational velocity. The RDI DVL is part of the standard sensor suite on the Iver3 AUV installed by the manufacturer; the MicroStrain AHRS was integrated into the nose cone of the Iver3 AUV by the authors. An auxiliary inertial measurement unit (IMU) sensor, a KVH 1775 (KVH Industries, Middletown, RI, USA) fiber-optic gyroscope (FOG), which measures angular rate, was installed inside the Iver3 for these experimental trials. The KVH FOG, which is more accurate than the MicroStrain MEMS gyro, was used as an external source of “ground truth” for the angular rate DOFs during validation. The sensors used for AID and their noise characteristics are listed in Table 3.

7.1. Dive mission plans

Three experimental trials were conducted:

- Dive 86 was a waypoint-following undulate mission with relatively high variation in motion.
- Dive 87 was a waypoint-following mission with a lawnmower pattern at constant depth
- Dive 88 was the same mission as Dive 86.

An overview of the three dives is given in Figure 3. While it may have been ideal to utilize open-loop control

验证实验结果中参数估计收敛到“真实”参数集的情况。

在本研究中，我们模拟了AID过程5000秒的植物运动。所有可用的控制输入都在开环下被激励以实现PE条件（134）。输入信号的摘要见表1。仿真研究中使用的一组自适应增益列在表2中。我们注意到，使用不同的增益值，可能实现更快速的参数收敛，并且自适应增益的选择是自适应系统理论中的一个未解决问题。

在姿态和速度信号中加入了高斯噪声，以模拟 JHU Iver3 AUV 传感器套件预期的测量噪声。假设所有传感器的工作频率相同，为 10 Hz，这也是 JHU Iver3 实际使用的 MicroStrain 3DM-GX5-25 姿态和航向参考传感器（AHRS）的采样频率。本次仿真研究中使用的特定噪声特性见表 3。这些数值是根据制造商提供的数据表和台式测试确定的。

图 1 表明，只要满足第 5.3 节中定义的 PE 条件 (134)，在仿真中估计参数会收敛到“真实”参数。我们注意到，为了计算估计参数与非唯一“真实”参数集合之间的参数误差，我们使用了相对于一个已知参数的缩放因子。

表 1. 仿真研究控制输入。

Actuator	Training data
Propeller	$3.5 + 4 * \sin(0.7 t) N$
R Fin	$5^{\circ} + 10^{\circ} * \sin(0.4 t)$
T Fin	$10^{\circ} * \sin(0.2 t)$
L Fin	$5^{\circ} + 10^{\circ} * \sin(0.4 t)$
B Fin	$10^{\circ} * \sin(0.2 t)$

表 2. 模拟研究 AID 适应收益。

$\mathcal{A}$	$= \mathbb{I}$
Γ_m	$= 1000 \mathbb{I}$
Γ_d	$= 8000 \mathbb{I}$
$\Gamma_{\mathcal{G}}$	$= \text{diag}([1, 20, 20, 20]^T)$
Γ_a	$= 500 \mathbb{I}$

7. 实验结果

本节报告了三次海上实地试验的结果，以实验性评估上文所述的自适应识别（AID）参数估计器。我们使用了约翰·霍普金斯大学（JHU）的 Iver3 自主水下航行器（AUV）（L3 OceanServer，美国马萨诸塞州福里弗）获得的实验数据，如图 2 所示。我们于2019年11月11日在马里兰州切萨皮克湾部署了 JHU Iver3 AUV，并报告了三次下潜（第86、87和88次下潜）的数据，这些下潜在第7.1节中有所描述。

在这些实验中，Iver3 AUV 在“航点跟随”控制下运行，其中用户输入了一组预编程的航点，Iver3 专有控制器确定了跟随航点所需的具体控制输入。

为了使用AID估算Iver3 AUV的参数，我们使用了来自MicroStrain 3DM-GX5-25（Parker LORD MicroStrain，美国佛蒙特州Willston）的信号，该设备是一种微机电系统（MEMS）姿态与角速度参考系统（AHRS），用于姿态和角速度测量；以及来自Teledyne RDI Explorer（Teledyne RDInstruments，美国加利福尼亚州Poway）600 kHz相控阵多普勒速度计（DVL）的信号，用于平移速度测量。RDI DVL是由制造商安装在Iver3 AUV上的标准传感器套件的一部分；MicroStrain AHRS由作者集成在Iver3 AUV的头锥内。为了这些实验试验，还在Iver3内部安装了一个辅助惯性测量单元（IMU）传感器，即KVH 1775（KVH Industries，美国罗德岛州Middletown）光纤陀螺仪（FOG），用于测量角速度。KVH FOG比MicroStrain MEMS陀螺仪更为精确，在验证过程中用作角速度自由度的“地面真值”外部来源。用于AID的传感器及其噪声

7.1. 潜水任务计划

进行了三次实验试验：

- 潜水86是一个遵循航点的波动任务，运动变化相对较大。
- 潜水87是一个遵循航点的任务，采用恒深度的割草机模式
- 潜水88与潜水86的任务相同。

图 3 给出了这三次潜水的概述。虽然理想情况下可能使用开环控制

标签第3节 测量来源、精度及在报告R中的使用

State	Source	Measurement	Simulation	Experimental	Experimental
		Standard Deviation	Results	Results (training)	Results (validation)
Attitude	MicroStrain MSF	0.25°	X	X	
Angular velocity	MicroStrain MST	0.1°/s		X	
Angular velocity	KVH 1775 FOG	0.01°/s	X		X
Trans velocity	RDI 600 kHz DVL	1.3 cm/s	X	X	X

结果。

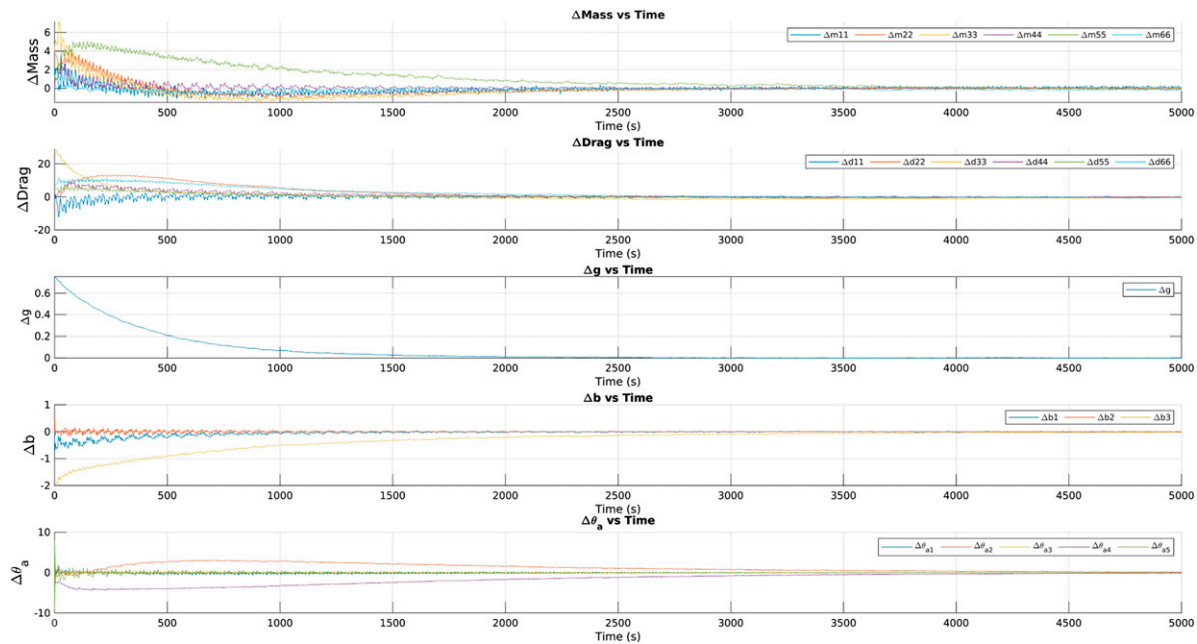


Figure 1. Simulation Study: Difference between true and estimated parameters (Y-Axis) versus time in seconds (X-axis). This Figure shows that the AID-estimated parameters converge to the true parameter set in simulation when a PE condition is satisfied.

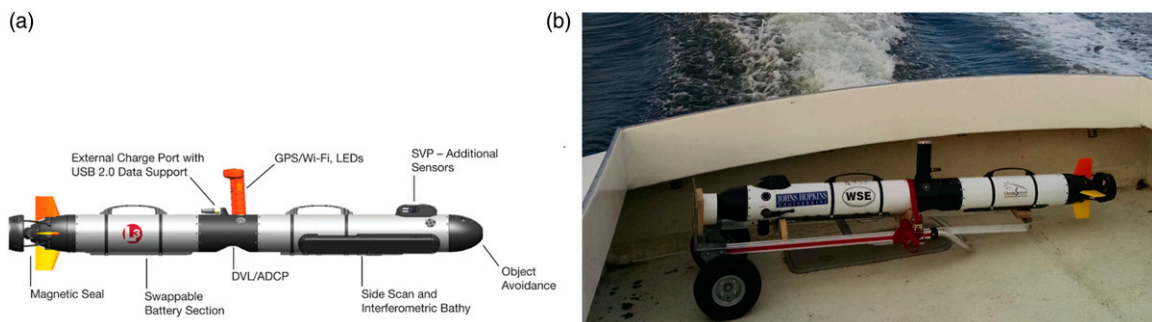


Figure 2. The Iver3 AUV is an underactuated AUV whose control authority is provided by the commanded rotational speed of its ducted propeller and commanded angles for the four red/yellow tail fins, all located at the stern of the vehicle. The 100 m depth-rated Iver3 AUV is one of several commercially available small AUVs designed for oceanographic survey operations including biological, physical-oceanographic, and bathymetric survey missions.

signals for these experiments, as we did in simulation, it would have been infeasible in practice when operating in open water with an untethered UV. Instead, we utilized the manufacturer-provided mission planning tool to enter pre-programmed waypoints for the proprietary controller to follow.

The waypoint-following undulate missions (Dives 86 and 88) were designed to excite the UV dynamics. The UV attempted to follow a series of zig-zagging waypoints while constantly undulating between 1 m and 5 m depths at a fixed 20° pitch angle. The constant-depth lawnmower mission (Dive 87), where the UV followed a series of waypoints in a rectangular grid, was designed to emulate typical seafloor mapping with a multi-beam sonar.

While Dives 86 and 88 employed the same mission, the control inputs determined by the manufacturer's controller

were not identical. External factors such as the UV's initial position and the ambient water current resulted in different control inputs when following the same mission waypoints.

7.2. Experimental evaluation methodology

Our approach to evaluating the performance of the new AID was a “training and validation” procedure, in which we identified the parameters using the AID algorithm on experimental data from one dive, termed the “training” data, and then used those parameters in forward simulations of different dives, termed the “validation” data. We compared the resulting 6-DOF simulated model velocities to the actual experimentally-observed 6-DOF vehicle velocities. Since the control algorithms of the Iver3 feedback control system are not disclosed by the manufacturer, we used

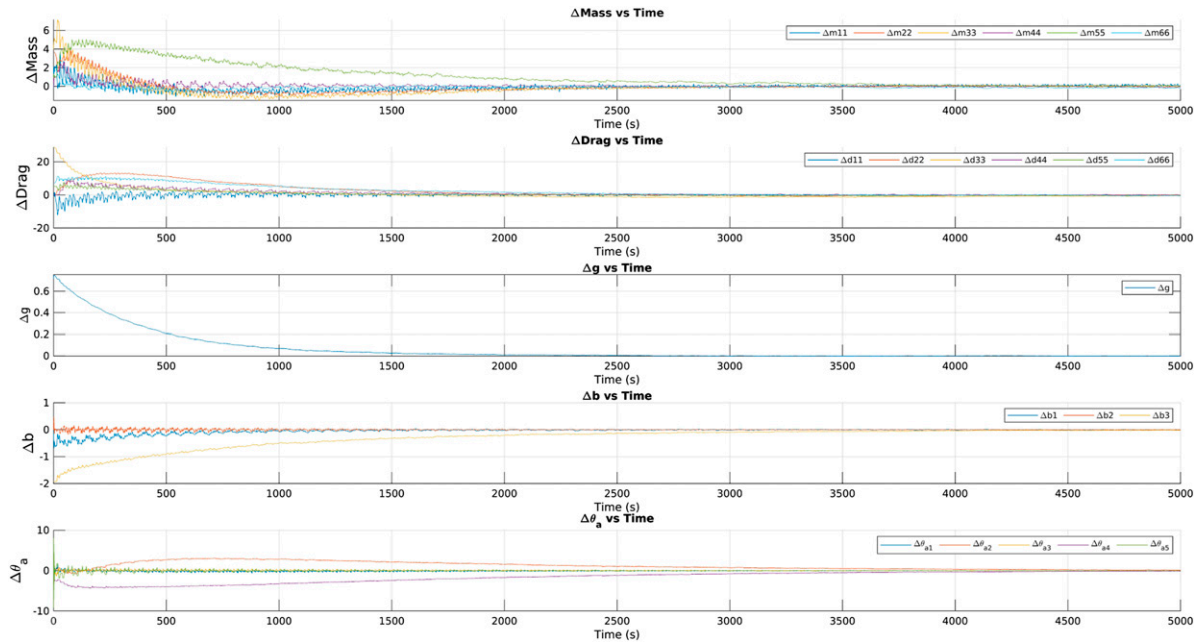


图1. 仿真研究：真实参数与估计参数的差异（Y轴）与时间（秒）（X轴）的关系。该图显示，当满足PE条件时，AID估计的参数在仿真中会收敛到真实参数集。

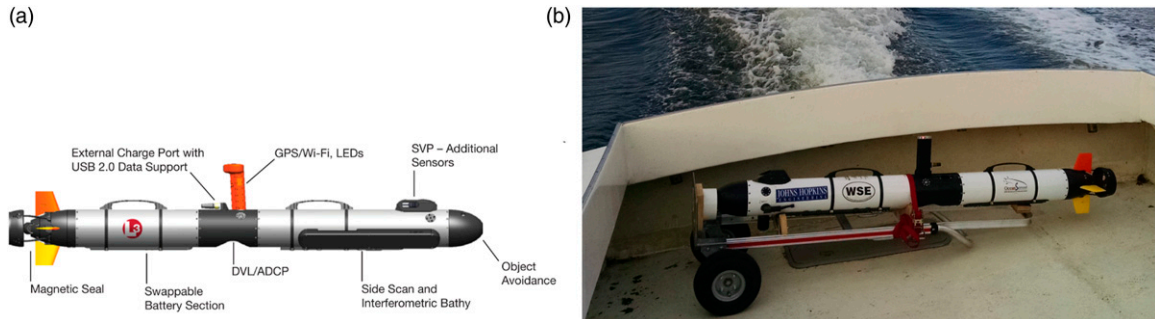


图 2. Iver3 AUV 是一种欠驱动的自主水下航行器，其控制力来源于其管道螺旋桨的指令转速以及位于车辆尾部的四个红色/黄色尾鳍的指令角度。Iver3 AUV 额定潜深为 100 米，是几种商业可获得的小型自主水下航行器之一，设计用于海洋调查任务，包括生物、物理海洋学和水深测量调查任务。

这些实验的信号，就像我们在模拟中所做的那样，在实际使用中在开放水域操作一个无缆UV时是不可行的。相反，我们使用了制造商提供的任务规划工具来输入预编程的航点，以供专有控制器遵循。

航点跟随的起伏任务（潜水86和88）旨在激发水下航行器的动力学。水下航行器试图在固定20°的俯仰角下，在1米到5米的深度之间不断起伏，同时跟随一系列之字形航点。恒深割草机式任务（潜水87）中，水下航行器按照矩形网格中的一系列航点行进，旨在模拟使用多波束声呐进行的典型海底测绘。

虽然潜水器86号和88号执行相同的任务，但控制输入由制造商的控制器决定

并不相同。外部因素，如无人潜航器的初始位置和环境水流，导致在遵循相同任务航点时控制输入不同。

7.2. 实验评估方法

我们评估新型AID性能的方法是“训练与验证”程序，在该程序中，我们使用来自一次潜水的实验数据（称为“训练”数据）通过AID算法识别参数，然后将这些参数用于不同潜水的前向模拟，这些不同潜水的实验数据称为“验证”数据。我们将由此得到的6自由度模拟模型速度与实际实验观测到的6自由度车辆速度进行了比较。由于Iver3反馈控制系统的控制算法未由制造商公开，我们使用了

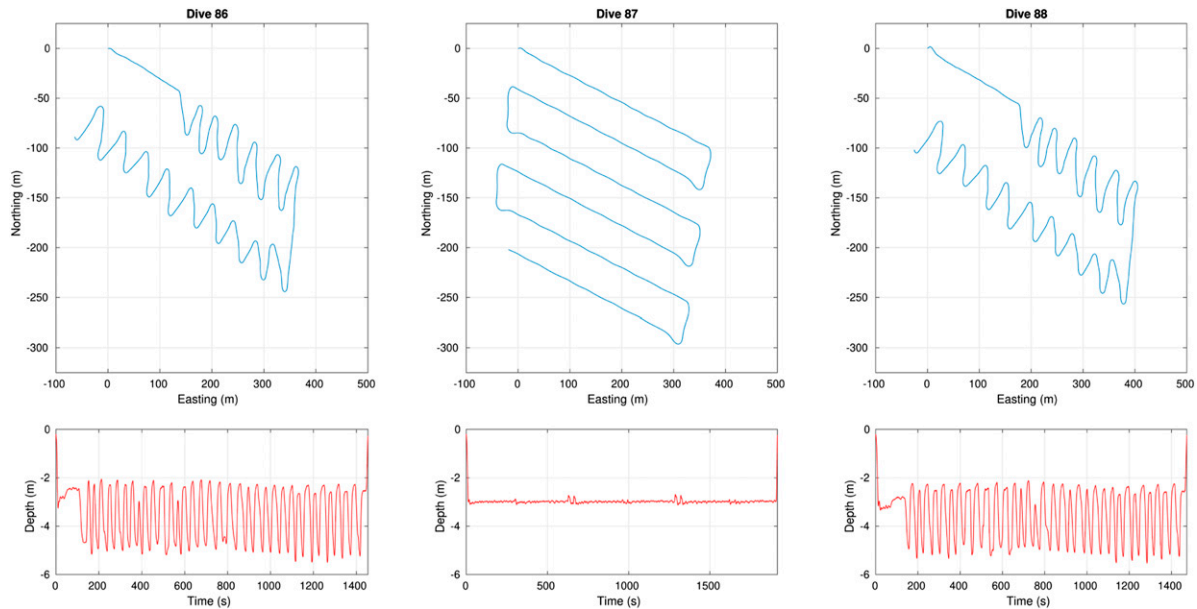


Figure 3. Overview of the three dives conducted in the Chesapeake Bay using the Iver3. The translational position during each dive as estimated by the front seat of the Iver3 is shown above in blue. The depth profile of each dive is shown below in red.

the experimentally-recorded control signals as inputs to the simulation of each dive. We also performed a “self-validation” procedure by computing a forward simulation of the training dive using its own estimated parameters.

The Iver3 AUV control inputs (fin angles and propeller angular velocity) and relevant components of the vehicle state (attitude, translational velocity, and angular velocity) were logged and time-stamped in real-time to the Iver3 AUV’s on-board CPU during the experimental trials. These data were used in post-processing to evaluate the performance of the proposed AID approach. No filtering of any sensor signal was used, the adaptation gains were held constant across all simulations, and the nonlinear differential equations which make up the AID were evaluated using simple Euler integration.

This AID method is well suited for online parameter estimation—assuming that the programming implementation is capable of handling asynchronous sensor readings, the sensor alignments are calibrated, and suitable adaptation gains are chosen. The appropriate adaptation gains can be found using simulation studies or data from previous experiments.

7.2.1. Adaptation gains. During the training procedure, we empirically selected adaptation gains. The adaptation gains were then held constant throughout the self-validation and validation procedure for all dives reported below. We note that the AID reported here is not particularly sensitive to the choice of adaptation gains; that is, small changes in the magnitude of a gain did not result in large changes in performance of the AID.

The adaptation gains for the AID were tuned to maximize performance on the training data set, Dive 86, with the AID active only when the UV was fully submerged. We

found Dive 86 to contain over 24 uninterrupted minutes of maneuvering below the surface. The adaptation gains are given in Table 4.

7.2.2. AID initialization. The parameter estimates for the AID were initialized to values given in Table 5.

7.3. AID performance

We report the performance of the AID as the mean absolute error (MAE), where the error is the difference between the simulated plant velocity resulting from the AID parameters and the velocity measured via the DVL and KVH gyro.

Table 6 reports the MAE when using the parameters estimated from the data of Dive 86 (the “training data”) on a forward simulation of the same dive and of Dives 87 and 88 (the “validation data”). A plot of the body velocity during the validation of Dive 88 is shown in Figure 4.

The parameter set $\hat{\theta}$ that was found after training on Dive 86 is reported in Table 7. As explained in Section 4.3, the dynamics of the AID plant converges to that of the real plant, but the true parameter set is not unique. Moreover, this parameter set is only applicable to the particular parameterization of the plant and actuator model satisfying (34).

7.4. Discussion

The results presented here in Table 6 and Figure 4 show that the ability of a forward simulation of the plant model to match experimentally observed plant performance depends highly on the model parameter values, a well-known fact that has been reported previously, for example in Harris et al. (2018); McFarland and Whitcomb (2021).

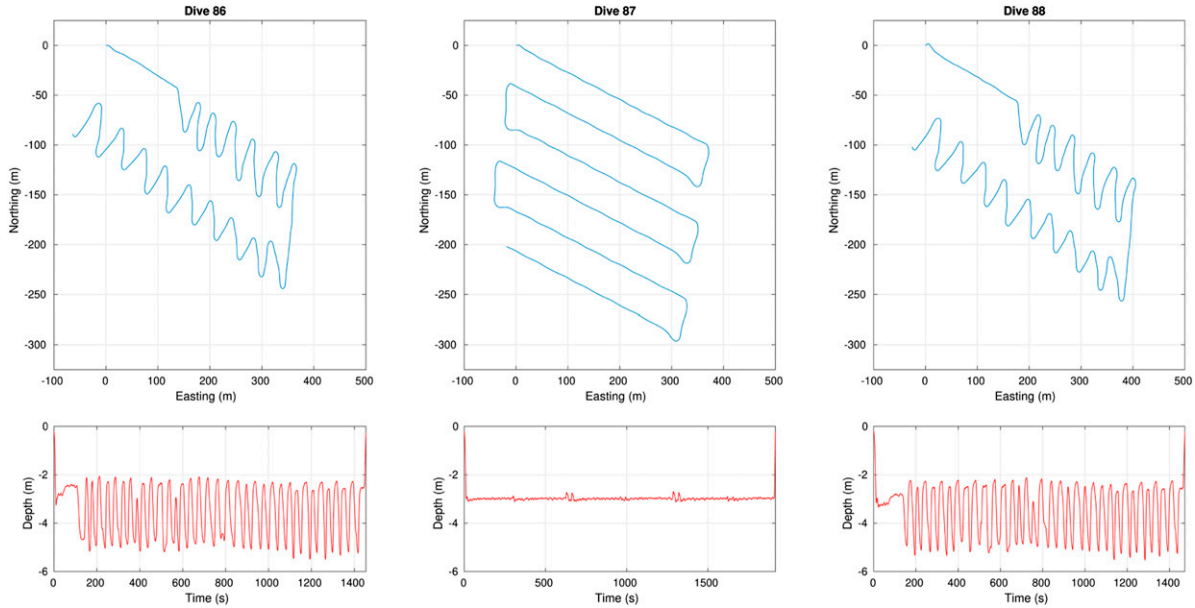


图3. 在切萨皮克湾使用 Iver3 进行的三次潜水概览。图中上方蓝色显示了由 Iver3 前座估算的每次潜水的平移位置。下方红色显示了每次潜水的深度曲线。

将实验记录的控制信号作为每次潜水模拟的输入。我们还通过使用其自身估计的参数计算训练潜水的前向模拟，执行了“自我验证”过程。

Iver3 AUV 的控制输入（鳍角和螺旋桨角速度）以及车辆状态的相关分量（姿态、平移速度和角速度）在实验过程中实时记录并加时间戳到 Iver3 AUV 的车载 CPU 中。这些数据在后处理过程中用于评估所提出的 AID 方法的性能。未对任何传感器信号进行滤波，所有仿真中的自适应增益保持不变，并且组成 AID 的非线性微分方程使用简单的欧拉积分进行求解。

这种AID方法非常适合在线参数估计——假设编程实现能够处理异步传感器读数，传感器对齐已校准，并且选择了合适的自适应增益。可以通过仿真研究或以前实验的数据来找到适当的自适应增益。

7.2.1. 适应增益。在训练过程中，我们通过经验选择了适应增益。随后，在下文报告的所有潜水的自我验证和验证过程中，适应增益保持不变。我们注意到，此处报告的AID对适应增益的选择并不特别敏感；也就是说，增益幅度的微小变化并不会导致AID性能的显著变化。

AID 的自适应增益经过调校，以在训练数据集 Dive 86 上最大化性能，且仅当 UV 完全潜入水中时才激活 AID。我们

发现在Dive 86中，水下连续机动超过24分钟。适应增益见表4。

7.2.2. AID 初始化。AID 的参数估计值初始化为表 5 中给出的值。

7.3. AID 性能

我们报告了AID的性能，作为平均绝对误差（MAE），其中误差是由AID参数产生的模拟植物速度与通过DVL和KVH陀螺仪测量的速度之间的差异。

表6报告了使用从潜水86的数据（“训练数据”）中估计的参数对同一次潜水以及潜水87和88（“验证数据”）的前向模拟时的平均绝对误差（MAE）。潜水88验证过程中身体速度的图示见图4。

在 Dive 86 上训练后得到的参数集 $b\theta$ 列在表 7 中。如第 4.3 节所述，AID 装置的动力学会收敛到真实装置的动力学，但真实参数集并非唯一。此外，该参数集仅适用于满足 (34) 的特定装置和执行器模型参数化。

7.4. 讨论

表6和图4中显示的结果表明，植物模型正向仿真匹配实验观察到的植物性能的能力在很大程度上取决于模型参数值，这一众所周知的事实此前已有报道，例如在Harris等人（2018）；McFarland和Whitcomb（2021）中。

In Section 6, we showed that the adaptively identified parameters converge to the true parameters from an arbitrary initialization in a simulation study (in which the simulated “true” parameters are known) when the PE condition (134)

Table 4. Experimental Study AID Adaptation Gains.

A	$= \text{diag}([6, 7.5, 7.5, 6, 0.75, 0.75]^T)$
Γ_m	$= \text{diag}([34, 2, 2, 2, 34, 34]^T)$
Γ_d	$= \text{diag}([80, 23, 23, 23, 80]^T)$
Γ_G	$= \text{diag}([7.5, 6, 6, 60]^T)$
Γ_a	$= \text{diag}([0.75, 5, 10, 1.5, 10, 5 \times 10^{-12}]^T)$

Table 5. Parameter Estimate Initial Values.

$\hat{\theta}_m(t_0)$	$= [45, 70, 70, 20, 110, 110]^T$
$\hat{\theta}_d(t_0)$	$= [2.73, 245, 245, 0.05, 70, 70]^T$
$\hat{\theta}_G(t_0)$	$= [6, 1.2, 1.0, -11.7353]^T$
$\hat{\theta}_a(t_0)$	$= [0.1, 1.0, 2.0, 0.1, 1.5, 5.8 \times 10^{-6}]^T$

is satisfied. However, we cannot compare the experimentally identified parameters to the true model parameters for two reasons:

- (1) We do not know the true Iver3 AUV model parameters *a priori*.
- (2) As observed in Section 4.5, the “true” parameters are not unique, but may be any element of the set $P(\theta)$ (37).

Thus, the experimental validity of this AID algorithm can be evaluated only by the ability of the estimated parameters to predict the UV’s velocity in forward simulation. As proven in Section 4.5.2, any parameter vector in the set $P(\theta)$ (37) will result in identical vehicle model input-output dynamics.

The results reported in Figure 4 show that the experimentally identified model (Table 7) performs far better than the model using the arbitrarily chosen initial parameter set (Table 5). Further, these results show that the identified parameter set performs approximately as well on the validation data as it does on the training data. We note that the

Table 6. Mean Absolute Error of Simulated 6-DOF UV Plant Using Identified Parameters.

DOF	Training Dive 86	Validation Dive 87	Validation Dive 88
v_1	0.0419 m/s	0.0394 m/s	0.0396 m/s
v_2	0.0439 m/s	0.0619 m/s	0.0428 m/s
v_3	0.0294 m/s	0.0117 m/s	0.0267 m/s
v_4	0.0464 rad/s	0.0284 rad/s	0.0447 rad/s
v_5	0.0288 rad/s	0.0085 rad/s	0.0289 rad/s
v_6	0.0238 rad/s	0.0248 rad/s	0.0245 rad/s

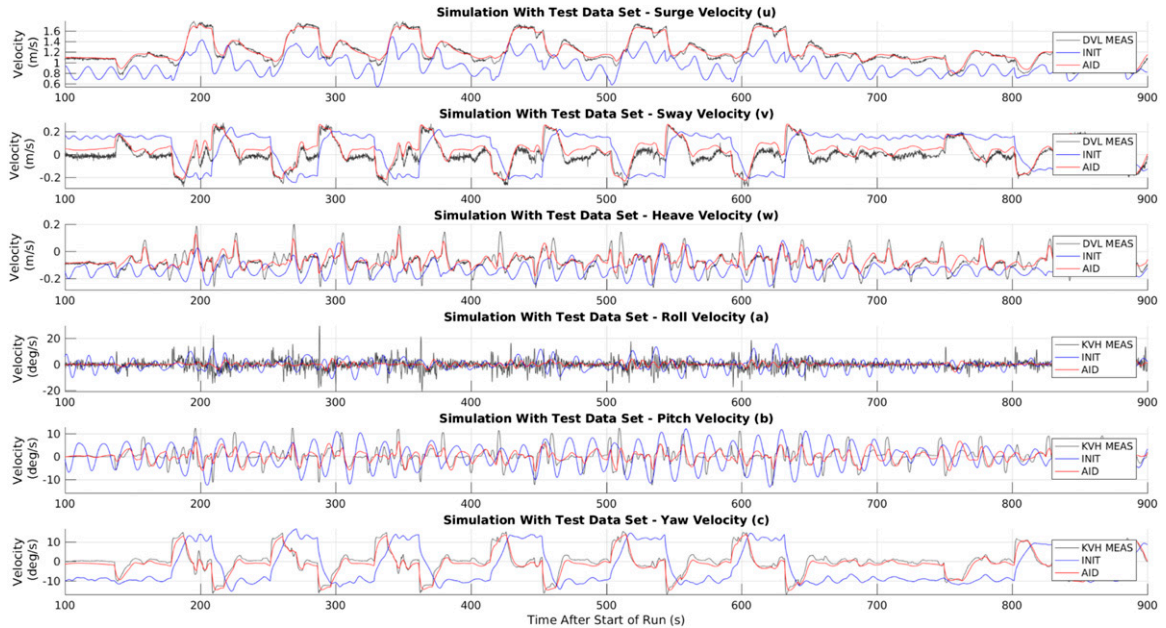


Figure 4. Validation results of Dive 88 using parameters identified during Dive 86. Gray traces are (unfiltered) translational and rotational velocities as measured by the DVL and KVH gyroscope. Blue traces are the plant velocities in forward simulation using the parameters as they were initialized in the AID. Red traces are plant velocities in forward simulation using the parameters identified via the new AID method.

在第6节中, 我们展示了, 在仿真研究中(其中已知模拟的“真实”参数), 当满足PE条件(134)时, 自适应识别的参数从任意初始化会收敛到真实参数。

表4. 实验研究 AID 适应收益。

A	$= \text{diag}([6, 7.5, 7.5, 6, 0.75, 0.75]^T)$
Γ_m	$= \text{diag}([34, 2, 2, 2, 34, 34]^T)$
Γ_d	$= \text{diag}([80, 23, 23, 23, 80]^T)$
Γ_G	$= \text{diag}([7.5, 6, 6, 60]^T)$
Γ_a	$= \text{diag}([0.75, 5, 10, 1.5, 10, 5 \times 10^{-12}]^T)$

表5. 参数估计初始值。

$\hat{\theta}_m(t_0)$	$= [45, 70, 70, 20, 110, 110]^T$
$\hat{\theta}_d(t_0)$	$= [2.73, 245, 245, 0.05, 70, 70]^T$
$\hat{\theta}_G(t_0)$	$= [6, 1.2, 1.0, -11.7353]^T$
$\hat{\theta}_a(t_0)$	$= [0.1, 1.0, 2.0, 0.1, 1.5, 5.8 \times 10^{-6}]^T$

表6. 使用识别参数模拟的6自由度水下机器人植物的平均绝对误差。

DOF	Training Dive 86	Validation Dive 87	Validation Dive 88
v_1	0.0419 m/s	0.0394 m/s	0.0396 m/s
v_2	0.0439 m/s	0.0619 m/s	0.0428 m/s
v_3	0.0294 m/s	0.0117 m/s	0.0267 m/s
v_4	0.0464 rad/s	0.0284 rad/s	0.0447 rad/s
v_5	0.0288 rad/s	0.0085 rad/s	0.0289 rad/s
v_6	0.0238 rad/s	0.0248 rad/s	0.0245 rad/s

感到满意。然而, 由于两个原因, 我们无法将实验确定的参数与真实模型参数进行比较:

- (1) 我们事先不知道真正的 Iver3 AUV 模型参数。
- (2) 如第 4.5 节所观察到的, “真实” 参数并非唯一, 而可以是集合 $P(\theta)$ (37) 的任何元素。

因此, 这个AID算法的实验有效性只能通过估计参数预测UV在前向仿真中的速度的能力来评估。正如在第4.5.2节中证明的那样, 集合 $P(\theta)$ (37)中的任何参数向量都会导致相同的车辆模型输入输出动态。

图4中报告的结果表明, 实验中识别的模型(表7)的表现远优于使用随意选择的初始参数集(表5)的模型。此外, 这些结果显示识别的参数集在验证数据上的表现大致与在训练数据上的表现相当。我们注意到,

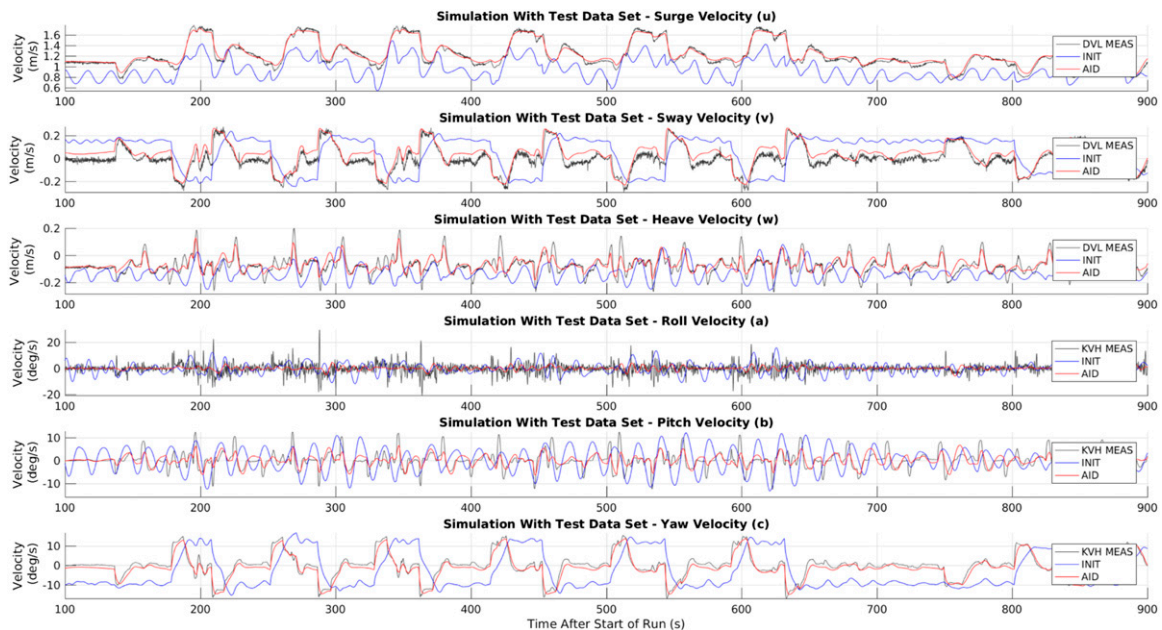


图4. 使用在潜水86中识别的参数对潜水88的验证结果。灰色曲线是DVL和KVH陀螺仪测量的(未滤波的)平移速度和旋转速度。蓝色曲线是在使用AID中初始化的参数进行前向模拟时的系统速度。红色曲线是在使用通过新的AID方法识别的参数进行前向模拟时的系统速度。

Table 7. Estimated Parameters for a Plant and Actuator Model (34) of the Iver3 3026, With Components Described in Section 4.3.

Param	Value	Param	Value	Param	Value	Param	Value
m_{11}	50.5 kg	d_{11}	-15.5 kg/m	g	3.5 N	θ_{a3}	0.68
m_{22}	70.5 kg	d_{22}	243.7 kg/m	b_1	-0.73 N·m	θ_{a4}	1.44
m_{33}	69.5 kg	d_{33}	242.8 kg/m	b_2	0.80 N·m	θ_{a5}	0.56
m_{44}	19.8 kg·m ²	d_{44}	-6.98 kg·m ²	b_3	-12.5 N·m	θ_{a6}	7.7×10^{-6} kg·m
m_{55}	109.2 kg·m ²	d_{55}	-80.9 kg·m ²	θ_{a1}	1.3		
m_{66}	109.9 kg·m ²	d_{66}	60.5 kg·m ²	θ_{a2}	0.0048		

angular-velocity data obtained by the more accurate KVH sensor were not used in the training of the AID and were used only to provide as close to a source of “ground truth” as is available for angular velocity.

We observed that the performance of the identified model is worst in sway and roll DOF. We believe this phenomenon is caused by the lack of precisely controlled excitation in these DOF.

8. Conclusion

This paper reports the theory and experimental evaluation of a novel adaptive identification (AID) algorithm for the simultaneous estimation of plant-model parameters and actuator parameters for underactuated vehicles in six degrees-of-freedom (DOF).

The simulation and experimental results reported herein indicate that it is feasible to use the AID algorithm for parameter estimation of underactuated underwater vehicles (UVs) in 6-DOF. Underactuated UVs are by far the most common class of UVs, but parameter estimation is more difficult for underactuated UVs than for fully actuated UVs because the reduced actuation necessarily reduces the plant excitation that can be induced from the control inputs. Additionally, underactuated UVs are often controlled with hydrodynamic control surfaces which are difficult to characterize.

The theoretical results reported herein provide the first reported AID algorithm with proof of convergence of the parameter estimates to the true set of parameters for this class of vehicles when a persistence of excitation condition is satisfied. Most previously reported AID approaches for this class of systems only show stability and boundedness of the parameter estimates.

The simulation and experimental results show that parameters estimated using this AID can converge with a “single training set” containing on the order of 1000 s of free-motion AUV dive data. The performance of the resulting model was verified in both self-validation and cross-validation. Additionally, this AID algorithm does not require access to body linear and angular acceleration signals, which can offer an advantage over other parameter-estimation methods.

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Appendix A

UV control inputs

This section describes a model of actuator forces and moments represented by the term $\tau(v, \varphi, \zeta)$ defined in (23), for a class of torpedo-shaped UVs which includes the JHU Iver3. The JHU Iver3 is actuated with a combination of control surfaces and a propeller, as is common in many aerial and underwater vehicles.

We note that the AID described and utilized herein requires the *value* of the control input signals, ζ , but it is completely agnostic to the specific control law utilized by the UV.

表7. Iver3 3026的植物和执行器模型(34)的估计参数, 组件在第4.3节中描述。

Param	Value	Param	Value	Param	Value	Param	Value
m_{11}	50.5 kg	d_{11}	-15.5 kg/m	g	3.5 N	θ_{a3}	0.68
m_{22}	70.5 kg	d_{22}	243.7 kg/m	b_1	-0.73 N·m	θ_{a4}	1.44
m_{33}	69.5 kg	d_{33}	242.8 kg/m	b_2	0.80 N·m	θ_{a5}	0.56
m_{44}	19.8 kg·m ²	d_{44}	-6.98 kg·m ²	b_3	-12.5 N·m	θ_{a6}	7.7×10^{-6} kg·m
m_{55}	109.2 kg·m ²	d_{55}	-80.9 kg·m ²	θ_{a1}	1.3		
m_{66}	109.9 kg·m ²	d_{66}	60.5 kg·m ²	θ_{a2}	0.0048		

通过更精确的 KVH 传感器获得的角速度数据未用于 AID 的训练, 只用于提供尽可能接近角速度“真实来源”的数据。

我们观察到所识别模型在横摆和横滚自由度的性能最差。我们认为这一现象是由于在这些自由度中缺乏精确控制的激励所致。

8. 结论

本文报道了一种新型自适应识别 (AID) 算法的理论和实验评估, 该算法用于六自由度 (DOF) 欠驱动飞行器的植物模型参数和执行器参数的同时估计。

本文报告的仿真和实验结果表明, 使用 AID 算法对 6 自由度受控不足水下航行器 (UVs) 进行参数估计是可行的。受控不足的水下航行器是迄今为止最常见的一类水下航行器, 但对受控不足的水下航行器进行参数估计比对全驱动水下航行器更困难, 因为驱动力的减少必然降低了可从控制输入中产生的系统激励。此外, 受控不足的水下航行器通常由水动力控制面控制, 而这些控制面难以表征。

本文报道的理论结果提供了首个已报告的自适应参数辨识 (AID) 算法, 并证明在满足持久激励条件时, 该类车辆的参数估计会收敛到真实参数集。此前报道的大多数针对该类系统的自适应参数辨识方法仅展示了参数估计的稳定性和有界性。

仿真和实验结果表明, 使用该 AID 估计的参数可以通过包含约 1000 次自由运动 AUV 下潜数据的“单一训练集”收敛。所得模型的性能在自我验证和交叉验证中均得到了验证。此外, 该 AID 算法不需要获取机体线性和角加速度信号, 这相对于其他参数估计方法具有一定优势。

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附录 A

紫外线控制输入

本节描述了一种执行器力和力矩的模型, 该模型由公式(23)中定义的项 $\tau(v, \varphi, \zeta)$ 表示, 适用于包括 JHU Iver3 在内的一类鱼雷形无人潜航器。JHU Iver3 采用舵面和螺旋桨组合方式驱动, 这在许多空中和水下车辆中很常见。

我们注意到, 此处描述和使用的 AID 需要控制输入信号 ζ 的值, 但它对 UV 所采用的具体控制律完全不敏感。

We define the following coordinate frames for each fin:

- V – Vehicle coordinate frame, centered at the UV's center of pressure (CP) with positive x forward, positive y starboard, and positive z down.
- F – Fin coordinate frame, centered at the fin CP when the fin is at commanded fin angle δ , with the x -axis along the chord line of the fin and the y -axis pointing away from the center line of the vehicle.
- F_0 – Fin coordinate frame at $\delta = 0$.
- W – Flow coordinate frame, corresponding to the flow of water across the fin, where α is the angle of attack of the incident flow with respect to the chord of the fin.

We note that the commanded fin angle, δ , is not the fin angle of attack to incident flow, α , so that the F and W frames are generally not coincident. The position of the i th fin in the vehicle frame is specified by ${}^V p_i \in \mathbb{R}^3$, the vector from the vehicle's center of gravity (CG) to the CP of the i th fin, and ${}^V \phi_i \in \mathbb{R}$ is the angular position of the i th fin in the vehicle frame.

We define the transformations between coordinate frames of each fin using rotation matrices $R \in SO(3)$ as follows:

$${}^V R_i = {}^{V_{F_0}} R_i {}^{F_0} R_i \in SO(3) \quad (165)$$

$${}^V R_i = {}^F R_i {}^F W R_i \in SO(3) \quad (166)$$

where the notation ${}^V R_i$ is the transformation from the fin frame to the vehicle frame and ${}^F W R_i$ is the transformation from the flow frame to the vehicle frame. The individual transformations are defined as

$${}^{V_{F_0}} R_i = R_x(V_{\phi_i}) \quad (167)$$

$${}^{F_0} R_i = R_y(\delta_i) \quad (168)$$

$${}^F W R_i = R_y^T(\alpha_i), \quad (169)$$

where $R_x, R_y \in SO(3)$ are elemental rotations about the x - and y -axis, respectively. The form of (167) is assumed because, for the JHU Iver3 used to obtain the experimental results presented in Section 7 and many other UVs, the x -axis of the fin frame aligns with the x -axis of the vehicle frame.

The velocity of the i th fin through the water at the fin CP in vehicle coordinates is

$${}^V \dot{p}_i = v + (\omega)^\wedge \cdot {}^V p_i, \quad (170)$$

and the velocity of the i th fin in the fin frame F is thus

$$F \dot{p}_i = {}^F R_i^T {}^V \dot{p}_i. \quad (171)$$

Assuming flow along the span of the fin foil does not affect the lift or drag, we use a projection matrix to find the flow along the x - and z -axes

$$F \dot{p}_{xz_i} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} F \dot{p}_i, \quad (172)$$

from which we can find the angle of attack, α , as

$$\alpha_i = \text{atan2}({}^F \dot{p}_{z_i}, {}^F \dot{p}_{x_i}), \quad (173)$$

where atan2 is the four-quadrant arc-tangent function.

We then compute the hydrodynamic lift and drag force acting on each fin as

$$L(\alpha_i) = \frac{1}{2} \rho A C_L(\alpha_i) \left\| {}^F \dot{p}_{xz_i} \right\|^2 \quad (174)$$

$$D(\alpha_i) = \frac{1}{2} \rho A C_D(\alpha_i) \left\| {}^F \dot{p}_{xz_i} \right\|^2, \quad (175)$$

where A is the surface area of the fin. The lift and drag coefficients of the fin, $C_L(\alpha_i)$ and $C_D(\alpha_i)$, respectively, are a function of the angle of attack. We parameterize these functions as polynomials. The lift coefficient of each fin is modeled as a two-parameter cubic polynomial:

$$C_L(\alpha_i) = \theta_{a1} \alpha_i + \theta_{a2} \alpha_i^3. \quad (176)$$

The drag coefficient of each fin is modeled as a three-parameter quadratic polynomial:

$$C_D(\alpha_i) = \theta_{a3} + \theta_{a4} \alpha_i^2 + \theta_{a5} \alpha_i^4. \quad (177)$$

These polynomial lift and drag coefficient models approximate the lift and drag coefficients from a symmetric NACA foil shape.

The resultant force vector in the flow frame W is ${}^W f = -[D \ 0 \ L]^T$. Thus, the force vector from the i th fin in the vehicle frame is

$$V_{f_i} = {}^V R_i {}^W f_i, \quad (178)$$

and the force and moment vector from the i th fin is

$$\tau_i = \begin{bmatrix} {}^V f_i \\ (V r_i)^\wedge \cdot {}^V f_i \end{bmatrix}. \quad (179)$$

The total force and moment vector on a typical under-actuated torpedo-like UV with a total of N control surfaces and one propeller is thus

$$\tau(v, \phi, \zeta) = \sum_i^N \tau_i(v, \phi, \zeta) + [\theta_{a6} \omega_p^2(t) \ 0_{1 \times 5}]^T, \quad (180)$$

where θ_{a6} is the propeller thrust coefficient relating radial velocity ω_p to axial thrust T :

$$T = \theta_{a6} \omega_p^2. \quad (181)$$

This simplified thruster model is a reasonable assumption for class of UVs because the propeller is ducted with a

我们为每个鳍定义以下坐标系：

- V – 车辆坐标系，以无人艇的压力中心 (CP) 为中心，x轴正方向向前，y轴正方向向右舷，z轴正方向向下。
- F – 鳍坐标系，当鳍处于指令的鳍角 δ 时，以鳍的质心 CP 为中心，x 轴沿鳍的弦线方向，y 轴指向远离飞行器中心线的方向。
- F0 – 在 $\delta = 0$ 处的鳍坐标系。
- W – 流动坐标系，对应于水流通过鳍的流动，其中 α 是入流相对于鳍弦线的攻角。

我们注意到，指令的尾翼角 δ 并不是遇到的气流尾翼攻角 α ，因此 F 和 W 坐标系通常不重合。第 i 个尾翼在载具坐标系中的位置由 ${}^V p_i \in \mathbb{R}^3$ 指定，即从载具质心 (CG) 到第 i 个尾翼受力点 (CP) 的向量，而 ${}^V f_i \in \mathbb{R}^3$ 是第 i 个尾翼在载具坐标系中的角位置。

我们使用旋转矩阵 $R \in SO(3)$ 来定义每个鳍的坐标系之间的变换，如下所示：

$${}^V R_i = {}^{V_0} R_i {}^{F_0} R_i \in SO(3) \quad (165)$$

$${}^V R_i = {}^V R_i {}^F R_i \in SO(3) \quad (166)$$

其中符号 ${}^V R_i$ 表示从鳍框架到车辆框架的变换，而 ${}^V R_i$ 表示从流动框架到车辆框架的变换。各个变换定义如下

$${}^{V_0} R_i = R_x(V_{\phi_i}) \quad (167)$$

$${}^{F_0} R_i = R_y(\delta_i) \quad (168)$$

$${}^F R_i = R_y^T(\alpha_i), \quad (169)$$

其中 $R_x, R_y \in SO(3)$ 分别是绕 x 轴和 y 轴的基本旋转。采用 (167) 的形式是因为，对于用于获得第 7 节中实验结果的 JHU Iver3 以及许多其他水下车辆，鳍架的 x 轴与车辆坐标系的 x 轴对齐。

在车辆坐标系中，第 i 个鳍在鳍重心处穿过水的速度为

$${}^V \dot{p}_i = v + (\omega)^\wedge \cdot {}^V p_i, \quad (170)$$

因此，第 i 个鳍在鳍坐标系 F 中的速度为

$$F \dot{p}_i = {}^V R_i^T \cdot {}^V \dot{p}_i. \quad (171)$$

假设沿翅片翼型的展长方向的流动不会影响升力和阻力，我们使用投影矩阵来求沿 x 轴和 z 轴的流动

$$F \dot{p}_{xz_i} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} F \dot{p}_i, \quad (172)$$

从中我们可以找到攻角， α ，如

$$\alpha_i = \text{atan2}(F \dot{p}_{zi}, F \dot{p}_{xi}), \quad (173)$$

其中 atan2 是四象限反正切函数。

然后我们计算作用在每个鳍上的流体动力升力和阻力

$$L(\alpha_i) = \frac{1}{2} \rho A C_L(\alpha_i) \|F \dot{p}_{xz_i}\|^2 \quad (174)$$

$$D(\alpha_i) = \frac{1}{2} \rho A C_D(\alpha_i) \|F \dot{p}_{xz_i}\|^2, \quad (175)$$

其中 A 是鳍片的表面积。鳍片的升力系数和阻力系数，分别为 $C_L(\alpha_i)$ 和 $C_D(\alpha_i)$ ，是攻角的函数。我们将这些函数参数化为多项式。每个鳍片的升力系数被建模为一个两参数的三次多项式：

$$C_L(\alpha_i) = \theta_{a1} \alpha_i + \theta_{a2} \alpha_i^3. \quad (176)$$

每个鳍的阻力系数被建模为三参数二次多项式：

$$C_D(\alpha_i) = \theta_{a3} + \theta_{a4} \alpha_i^2 + \theta_{a5} \alpha_i^4. \quad (177)$$

这些多项式升力和阻力系数模型近似于对称 NACA 翼型的升力和阻力系数。

在流动坐标系 W 中的合力向量为 ${}^W f_i = \frac{1}{4} D_i \mathbf{0} L_i^T$ 。因此，第 i 个鳍在车辆坐标系中的力向量为

$$V_{f_i} = {}^V R_i {}^W f_i, \quad (178)$$

来自第 i 个鳍的力和力矩向量是

$$\tau_i = \begin{bmatrix} V_{f_i} \\ (V_{r_i})^\wedge \cdot V_{f_i} \end{bmatrix}. \quad (179)$$

因此，对于一个典型的欠驱动鱼雷型无人潜航器，其总共有 N 个控制面和一个推进器，总力和力矩向量为

$$\tau(v, \varphi, \zeta) = \sum_i^N \tau_i(v, \varphi, \zeta) + [\theta_{a6} \omega_p^2(t) \quad \mathbf{0}_{1 \times 5}]^T, \quad (180)$$

其中 θ_{a6} 是螺旋桨推力系数，将径向速度 ω_p 与轴向推力 T 相关联：

$$T = \theta_{a6} \omega_p^2. \quad (181)$$

这个简化的推进器模型对于这一类无人水下航行器是一个合理的假设，因为螺旋桨是带导管的

high jet velocity compared to the advance velocity of the vehicle Newman (1977).

We have described a model of actuator forces and moments which act on the UV, and this model is expressed as $\tau(v, \varphi, \xi)$ with control inputs

$$\xi = [\delta_1 \quad \delta_2 \quad \delta_3 \quad \delta_4 \quad \omega_p]^T. \quad (182)$$

While $\tau(v, \varphi, \xi)$ is a nonlinear function of v, φ , and ξ , it is linear in the actuator parameters θ_a , defined in this case as

$$\theta_a = [\theta_{a1} \quad \theta_{a2} \quad \theta_{a3} \quad \theta_{a4} \quad \theta_{a5} \quad \theta_{a6}]^T \in \mathbb{R}^6. \quad (183)$$

Appendix B

Parameter convergence

B.1. Barbalat's Lemma

[Slotine and Li (1991) Lemma 4.2]

If the differentiable function $f(t)$ has a finite limit as $t \rightarrow \infty$, and if $\dot{f}$ is uniformly continuous, then $\dot{f}(t) \rightarrow 0$ as $t \rightarrow \infty$

B.2. Uniform Continuity of $\Delta \dot{v}$

We observe that $\Delta \dot{v}$ is composed of sums and products of bounded functions which are UC in time, since examining its expression in (121) term-by-term:

- $\mathcal{A}$ is constant and bounded by assumption;
- Δv is UC in time from the boundedness of $\Delta \dot{v}$ (77), which was shown in Section 5.2;
- $\hat{\theta}$ is UC in time from the boundedness of $\dot{\hat{\theta}}$ (64), which was shown in Section 5.2;
- $M(\hat{\theta})^{-1}$ is UC in time from the boundedness of its derivative

$$\frac{d}{dt} [M(\hat{\theta})^{-1}] = -M(\hat{\theta})^{-1} \left[\frac{d}{dt} M(\hat{\theta}) \right] M(\hat{\theta})^{-1} \quad (184)$$

$$= -M(\hat{\theta})^{-1} \dot{M}(\hat{\theta}) M(\hat{\theta})^{-1} \quad (185)$$

since, as shown in Section 5.2, $M(\hat{\theta})^{-1}$ exists and is bounded, and $\dot{\hat{\theta}}$ is bounded; and

- $\mathbb{W}(\dot{v}(\theta^*), v, \varphi, \xi)$ is a UC function of $\dot{v}(\theta^*), v, \varphi, \xi$, where
 - $\dot{v}(\theta^*)$ is the product of constants $\theta^*, M(\theta^*)^{-1}$ and the function $W_v(v, \varphi, \xi)$, which is assumed UC in v, φ , and ξ ,
 - v is UC in time from the boundedness of $\dot{v}(\theta^*)$ (46), which was discussed in Section 5.2,
 - φ is UC in time since its derivative (8) is locally bounded given the assumed boundedness of η, v , and
 - ξ is UC in time by assumption.

Thus, we can conclude that $\Delta \dot{v}$ is UC in time under the above assumptions.

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我们已经描述了作用于UV的执行器力和力矩模型, 该模型表示为 $\tau(v, \varphi, \xi)$, 并具有控制输入

$$\xi = [\delta_1 \quad \delta_2 \quad \delta_3 \quad \delta_4 \quad \omega_p]^T. \quad (182)$$

虽然 $\tau(v, \varphi, \xi)$ 是 v, φ 和 ξ 的非线性函数, 但它在执行器参数 θ_a 中是线性的, 在这种情况下定义为

$$\theta_a = [\theta_{a1} \quad \theta_{a2} \quad \theta_{a3} \quad \theta_{a4} \quad \theta_{a5} \quad \theta_{a6}]^T \in \mathbb{R}^6. \quad (183)$$

附录 B

参数收敛

B.1. 巴尔巴拉的引理

[Slotine 和 Li (1991) 引理 4.2]

如果可微函数 $f(t)$ 在 $t \rightarrow \infty$ 时有有限的极限, 并且 $\dot{f}$ 是一致连续的, 那么 f 在 $t \rightarrow \infty$ 时趋于 0

B.2. Δ_v 的一致连续性

我们观察到 Δ_v 是由有界函数的和与积组成的, 这些函数在时间上是一致连续的, 因为逐项检查其在 (12) 中的表达式:

- A 是常数, 并且根据假设是有界的;
- Δ_v 由于 $\Delta_v(77)$ 的有界性是时间上均匀连续的, 这在第5.2节中已经展示;
- $b\theta$ 由于 $b\theta(64)$ 的有界性是时间上均匀连续的, 这在第5.2节中已经展示;
- $M\delta b\theta P^{-1}$ 由于其导数的有界性是时间上均匀连续的

$$\frac{d}{dt} [M(\hat{\theta})^{-1}] = -M(\hat{\theta})^{-1} \left[\frac{d}{dt} M(\hat{\theta}) \right] M(\hat{\theta})^{-1} \quad (184)$$

$$= -M(\hat{\theta})^{-1} \dot{M}(\hat{\theta}) M(\hat{\theta})^{-1} \quad (185)$$

由于如第5.2节所示, $M\delta b\theta P^{-1}$ 存在且有界, 且 $b\theta$ 有界; 并且

- $\mathbb{W}(\dot{v}(\theta^*), v, \varphi, \xi)$ is a UC function of $\dot{v}(\theta^*), v, \varphi, \xi$, where
 - $\dot{v}(\theta^*)$ is the product of constants $\theta^*, M(\theta^*)^{-1}$ and the function $\mathbb{W}_v(v, \varphi, \xi)$, which is assumed UC in v, φ , and ξ ,
 - v is UC in time from the boundedness of $\dot{v}(\theta^*)$ (46), which was discussed in Section 5.2,
 - φ is UC in time since its derivative (8) is locally bounded given the assumed boundedness of η, v , and
 - ξ is UC in time by assumption.

因此, 我们可以在上述假设下得出结论, Δ_v 在时间上是 UC 的。

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